Pamrevlumab plus nab-paclitaxel/gemcitabine (Pam + GA) as first- and second-line therapy in metastatic pancreatic cancer (mPDAC): Results from Precision Promise (PrP) Bayesian platform trial.

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Background: Pamrevlumab (Pam) is a fully human recombinant monoclonal antibody against connective tissue growth factor. Early clinical data with Pam plus chemotherapy showed a favorable safety profile and potential efficacy in PDAC. In PrP, Pam + GA was tested as first line (Line 1) and second line (Line 2) therapy for mPDAC vs GA. PrP is a phase 2/3, innovative Bayesian adaptive platform trial sponsored by the Pancreatic Cancer Action Network testing multiple experimental arms efficiently against common controls (1). **Methods:** Randomization is 70% (adaptive amongst experimental arms in stage 1) and 15%:15% amongst two control arms (GA, mFOLFIRINOX). Pam + GA graduates from stage 1 to 2 if the Bayesian predictive power (PP) of eventual success is ≥ 35% for one of the arm's signatures [Line 1, Line 2 or Line 1 & 2 (All)]. All participants (pts) are followed for 12 months (mos) after last pt randomized and treated in Pam + GA. Stages 1 and 2 are combined for final analysis. Efficacy is defined by overall survival (OS) hazard ratio (HR, experimental vs control), by a Bayesian statistical model. Superiority (HR < 1) is claimed at final analysis if the Bayesian probability of superiority is ≥ 98%. **Results:** Pam + GA entered PrP in Jun 2021. Between 6/2021 – 1/2023, at 23 US sites, 317 pts were treated and are included in the mITT analysis. 213 pts were treated in the Pam + GA arm (102 Line 1; 111 Line 2), 45 in the GA arm (23 Line 1; 22 Line 2), 31 in the mFOLFIRINOX arm and 28 in other experimental arms. All treated pts before and during Pam + GA enrollment were in the Bayesian model, with outcomes adjusted via a time machine to increase the trial's statistical power. Baseline characteristics were balanced across the arms. Pam + GA met criteria to graduate to stage 2 in Sep 2022 in the All signature (Line 1 & 2). At final analysis, Pam + GA did not meet the OS primary endpoint [model estimated HR: 1.18 (95% credible interval, 0.88, 1.56), posterior Pr(HR < 1) = 0.14, below specified ≥ 0.98]. No benefit was seen in PFS nor ORR (Table). No new safety signals were seen. **Conclusions:** PrP, the first Bayesian platform trial in mPDAC, performed as designed; Pam + GA did not improve OS versus GA in Line 1 and Line 2 mPDAC. The novel Bayesian design warrants further study in mPDAC to enhance efficiency of drug development. Future designs should explore different strategies to allocate patients to control arm(s) vs experimental arms and accommodate novel agents intended to benefit subsets of mPDAC. 1. Picozzi, ASCO TPS 4188, 2022. Clinical trial information: NCT04229004. Research Sponsor: Pancreatic Cancer Action Network; FibroGen, Inc.

	Line 1			Line 2		
	GA	Pam + GA	Hazard Ratio	GA	Pam + GA	Hazard Ratio
Model Estimated mOS (mos)	11.3	9.7	1.18 (95% CI* 0.88, 1.56)	7.8	6.6	1.18 (95% CI* 0.88, 1.56)
mPFS (mos)	5.3	5.9	0.64 (95% CI^ 0.36, 1.14)	7.0	3.9	1.35 (95% CI^ 0.78, 2.33)
ORR (%)	26.1	35.3		4.5	9.0	

*Credible Interval.
^Confidence Interval.

LAPIS: Randomized phase 3 trial of chemotherapy (CTX) with and without pamrevlumab (PAM) for locally advanced pancreatic cancer (LAPC).

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Background: PAM improved response, surgical eligibility, and resection rates when combined with CTX in phase 1/2 LAPC trials (NCT01181245; NCT02210559). LAPIS (NCT03941093) evaluated efficacy and safety of PAM + CTX in unresectable LAPC. **Methods:** LAPIS was a global, double-blind, placebo-controlled phase 3 trial of adults with treatment-naïve, -confirmed unresectable LAPC. Patients received (randomized 1:1) PAM (35 mg/kg Q2W) or placebo (PBO) plus CTX (gemcitabine + nab-paclitaxel [GnP] or FOLFIRINOX [FFX] per standard protocol) for up to six 28-day cycles. based on changes in -9, on FDG-PET, resectability, with final surgical decision made by surgeon. Primary endpoint: overall survival (OS); secondary endpoints: event-free survival (EFS), progression-free survival (PFS), and objective response rate (ORR. Safety (including treatment-emergent adverse events [TEAEs]) was evaluated in patients who received treatment. **Results:** 143 and 141 patients (median [range] age, 65.0 [31–90] yrs; 53.2% male) were randomized to PAM and PBO arms. 142 and 141 received treatment: GnP, 114 (79.7%) and 112 (79.4%); FFX, 28 (19.6%) and 29 (20.6%). PAM (64.8%) and 96 PBO (68.1%) patients completed 6 treatment cycles. . Survival similar for PAM and PBO arms (Table) and did not differ by. For PAM + FFX vs PBO + FFX, respiratory, thoracic and mediastinal disorders (39.3% vs 27.6%) and skin and subcutaneous tissue disorders (53.6% vs 37.9%) occurred >10% more with PAM; TEAE frequencies were similar with PAM + GnP and PBO + GnP. One treatment-related death occurred in PAM arm. **Conclusions:** LAPIS contained important LAPC trial innovations: pre/post-CTX FDG-PET imaging, objective criteria for surgical intervention, external surgical review panel, and composite EFS measurement. Addition of PAM to CTX was not associated with additional toxicity but did not improve survival outcomes for unresectable LAPC. Clinical trial information: NCT03941093. Research Sponsor: None.

LAPIS endpoints.

	PAM arm (n=143)	PBO arm (n=141)	
OS, months, median (95% CI)	events=118 (82.5%) 17.25 (15.47, 18.89)	events=112 (79.4%) 17.94 (14.59, 20.34)	HR (95% CI) 1.08 (0.83, 1.41) p=0.55
EFS,[a] months, median (95% CI)	events=99 (69.2%) 5.72 (5.59, 6.01)	events=102 (72.3%) 5.78 (5.62, 6.37)	HR (95% CI) 1.05 (0.78, 1.39)
PFS, months, median (95% CI)	events=48 (33.6%) 9.36 (7.75, 11.79)	events=44 (31.2%) 9.40 (7.69, 10.84)	HR (95% CI) 1.01 (0.65, 1.56)
ORR, n (%)	43 (30.1)	64 (45.4)	OR (95% CI) 0.50 (0.31, 0.82)
Complete response	0	0	
Partial response	43 (30.1)	64 (45.4)	

CI, confidence interval; HR, hazard ratio; OR, odds ratio.
[a]Earliest of 1) failure to achieve local disease-free status at end of treatment and/or after surgery; 2) local or distant recurrence/progression; or 3) death.

Effect of baseline geriatric and quality of life assessments on treatment outcomes in ECOG-ACRIN EA2186 (GIANT): A randomized phase II study of gemcitabine and nab-paclitaxel compared with 5-fluorouracil, leucovorin, and liposomal irinotecan in older patients with treatment-naïve metastatic pancreatic cancer.

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Background: Data is lacking to guide the care of vulnerable older adults (OA) with newly diagnosed metastatic pancreatic adenocarcinoma (mPDAC). EA2186 trial demonstrated poor outcomes among vulnerable OA with mPDAC treated with dose-reduced chemotherapy. To understand the factors driving treatment outcomes in this patient population, we analyzed the correlation between baseline geriatric and quality of life (QOL) assessments and treatment outcomes. **Methods:** Vulnerable OA ≥70 yo with mPDAC, ECOG PS 0-2 were enrolled. Vulnerability was defined by screening geriatric assessment (GA) demonstrating mild abnormalities in function, comorbidities, cognition, or age≥80y. Pts were randomized to Arm A: Gemcitabine (1000mg/m2) + Nab-Paclitaxel (125mg/m2) q14 days or Arm B: 5-Fluorouracil (2400mg/m2 46hr) + Leucovorin (400mg/m2) + Liposomal Irinotecan (50mg/m2) q14 days. GA and QOL evaluations were completed at baseline and 3 time points. Secondary endpoints of the study included evaluating the correlation between baseline GA, QOL and treatment outcomes. Regression models were used to evaluate the associations between baseline GA and QOL factors, survival and grade 3 or higher toxicity, with 80% power to detect doubling in grade 3 toxicity for GA/QOL measures. **Results:** 176 pts (88 per arm) enrolled with median age 77 (range 70-90), 24% ECOG-0, 64% ECOG-1 and 12% ECOG-2. Pts were deemed vulnerable by cognition (46%), age (36%) or comorbidities (31.4%), with 35% meeting vulnerability criteria in ≥2 domains. No significant difference was seen in median OS (4.7 vs. 4.4 months; p=0.72) or ≥grade 3 toxicity rate (45.6% vs. 58.7%; p=0.10) between arms A and B, respectively. Strong correlation was found between OS and baseline instrumental activities of daily living score (HR 0.84; p=0.02), nutritional scores (HR 0.82; p<0.0001), depression scores (HR 1.07; p=0.02), and scores of all QOL measures (HR 0.98; p<0.0001). No correlation was found between OS and comorbidity, cognition, and Activities of Daily Living scores. After adjustment for age and PS, only baseline WBC level (OR=0.35; p=0.0054), and depression scores (OR=1.20 per score unit; p=0.021) and to a lesser extent FACT-G score (OR=0.98 per score unit; p=0.061) were found to correlate with rates of ≥grade 3 toxicity. **Conclusions:** Baseline GA and QOL factors among vulnerable older adults with mPDAC correlate strongly with survival and treatment tolerance. Supportive care to address these factors may favorably affect outcomes in this patient population. Clinical trial information: NCT04233866. Research Sponsor: National Cancer Institute.

Assessing socioeconomic disparities and the likelihood of multimodal treatment utilization in patients with pancreatic cancer using SEER Medicare data.

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Background: Optimal treatment for pancreatic cancer necessitates multimodal therapies combining surgery, chemotherapy, and radiation. Socioeconomic determinants of health may influence access to these treatments, leading to disparities in outcomes. This study examines the association between socioeconomic status (SES) and the likelihood of receiving multimodal treatment among pancreatic cancer patients using SEER Medicare data. **Methods:** Our study population consists of elderly persons diagnosed with pancreatic cancer from the 2008–2017 SEER-Medicare dataset. A multinomial logistic regression controlling for sociodemographic characteristics, cancer stage, comorbidities, geographic region, and ecological measures of urbanicity and SES was used to assess the likelihood of receiving one of three mutually exclusive treatment categories: no treatment, one modality (surgery, chemotherapy, or radiotherapy), or multiple modalities (different combinations of surgery, chemotherapy, and radiotherapy), with multiple modalities as the reference group. Survival analysis was conducted using Kaplan-Meier curves (KMC) and Cox proportional hazards (CPH) models. **Results:** Adjusted analyses showed that among stages 0–3 patients, living in areas with a high percentage of individuals without a high school education (non-HS) was associated with increased odds of receiving no treatment (OR = 1.26, $p < 0.01$) and one modality (OR = 1.20, $p < 0.01$) versus multiple modalities. Low median household income was linked to higher odds of no treatment (OR = 1.30, $p < 0.01$) and one modality (OR = 1.16, $p < 0.05$). This pattern persisted in stage 4 patients, where high non-HS percentage increased odds of no treatment (OR = 1.38, $p < 0.01$), and low median income was associated with higher odds of no treatment (OR = 1.25, $p < 0.05$). Non-Hispanic Black patients had higher odds of receiving no treatment in both stages 0–3 (OR = 1.26, $p < 0.05$) and stage 4 (OR = 1.91, $p < 0.01$) compared to Non-Hispanic Whites. Being married was associated with reduced odds of receiving no treatment for stages 0–3 (OR = 0.53, $p < 0.01$) and stage 4 (OR = 0.55, $p < 0.01$). Lower SES and being Non-Hispanic Black are associated with decreased likelihood of receiving multimodal treatment across all stages. Patients receiving multimodal therapies demonstrated improved survival on our KMC analysis, and CPH models indicated lower mortality for those receiving multiple modalities. **Conclusions:** Socioeconomic disparities significantly influence multimodal treatment utilization in pancreatic cancer. Lower SES indicators—high non-HS percentages, low median income—and being Non-Hispanic Black are associated with decreased use of multimodal therapies and poorer survival. Addressing these disparities is essential to ensure equitable access to optimal treatments and improve survival rates. Research Sponsor: American Cancer Society; Medical College of Wisconsin.

Social determinants of health and financial burden among patients with pancreatic cancer at a tertiary cancer center.

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Background: Financial toxicity (FT) represents a challenge for cancer patients that impacts treatment decisions, patient satisfaction, and quality of life. Social determinants of health (SDoH) screening may predict who is at risk of FT and could potentially benefit from financial interventions. **Methods:** This is a retrospective study of all cancer patients presenting to Mayo Clinic sites between Sept 2023 – Jun 2024. Patients completed SDoH questionnaires and were given risk scores of Low Risk, Medium Risk, and High Risk for SDoH categories: Financial Strain Risk, Transportation Risk, and Food Insecurity Risk. Patient financial data including active debt amounts, insurance coverage, and financial aid were collected. Patients with pancreatic ductal adenocarcinoma (PDAC) were compared to those with pancreatic neuroendocrine tumors (PNET) and patients with other cancers (Others). Categorical variables were compared with Pearson's Chi-squared test, continuous variables with Kruskal-Wallis rank sum test. P values $<$ 0.05 were considered significant. **Results:** Data was obtained from 88,039 patients, 2,607 of whom had PDAC and 449 had PNET. Compared to PDAC patients, PNET patients and Others were more likely to have Medium/High risk scores for Financial Strain Risk (6% vs 6.5% and 8.5%, respectively) and Food Insecurity (1.7% vs 3.8% and 3.2%) (Table). There was no statistically significant difference in median medical debt and need of financial assistance between the groups. Among PDAC patients, there were differences in distributions among race, primary language spoken, and patient electronic record access depending on Financial Risk categories. African American patients had the greatest proportion of Medium/High risk (12.9%) compared to Asian (4.4%) and White (5.8%) patients (P$<$0.001). Among patients with English as their primary language, 64.9% had Low Risk scores compared to 27.7% with other primary languages (P$<$0.001). Patients with electronic record access were more likely to report High Risk scores (6.9%) compared to those without access (1.6%). **Conclusions:** Per patient responses collected, PDAC patients presenting to our center have lower risk of financial strain and similar medical debt levels compared to the remainder of cancer patients. Among PDAC patients, higher financial strain was experienced by non-whites (7.9% vs 5.8%). Research Sponsor: None.

Diagnosis	PDAC (N=2607)	PNET (N=449)	Other (N=84983)	p value
Current Financial Risk (Grouped)				**< 0.001**
- Low Risk	1662 (63.8%)	304 (67.7%)	59298 (69.8%)	
- Medium/High Risk	157 (6.0%)	29 (6.5%)	7209 (8.5%)	
- Unknown/Missing	788 (30.2%)	116 (25.8%)	18476 (21.7%)	
Food Insecurity Risk				**< 0.001**
- Low Risk	1764 (67.7%)	308 (68.6%)	63235 (74.4%)	
- Medium Risk	45 (1.7%)	17 (3.8%)	2703 (3.2%)	
- Unknown/Missing	798 (30.6%)	124 (27.6%)	19045 (22.4%)	
Median Active Debt in USD	1102.230	1020.010	1141.135	**0.593**

Racial disparities in receipt of guideline concordant pancreatic cancer care among older adults in the US.

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Background: Patients racialized as Black experience higher incidence and mortality from pancreatic cancer (PC). The aim of this study was to examine differences in receipt of guideline concordant care (GCC) among older adults with PC racialized as Black, White, or Hispanic ethnicity. **Methods:** Patients 65 years and older with incident PC racialized as White, Black, or Hispanic ethnicity were identified in the SEER-Medicare database from 2004-2019. The primary outcome was receipt of GCC (stage I/II: surgery+chemotherapy+/-radiation; stage III: chemotherapy+/-surgery+/-radiation; stage IV: chemotherapy+/-radiation). Multivariable logistic regression was used to identify factors associated with GCC. Oaxaca-Blinder decomposition was used to examine the contribution of measured and unmeasured variables to racial disparities in receipt of GCC. **Results:** Of 12,772 patients included, 85.5% of patients racialized as White (n=10,915), 7.8% Black (n=992), and 6.8% Hispanic ethnicity (n=865). Patients racialized as Hispanic ethnicity and Black were more often dual-eligible for Medicare/Medicaid than patients racialized as White (38.6%, 26.9%, and 7.5%, respectively). In total, 56.3% received GCC. On adjusted analysis, patients racialized as Black were less likely to receive GCC for early stage (stage I/II) disease compared to patients racialized as White (OR 0.66; 95% CI 0.53-0.83), but not for stage III (OR 0.65; 95% CI 0.41-1.01) or stage IV (OR 0.87; 95% CI 0.66-1.14) disease. No difference in GCC was observed among patients of Hispanic ethnicity when compared to patients racialized as White. Increased age and dual-eligibility were associated with decreased likelihood of GCC across stages. Differences in receipt of GCC among patients racialized as Black and White remain largely unexplained. Of the measured factors, comorbidities, dual-eligibility, and area level poverty contributed most to the disparity in GCC (Table). **Conclusions:** Just over half of an insured population with PC receive GCC. Despite an increased focus on equity in cancer care, Black-White racial disparities in receipt of GCC are prevalent, particularly among patients with early-stage disease. Further studies are critical to identify and address the unmeasured factors driving treatment disparities. Research Sponsor: U.S. National Institutes of Health; T32 CA 009599; MD Anderson Cancer Center; P30 CA 016672.

Contributing factors to racial disparities in guideline concordant care for patients with pancreatic cancer.

	Stage I/II	Stage III	Stage IV
Difference (White-Black)	11.6%	9.4%	4.1%
Explained			
Age	-3.0%	-2.0%	-1.0%
Sex	0.1%	0.8%	0.1%
Dual-Eligibility	2.0%	3.1%	0.8%
Charlson Comorbidity Index	1.3%	0.2%	0.7%
Area Level Poverty	2.2%	-0.8%	0.7%
Hospital/Practice Volume	1.2%	-0.3%	-0.03%
Region	0.4%	0.9%	-0.2%
Year of Diagnosis	-0.1%	0.5%	0.1%
Unexplained	7.5%	7.0%	3.0%

Positive values indicate these factors increased the measured disparity, while negative values indicate these decreased the measured disparity.

Association of malnutrition at diagnosis with survival among older adults with pancreatic cancer.

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Background: Malnutrition is common among older adults newly diagnosed with pancreatic cancer. However, the implications of malnutrition among this vulnerable population are poorly understood. We sought to understand the association between malnourishment at diagnosis and overall survival (OS) among older adults with pancreatic cancer. **Methods:** We included adults aged ≥ 60 years with newly diagnosed pancreatic cancer, who were enrolled in the University of Alabama at Birmingham (UAB) Cancer and Aging Resilience Evaluation (CARE) registry between October 2017 and April 2024. To assess nutritional status, we administered an abridged version of the Patient-Generated Subjective Global Assessment (abPG-SGA) to patients at baseline prior to receiving chemotherapy. Using abPGSGA scores, we categorized patients as either normal (score ≤ 5) or malnourished (score ≥ 6). All participants were followed until death or until the cutoff date of July 10, 2024. We employed Kaplan-Meier survival analysis to estimate overall survival (OS) for both malnourished and non-malnourished groups and compared them using log-rank test. Additionally, we built multivariable Cox proportional hazards models to test the association between malnutrition and OS, while controlling for age at diagnosis, sex, race/ethnicity, and cancer stage. **Results:** A total of 274 patients were included; the median age at diagnosis was 70 (interquartile range, IQR: 65-75) years with 52% males, 75% non-Hispanic White and 52% with Stage IV disease at diagnosis. At baseline, 59% of patients were malnourished, and the median abPGSGA score was 8 (IQR 4-14). Patients who were malnourished had a similar demographic and clinical characteristics as compared to those without malnutrition. Over a median follow up of 36 months, a total of 202 patients (74 %) died. The median overall survival (OS) was significantly lower for malnourished patients at 13 months compared to 21 months for non-malnourished patients (log-rank test, $p < .001$). After adjusting for potential age, sex, race/ethnicity and cancer stage, malnutrition at diagnosis remained independently associated with worse survival outcomes (adjusted HR: 1.762; 95% CI: 1.316−2.358; $p < .001$). **Conclusions:** Our study reports that malnutrition at baseline is significantly associated with worse overall survival in older adults with pancreatic cancer. These findings underscore the importance of early nutritional assessment and intervention to enhance outcomes in this population. Research Sponsor: None.

Redefining the use of a first-line FOLFIRINOX-like regimen in older patients with metastatic pancreatic cancer.

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Background: Prospective elderly-specific data for treating pancreatic cancer are scarce. First-line treatment with triplet chemotherapy (e.g., FOLFIRINOX) is often too toxic for older adults (OA). An alternating regimen of FOLFOX and FOLFIRI (aFOLFIRINOX) has demonstrated efficacy and decreased toxicity in patients with colon cancer. This ongoing multicenter phase II trial evaluates the tolerability and efficacy of aFOLFIRINOX in OA with metastatic pancreatic adenocarcinoma (mPDAC). Herein we report the interim safety analysis results. **Methods:** Eligible patients (goal n=37) ≥65 years old with measurable untreated mPDAC, ECOG PS 0-2, receive standard doses of FOLFOX alternating with FOLFIRI every 2 weeks for up to 6 months. Patients can continue treatment thereafter at the discretion of their physician. The primary endpoint is rate of treatment-limiting grade ≥3 adverse events (TLAE) per CTCAE v5.0 in the first 8 weeks of therapy. These toxicities are defined as treatment-related grade 3 non-hematologic AE persisting > 2 weeks despite best supportive care, grade 3 anemia or thrombocytopenia, grade >3 neuropathy, or any grade > 4 AE. The study is powered to detect a 20% improvement in grade ≥3 TLAE: null vs alternate: 45% vs 65% of patients without TLAE at 8 weeks. Secondary endpoints include progression free & overall survival (PFS & OS) and objective response rate (ORR). FACT-G and PROMIS-10 quality of life (QOL) surveys and geriatric assessments are collected and will be correlated with treatment outcomes. **Results:** 21 eligible patients were included in the interim safety analysis: 11 male, 10 female; median age: 73 (range 65-82); 18 patients had ECOG PS 0-1. 16 (76%) patients did not experience TLAE as defined above. TLAE included neutropenia, thromboembolism, and hypokalemia, and are similar to reported toxicity with FOLFOX and FOLFIRI in this patient population. One patient died of their disease within 8 weeks of treatment initiation and another within 2 weeks due to pulmonary embolism deemed unrelated to treatment. Of the 18 patients evaluable for response, ORR was 11% (partial response: n =2), Disease Control Rate was 89% (stable disease: n =14). 7 patients completed 6 months of treatment on study. **Conclusions:** The interim safety analysis of this clinical trial fulfilled the predefined futility rule without excess treatment-limiting or unexpected toxicity among OA with mPDAC treated with first-line aFOLFIRINOX, suggesting this may be a reasonable treatment option in this patient population. Trial is ongoing for evaluation of efficacy data of this regimen. Clinical trial information: NCT05360732. Research Sponsor: None.

Optimizing clinical trial design to mitigate racial disparities in pancreatic cancer clinical trial enrollment.

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Background: Participation in pancreatic ductal adenocarcinoma (PDAC) clinical trials remains dismal among patients from minoritized groups. This study aimed to investigate if clinical trial design, including eligibility criteria, affects representation of minoritized populations. **Methods:** United States Phase I-IV PDAC clinical trials between 2000-2020 were extracted from the ClinicalTrials.gov database. The Surveillance, Epidemiology, and End Results (SEER) database was used to determine cancer incidence linked with years of trial enrollment. Enrollment fraction (EF) was defined as the number of trial participants divided by the incidence of PDAC in each racial and ethnic group. Representation Quotient (RQ) was defined as the proportion of trial participants divided by proportion of PDAC incidence for each subgroup, where underrepresentation was RQ <1.0. **Results:** 8451 patients from 228 unique PDAC clinical trials were analyzed. 55.7% (n=128) and 44.3% (n=102) trials reported racial and ethnic demographic data, respectively. Non-Hispanic Black (NHB), Non-Hispanic Asian/Pacific Islander (NHAPI), Non-Hispanic American Indian/Alaskan Native (NHAIAN), and Hispanic patients were underrepresented (Table). NHB patients were more likely to be underrepresented in trials for metastatic PDAC disease (89.4% vs. 73.8% of trials, p=0.035). NHB patients demonstrated an increase in median RQ (0.28 vs. 0.48) in trials from 2000-2010 vs. 2011-2020, while Hispanic patients had a decreased median RQ (0.36 vs. 0.25). Trials with renal insufficiency exclusion criteria without cutoffs or explicit methodology were more likely to accrue NHB patients (RQ 0.88 vs. 0.46, p=0.02), whereas trials that used a creatinine cutoff or the modified Cockcroft-Gault (CG) formula did not significantly affect RQ. Other eligibility criteria including cardiac history, HIV, Hepatitis B/C, and receipt of neoadjuvant therapy did not significantly affect RQ for any racial or ethnic group. Trial location and funding source did not affect RQ for any group. **Conclusions:** Systemic barriers persist in minority enrollment in PDAC clinical trials. Clinical trial design optimizing inclusive eligibility requirements is a first feasible step in increasing minority clinical trial enrollment. Research Sponsor: National Cancer Funding.

Racial and ethnic representation in pancreatic ductal adenocarcinoma clinical trials.

	Enrollment Fraction (Median, IQR)	Representation Quotient (Median, IQR)
NHW	0.28 (0.12-0.62)	1.17 (1.09-1.24)
NHB	0.15 (0-0.33)	0.47 (0-0.87)
NHAPI	0 (0-0.23)	0 (0-0.45)
NHAIAN	0 (0-0)	0 (0-0)
Hispanic	0.07 (0-0.21)	0.26 (0-0.68)

*Abbreviations: Interquartile Range (IQR), Non-Hispanic Black (NHB), Non-Hispanic Asian/Pacific Islander (NHAPI), Non-Hispanic American Indian/Alaskan Native (NHAIAN).

683 Poster Session

Disparities in outcomes of pancreatic cancer hospitalizations: A ten-year trend study.

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Background: Pancreatic cancer is the third leading cause of cancer mortality in the United States with 51,750 estimated deaths from Pancreatic cancer in 2024. Demographic variations have been reported with Pancreatic cancer. Our aim was to identify trends in demographic-specific disparities in inpatient pancreatic cancer outcomes. **Methods:** This is a retrospective longitudinal trends study using the Nationwide Inpatient Sample (NIS) database (2010-2019). We identified hospitalizations with pancreatic cancer using ICD-9 and ICD-10 codes. We highlight disparities in mortality outcomes stratified by age, gender and race. **Results:** There were 992,550 hospitalizations of pancreatic cancer patients between 2010 and 2019. Over the period, the mean age at hospitalization was similar at 68 years (P = 0.006). A majority of the admissions were elderly patients (≥65 years) identifying as males and Caucasians ($P < 0.01$). Most patients had Medicare and were in large and urban teaching hospitals ($P < 0.001$). Over the decade, there was an overall decline in inpatient mortality, from 8.2% in 2010 to 7.1% in 2019 (P trend = 0.004). Amidst the age subgroups, only the elderly patient group experienced a significant decline in mortality rate (8.8% to 7.3%; P trend = 0.004). The young adult population (<45 years) had a trend towards increasing mortality rate, but it was not significant (P trend= 0.139). On gender subgroup analysis, only males had a significant decline in mortality rate (P trend= 0.003). On racial subgroup analysis, only Caucasians had a significant decline in mortality trend (P trend= 0.004) during the study period. **Conclusions:** There was an overall improvement in the mortality outcome of pancreatic cancer hospitalizations. On sub-group analysis, significant decline in mortality over the decade were seen only in elderly, male and Caucasian patient groups. Targeted research is needed in understanding the root causes of these disparities in pancreatic cancer outcomes and develop effective intervention to bridge this gap.
Research Sponsor: None.

Trends in mortality outcomes of pancreatic cancer hospitalizations by demographic subgroups.

Mortality rate, %	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	P- Trend value
Overall	8.21	7.94	7.85	7.70	7.73	7.68	7.46	7.87	7.47	7.14	0.004
Young (<45 years)	4.45	5.55	5.23	6.08	6.27	5.32	3.63	6.84	6.95	6.35	0.14
Middle-aged (45- 64 years)	7.55	7.24	7.74	7.32	7.03	6.99	7.62	6.80	6.74	6.91	0.06
Elderly (>65 years)	8.81	8.44	8.04	8.00	8.22	8.19	7.54	8.48	7.87	7.29	0.004
Females	7.72	7.32	7.23	7.15	6.99	7.31	6.72	7.38	7.01	6.87	0.08
Males	8.67	8.57	8.44	8.22	8.44	8.04	8.13	8.32	7.90	7.38	0.003
Whites	7.94	7.49	7.48	7.39	7.81	7.29	6.81	7.39	7.19	6.80	0.004
Blacks	9.87	9.40	9.38	9.15	8.35	8.13	9.40	9.01	8.55	8.41	0.07
Hispanics	7.10	8.99	8.02	6.85	6.66	8.80	7.54	7.43	7.10	6.14	0.05

Utility of circulating tumor DNA in patients with unresectable pancreatic cancer using a patient-specific panel: ARTEMIS-PC study.

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Background: Circulating tumor DNA (ctDNA) analysis has proven to be a highly accurate method for assessing treatment efficacy and detecting molecular residual disease (MRD) in cancer patients, outperforming tumor markers and imaging modalities. However, the ctDNA detection rate is reported to be low, and its utility remains unclear in pancreatic cancer (PC). We conducted the prospective "ARTEMIS-PC" study to evaluate the utility of a personalized, tumor-informed MRD assay in patients with unresectable PC. **Methods:** Eligible patients had histopathologically confirmed unresectable PC, were previously untreated, and had disease measurable per Response Evaluation Criteria in Solid Tumors v1.1. A patient-specific tumor-informed assay (Invitae Personalized Cancer Monitoring) was used for the detection of ctDNA. Blood samples were collected at structured pre- and post-systemic therapy timepoints. **Results:** A total of 99 patients were eligible for the ARTEMIS-PC study, and personalized panels were successfully created for 92 patients. The median age of patients was 70 years (range 45-81 years), with 43/92 being female. Thirty patients were stage III, and 62 stage IV. Median size of the primary tumor was 38 mm (range 13-100 mm). MRD positivity at enrollment was 88.0%, with 73.3% in stage III patients, and 95.2% in stage IV (p=0.007). During follow-up, ctDNA clearance was achieved in 33 patients (40.7%). ctDNA clearance was associated with a significantly higher objective response rate (61.5% vs. 17.6%, p=0.001) and disease control rate (100% vs. 64.7%, p=0.002) compared to those who did not achieve clearance. Patients who achieved ctDNA clearance had significantly longer progression-free survival (PFS) than those who did not (9.0 vs. 3.5 months, hazard ratio 0.2, $p<0.001$). The predictive performance of select biomarkers (variant allele frequency [VAF], CEA, CA19-9) was evaluated using area under the curve (AUC) for treatment response and disease control. For disease control, VAF showed superior predictive performance with AUCs of 0.84 at enrollment and 0.97 at both Week 4 and Week 8, compared to CEA (AUCs: 0.65, 0.73, 0.60) and CA19-9 (AUCs: 0.51, 0.57, 0.56). The predictive performance for treatment response was lower than AUC 0.75 for any of the biomarkers. Among patients who were MRD negative before treatment or became negative during follow-up (n=44), 45.5% (n=20) subsequently turned MRD positive. Of these, disease progression was confirmed by CT in 14 patients. The median time from MRD turning positive to CT-confirmed progressive disease was 88.5 days (range: -28 to 385 days). **Conclusions:** ctDNA clearance is a useful predictor of improved objective response, disease control, and PFS, with VAF emerging as the more accurate biomarker for disease control, underscoring its potential utility in treatment monitoring in patients with unresectable PC. Clinical trial information: UMIN000043561. Research Sponsor: None.

A systematic review of the impact of artificial intelligence in pancreatic cancer detection.

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Background: Pancreatic cancer is the fourth leading cause of cancer deaths in the United States, and early detection remains a significant challenge. Screening the general population is not feasible, but the rise of artificial intelligence (AI) has introduced new possibilities for improving early diagnosis and patient outcomes. **Methods:** A systematic literature search was conducted using PubMed, Google Scholar, and MEDLINE using MeSH terms Artificial intelligence or AI, and diagnosis and pancreatic carcinoma or pancreatic adenocarcinoma. Prisma guidelines were adhered to, and a total of 19 studies resulted, 6 of them were retrospective studies, 4 of them were literature reviews, and 4 of them were randomized trials. The rest were duplicates. The inclusion criteria for this study is AI being used in the diagnosis, only in pancreatic cancer, within the last 5 years, only in English, and only retrospective studies were included. **Results:** One study used Digital Imaging Processing (DIP) for analyzing Endoscopic ultrasound (EUS) images from 153 pancreatic cancer patients, yielding a sensitivity of 97.98% and a specificity of 94.32%. Two studies explored Computer Aided Diagnosis (CAD) models applied to PET/CT and EUS images, achieving a sensitivity of 95.23% and specificity of 97.51% in PET/CT scans and 83.3% and 93.3% in EUS images, respectively. Another study used a Faster R-CNN model to analyze CT images from 338 pancreatic cancer patients showed high diagnostic accuracy in much less time. Additionally, two studies utilized Natural Language Processing (NLP) for identifying family histories of pancreatic cancer and detecting pancreatic cysts, with the latter achieving sensitivity and specificity rates of 99.9% and 98.8%. **Conclusions:** The current strategies for early diagnosis of pancreatic cancer focus on serum biomarkers and EUS-guided Fine Needle Aspiration (EUS-FNA) and sensitivity varies but depends on the physician's expertise. AI is showing promise in improving pancreatic cancer diagnosis by enhancing early detection and accuracy. Techniques like deep learning, NLP-based models, Faster R-CNN, and CAD systems analyze medical data and images more effectively than manual methods. AI holds the potential to shape the future of pancreatic cancer diagnosis and improve patient outcomes.
Research Sponsor: None.

Impact of intrathecal targeted drug delivery (ITDD) on oral morphine equivalents and pain scores in advanced pancreatic cancer pain.

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Background: Pancreatic cancer accounts for approximately 3% of all cancers and affects ~64,000 adults in the United States each year. Seventy five percent of patients present with abdominal pain, leading to diagnosis. This is one of the most painful cancers, with pain severity strongly correlating to prognosis. Severe intractable abdominal pain negatively impacts quality of life. Traditionally, opioids have been used as the first line agent for the treatment of severe abdominal pain. However, opioids carry significant risk and often unwarranted side effects. With escalating doses to treat refractory pain there is also potential for unintentional overdose. Intrathecal targeted drug delivery (ITDD) provides pain medications directly to the intrathecal space limiting systemic side effects. Our study aimed to determine the impact ITDD has on pain scores as well as the amount of systemic opioid needed in patients with advanced pancreatic cancer. **Methods:** A retrospective review was conducted on 217 advanced cancer patients that underwent ITDD placement from March 2019 through July 2024. Thirty- five patients with a diagnosis of pancreatic cancer met inclusion criteria. Pain scores and oral morphine equivalent (OME) use was evaluated at baseline (prior to ITDD), at 1 month and 3 months post ITDD implant. A numeric scale of 0-10 was used to determine pain scores, while a comprehensive chart review and patient response was used to determine OME use. **Results:** We found that post ITDD placement, pain scores were reduced on average by 4.55 points at 1month and 4 points at 3-months after implantation. This was statistically significant with a two-paired t-test analysis ($p < 0.05$). Post ITDD placement, there was a 96.7% reduction in OME at 1-month ($p < 0.005$) and 98.1% reduction in OME at 3 months ($p < 0.04$). **Conclusions:** Despite the small sample size of patients, ITDD showed a significant reduction in pain scores and OME use compared to baseline with systemic opioid use only in patients with advanced pancreatic cancer. This reduction appeared to be maintained for 3 months post ITDD placement. Further studies conducted on a larger sample size would be beneficial to demonstrate statistical outcomes of ITDD therapy on pain scores and OME use. Research Sponsor: None.

Impact of long-term chemotherapy (CTx) on outcomes in pancreatic ductal adenocarcinoma (PDAC): A real-world UK multi-centre study.

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Background: PDAC is associated with poor outcomes with limited treatment options. Here, we present a multi-institutional review evaluating outcomes following short and long-term CTx with or without treatment breaks in PDAC patients (pts). **Methods:** All consecutive PDAC pts receiving > 3 CTx cycles between 2019 – 2023 at University College London Hospitals and Oxford University Hospitals were included. Treatment response, survival outcomes and predictors of clinical benefit were evaluated. Wilcoxon test, Kaplan-Meier and multivariate Cox regression models were performed. **Results:** Of the 213 screened pts, 127 eligible subjects met the study criteria. 1st, 2nd and 3rd line CTx were received by 127, 40 and 2 pts, respectively. 21 pts had resectable disease (9 remained relapse-free after adjuvant CTx and 8 pts alive at post 2 years follow-up surveillance). 106 pts had unresectable or metastatic disease and were selected for final analyses (12% borderline resectable (BR), 19% locally advanced (LA), 9% localised disease who developed metastases and 60% de novo metastatic pts). 7 pts (7%) pts underwent genetic profiling on pt request or previous clinical trial screening; *KRAS* aberrations (N = 4), actionable *PLAB2* / *BRCA2* mutations (N = 2). BR and LA pts (N = 33) achieved a median PFS1 (Progression Free Survival while on 1st line CTx) of 8.28 (95% Confidence Interval (CI), 5.49 – 15.11) and Overall Survival (OS) of 15.15 (95% CI, 9.46 – 26.41) months (mos). De novo metastatic pts (N = 64) attained a PFS1 of 6.64 (95% CI, 5.98 – 8.18) and OS of 9.30 (95% CI, 8.05 – 12.81) mos. Patient-specific factors in both cohorts comprising of age, gender, performance status, smoking history, co-morbidities were not associated with PFS1 and OS. Improved PFS was associated with ≥ 6 cycles of 1st line CTx (P = < 0.001), median duration of 1st CTx of ≥ 3.58 mos (P = < 0.001) and best response to 1st CTx (P = 0.007). A favourable OS was associated with > 1 line of CTx (P = < 0.001), ≥ 6 cycles of 1st line CTx (P = 0.009), median duration of 1st CTx of ≥ 3.58 mos (P = < 0.001) and best response to 1st CTx (P = 0.002). Subjects receiving ≥ 6 cycles of 1st CTx were younger than pts tolerating fewer cycles (median age 62.41 vs 72.25 years) (P = 0.04). In the localised disease group, improved PFS was associated with ≥ 6 cycles of 1st line CTx (P = 0.003), median duration of 1st CTx of ≥ 2.99 mos (P = 0.008) and local treatment after commencing 1st line CTx (P = < 0.001). These three factors were also associated with OS; P = 0.02, P = 0.02 and P = < 0.001, respectively. Median number and duration of 1st line CTx interruptions were not associated with PFS1 or OS in both cohorts. **Conclusions:** Despite challenges associated with long-term CTx, survival outcomes may be better in pts receiving CTx for longer periods and multiple treatment lines. Genetic profiling currently offers limited value but in carefully selected pts, localised treatment options may be associated with improved survival outcomes. Research Sponsor: None.

Clinical outcomes of immunotherapy in metastatic pancreatic carcinoma: A systematic review and meta-analysis.

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Background: Metastatic pancreatic carcinoma is one of the most challenging malignancies to treat. Traditional treatment modalities such as chemotherapy and radiation therapy have shown limited efficacy. This meta-analysis aims to investigate immunotherapy as a potential therapeutic option for metastatic pancreatic carcinoma. **Methods:** PubMed, Cochrane, Embase, and Clinicaltrials.gov were searched as per PRISMA guidelines. Data extraction was conducted independently by two reviewers, and any discrepancies were resolved through discussion. Pooled analysis was performed using the 'meta' package by Schwarzer et al. in the R programming language (version 4.16-2). **Results:** 834 patients from the 17 studies reporting immunotherapy outcomes in metastatic pancreatic carcinoma were included in this systematic review and meta-analysis. The studies included (There were) were 4 phase I (23.5%), 2 phase Ib (11.8%), 1 phase Ib/II (5.9%), and 5 phase II (29.4%), 1 retrospective study (5.9%) and 4 prospective studies (23.5%). The median age of the patients was 61 years and 55.6% (398/716) were male. Immunotherapies included were GVAX, tremelimumab, Mesothelin-specific Chimeric Antigen Receptor T cells, durvalumab, pembrolizumab, canerpaturev, Sotigalimab, nivolumab, CO-1.01, allogeneic natural killer cell immunotherapy, Anti-EGFR chimeric antigen receptor-modified T cells, bispecific antibody armed T cells, and clivatuzumab tetraxetan. The median follow-up duration was 17.8 months. The median overall survival was 7.9 months, and the median progressive-free survival was 4.6 months. The pooled overall response rate was calculated at 16% (95% CI 0.09-0.26, I2=70, $p<0.01$, n= 426) and the pooled partial response rate was 13% (95% CI 0.09-0.26, I2= 67%, $p<0.01$, n= 272). The pooled stable disease was calculated to be 28% (95% CI, 0.2-0.39, I2= 59%, $p<0.01$, n=277). The pooled progressive disease was 36% (95% CI, 0.23-0.52, I2=34%, p=0.22, n=68). The most common overall adverse effects were anemia, diarrhea, and thrombocytopenia. **Conclusions:** This meta-analysis demonstrates that novel immunotherapies have promising results on metastatic pancreatic cancer. Future new clinical trials with a follow-up on current early-phase results are needed. Research Sponsor: None.

A retrospective study of the prognostic value of early induction of S-1 postoperative adjuvant therapy for pancreatic cancer.

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Background: The usefulness of oral S-1 administration as adjuvant therapy (AC) after pancreatic cancer surgery has been demonstrated (1). Our study also suggested the importance of maintaining relative dose intensity or postoperative early induction (2). In a large Japanese data set, a comparison of induction before and after the postoperative 10 weeks showed a worse prognosis in the group which started 10 weeks after surgery (3). However, the number of patients of which early induction is possible are gradually increasing with recent improvements of surgical outcomes. In the present study, we retrospectively evaluated the survival benefit of AC which was started within 2 weeks after surgery. **Methods:** 129 patients with invasive pancreatic cancer who underwent radical surgery from January 2018 to May 2023 were enrolled in this retrospective study. The patients were divided into the following groups: 46 (36%) underwent induction within 2 weeks, 56 (43%) within 3-6 weeks, 13 (10%) after 7 weeks, and 14 (10%) without AC (4 with low renal function, 4 with deteriorating physical condition after surgery, 4 with not desired, and 2 with comorbid disease treatment). The background of each group was as follows (Described in the following order; induction within 2 weeks after surgery: 3-6 weeks: after 7 weeks: without AC), age 71: 71: 71: 78 (n.s.), pancreatic head cases 52: 68: 69: 50% (n.s.), BR cases 35: 23: 8: 21% (n.s.), NAC performed 86: 38: 15: 29% ($p < 0.001$), preoperative CEA $\geq$ 5.0ng/ml 24: 23: 8: 21% (n.s.), preoperative CA19-9 $\geq$ 37U/ml 48: 52: 54: 64% (n.s.), PV/SMV reconstruction 20: 16: 31: 14% (n.s.), postoperative complications (Clavien Dindo $\leq$3) 0: 16: 31: 43% ($p < 0.001$), Pathological stage 2 or higher 83: 65: 77: 79% (n.s.). **Results:** The need for reduction of S-1 dosage was 87: 89: 92% of the patients (n.s.), and the 6-month completion rate was 70: 71: 61% (n.s.). Median overall survival duration in each group was as follows: not reached: 49.5: 35.5: 48.6 months, and 1-year postoperative recurrence-free survival was 73: 66: 45: 71%, median recurrence free survival time was 34.0: 20.8: 10.3: 12.1 months, and induction within 2 weeks after surgery group significantly improved the recurrence-free survival in comparison with the group of after 7 weeks ($p < 0.05$). No significant factors were detected in multivariate analysis. **Conclusions:** These results suggest that early induction of AC as a result of improvement of surgical outcomes would contribute to postoperative recurrence-free survivals. 1. Uesaka et al, Lancet, 2016. 2. Matsushima et al. Anticancer Res, 2022. 3. Tomimaru et al, J Gastroenterol, 2023. Research Sponsor: None.

Overall survival (OS) of patients with metastatic pancreatic ductal adenocarcinoma (mPDAC) treated with first-line (1L) FOLFIRINOX (FFX): Bridging the gap between the NAPOLI 3 trial and real-world practice.

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Background: NALIRIFOX (liposomal irinotecan + 5-fluorouracil/leucovorin + oxaliplatin) demonstrated significantly improved OS compared to 1L gemcitabine + nab-paclitaxel in the phase 3 NAPOLI 3 trial (NCT04083235). The median OS (95% CI) in the NALIRIFOX intention-to-treat population (N= 383) was 11.1 (10.0, 12.1) months. However, the relative efficacy of NALIRIFOX versus FFX (irinotecan + 5-fluorouracil/leucovorin + oxaliplatin), another standard of care for 1L treatment of mPDAC, has not been studied. To further contextualize findings from NAPOLI 3 trial, the current study described OS among patients treated with 1L FFX in the real-world setting. **Methods:** Three cohorts of patients with mPDAC treated with 1L FFX were identified from the Flatiron Health database. The all-comer cohort included all adult patients with mPDAC treated with 1L FFX since January 2014. Among them, the trial-aligned cohort included patients with mPDAC treated with 1L FFX between January 1, 2020 and July 31, 2022 (to align with the time period of NAPOLI 3 trial) who met the eligibility criteria of NAPOLI 3. Finally, a subgroup of patients in the trial-aligned cohort who received modified FFX (mFFX) regimen composed the trial-aligned mFFX cohort. OS was assessed using Kaplan-Meier analysis. **Results:** The all-comer cohort consisted of 3,271 patients. The trial-aligned cohort included 219 patients, of whom 154 (70%) were treated with mFFX. The distribution of baseline characteristics was similar across three cohorts (Table). The median age of the trial-aligned cohort was 65 years old, with 46% of patients being female, 67% being White, and 47% having ECOG score of 1. The median OS (95% CI) in the all-comer cohort and trial-aligned cohort was 9.0 (8.5, 9.3) months and 9.1 (7.8, 10.9) months, respectively; the median OS (95% CI) was 8.6 (7.3, 10.5) months in the trial-aligned mFFX cohort. **Conclusions:** NALIRIFOX regimen in the NAPOLI 3 trial showed numerically improved OS compared to FFX, including mFFX, in the real-world setting. Analysis adjusting for baseline characteristics are warranted and will provide further insights for the comparative efficacy of the two regimens. Research Sponsor: Ipsen Corp.

Characteristics of real-world FOLFIRINOX cohorts.

	All-comer cohort (N=3,271)	Trial-aligned cohort (N=219)	Trial-aligned mFFX cohort (N=154)
Baseline characteristics			
Age, years, mean (SD)	63.5 (9.0)	64.6 (8.2)	65.1 (8.5)
Female, n (%)	1,377 (42.1)	100 (45.7)	68 (44.2)
White, n (%)	2,118 (64.8)	146 (66.7)	100 (64.9)
ECOG score = 1, n (%)	1,327 (48.9)*	103 (47.0)	77 (50.0)
Endpoint			
Median OS, months (95% CI)	9.0 (8.5, 9.3)	9.1 (7.8, 10.9)	8.6 (7.3, 10.5)

Abbreviations: CI, confidence interval; ECOG, Eastern Cooperative Oncology Group; OS, overall survival; SD, standard deviation.
*Proportion calculated based on 2,711 patients with ECOG available at baseline.

Real-world clinical outcomes of locally advanced unresectable and de-novo metastatic pancreatic ductal adenocarcinoma: A multicenter retrospective study from Saudi Arabia.

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Background: Pancreatic ductal adenocarcinoma (PDAC) has dismal clinical outcomes, even with existing treatment regimens. This study evaluates the clinical characteristics and outcomes of patients with locally advanced unresectable or de-novo metastatic PDAC in Saudi Arabia to provide real-world data that can be compared to international benchmarks. **Methods:** This is a retrospective, multicenter study. Patients diagnosed with unresectable locally advanced or de-novo metastatic PDAC between January 2015 and November 2023 were included. Data were collected from 10 oncology centers across Saudi Arabia. Patient characteristics, treatment regimens, and clinical outcomes, including progression-free survival (PFS) and overall survival (OS), were analyzed using univariate and multivariate analyses. **Results:** A total of 350 patients were identified, with a median age of 60 years at time of diagnosis, 63% of patients presenting with multiple metastatic sites, primarily in the liver (66.3%). Among patients tested, 7.1% had deficient mismatch repair (d-MMR), and 5.8% harbored BRCA mutations. FOLFIRINOX was the most common first-line treatment (55.1%), followed by gemcitabine plus nab-paclitaxel (15.1%). The median PFS for first-line treatment was 5.3 months, with FOLFIRINOX achieving the longest PFS (6.5 months). The median OS was 10.34 months for the entire cohort, with better survival outcomes observed in patients receiving FOLFIRINOX (12.3 months). Independent prognostic factors for PFS and OS included performance status, type of first-line regimen, and neutrophil-lymphocyte ratio (NLR). **Conclusions:** This real-world study confirms that clinical outcomes for locally advanced unresectable and metastatic PDAC in Saudi Arabia are consistent with international data, with trend of FOLFIRINOX showing superior outcomes over gemcitabine-based regimens. However, both treatments reflect the persistent poor prognosis of PDAC, underscoring the need for novel therapeutic strategies. Further research is warranted to optimize treatment selection and improve survival outcomes in this population. Research Sponsor: None.

Real-world dose intensity (DI) and time to adverse events (AEs) in a commercial database during first-line (1L) treatment of metastatic pancreatic adenocarcinoma (mPDAC) with FOLFIRINOX (FFX), FFX without 5FU bolus, and gemcitabine + nab-paclitaxel (GnP).

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Background: Onset of severe AEs during 1L FFX and GnP treatment for mPDAC negatively impacts clinical outcomes, requiring supportive care and/or treatment adjustment. Occasionally they lead to discontinuation. The 5FU bolus may be omitted from FFX (FFXnb) to increase tolerability. We examined DI and time to key AEs (anemia, neutropenia, thrombocytopenia, diarrhea, fatigue, nausea/vomiting, and neuropathy) during 1L FFX, FFXnb, and GnP. **Methods:** This retrospective study utilized Optum Market Clarity claims + EHR linked data. Inclusion criteria were: adults diagnosed with mPDAC between 1/1/2015 and 5/31/2023; initiated 1L FFX, FFXnb, or GnP (index date) within -14 to +90 days; ≥6 pre- and ≥1 month post-index enrollment; ≥1 AE of interest in 1L. Follow-up was index to earliest of death, disenrollment, start of 2L, or last administration + 14 days. DI was defined as cumulative dose divided by follow-up in months. Grade 3/4 hematologic AEs (hAEs) were measured from lab values. Ungraded AEs were sourced from EHR notes and claims diagnoses. FFX was defined as including 5FU bolus + infusion in the 1st cycle; FFXnb as 5FU infusion only. Kaplan-Meier (KM) methods estimated time AE. **Results:** Median follow-up in months for patients (pts) with hAEs was: FFX 4.2; mFFX 3.6; GnP 3.1. Median DI (mg/m2/month) for FFX vs FFXnb pts with hAEs was: 5FU 4567 vs 4581; leucovorin 632 vs 398; oxaliplatin 119 vs 142; irinotecan 253 vs 251. For GnP, DI was: gemcitabine 2230; nab-paclitaxel 269. The table below presents sample size (N) and median (95% CI) time to AE. Median time to anemia was numerically 2–4 weeks shorter for GnP and FFXnb vs FFX. Time to neutropenia was 8–12 days less for FFXnb vs GnP and FFX. Time to thrombocytopenia was 7 weeks less for GnP vs FFX and FFXnb. Time to neuropathy was nearly 4 weeks shorter for FFXnb vs FFX and GnP. **Conclusions:** While neutropenia is higher risk for FFX vs GnP, prophylactic growth factor is standard during FFX, which may result in the similar onset time vs GnP. Early onset of severe thrombocytopenia differentiates the GnP sample. Oxaliplatin DI was higher for FFXnb vs FFX pts, and may be associated with earlier onset of anemia, neutropenia, diarrhea, and neuropathy. Research is merited on interactions between dosing, AE timing, and treatment discontinuation. Research Sponsor: Ipsen Biopharmaceuticals, Inc.

Time to AE.

Grade 3/4 hAEs	FFX N	FFXnb N	GnP N	FFX median days (95% CI)	FFXnb median days (95% CI)	GnP median days (95% CI)
Anemia	47	25	97	64 (38–96)	36 (19–70)	47 (39–56)
Neutropenia	75	53	112	35 (25–47)	23 (14–43)	31.5 (18–42)
Thrombocytopenia	33	22	54	63 (35–79)	65.5 (32–94)	14 (13–34)
Ungraded AEs						
Diarrhea	216	264	297	27.5 (22–36)	17.5 (14–27)	27 (21–36)
Fatigue	211	267	418	34 (23–43)	27 (17–34)	27 (21–30)
Nausea/vomiting	337	403	561	6 (3–8)	9 (7–14)	9 (7–12)
Neuropathy	151	155	190	92 (70–112)	67 (48–79)	92 (80–111)

Real-world excess costs associated with hematologic adverse events (hAEs) during first-line (1L) treatment of metastatic pancreatic adenocarcinoma (mPDAC) with FOLFIRINOX (FFX), FFX without 5FU bolus, and gemcitabine+nab-paclitaxel (GnP).

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Background: During 1L treatment of mPDAC with FFX and GnP, hAEs can harm patients (pts) and accrue excess costs to the health system, i.e., costs beyond those of a treated pt without the hAE. This study examined excess total cost of care (TCoC) and components (total inpatient; outpatient transfusion, granulocyte colony-stimulating factor [G-CSF], and thrombopoietic growth factor [TGF]) for three key hAEs (anemia, neutropenia, thrombocytopenia) during 1L FFX, FFX without 5FU bolus (FFXnb), and GnP. **Methods:** This retrospective observational study utilized Optum Market Clarity claims + EHR linked data. Inclusion criteria were: adult pts diagnosed with mPDAC between 1/1/2015 and 5/31/2023; initiated 1L FFX, FFXnb, or GnP within -14 to +90 days (index date); ≥6 months pre-index enrollment. hAEs during 1L were detected from lab values and grouped all grades combined. Pts were divided into "hAE groups" and "ref. groups" with and without any of the three hAEs of interest, respectively. Pts were matched 1:1 without replacement based on follow-up time. Costs were measured from the day a lab value indicated an hAE until 30 days later and during the corresponding times for ref. pts, then standardized to per-pt-per-month (PPPM). This study did not adjust for differences in pt characteristics. **Results:** Means and differences (95% CI) in TCoC following hAE detection for each regimen and hAE of interest are presented (Table). PPPM TCoC was greater for all three regimens and hAEs, driven by 1.6–4.9 times higher inpatient costs. Inpatient costs were $5,614 and $5,665 higher for FFX and FFXnb neutropenia pts vs. ref. groups. Use of outpatient G-CSF at any time during 1L was lower for FFX and FFXnb neutropenia pts, at 76% and 63% vs. 83% and 75% in the ref. groups, respectively. Outpatient TGF and transfusions were rare. **Conclusions:** All hAEs showed excess PPPM TCoC across regimens, driven by inpatient costs. FFX and FFXnb neutropenia pts had lower overall utilization of outpatient G-CSF but higher inpatient costs following detection of neutropenia, which may represent an opportunity for improved pt management. Research Sponsor: Ipsen Biopharmaceuticals, Inc.

Excess TCoC of hAEs.

AE (any grade)	FFX, hAE group (N=199, 127, 157)	FFX, ref. group (N=199, 127, 157)	Diff. (95% CI)	FFXnb, hAE group (N=202, 106, 137)	FFXnb, ref. group (N=202, 106, 137)	Diff. (95% CI)	GnP, hAE group (N=406, 268, 270)	GnP, ref. group (N=406, 268, 270)	Diff. (95% CI)
Anemia TCoC, mean	30,561	22,310	8,251 (4,396–12,106)	31,866	23,380	8,487 (4,007–12,966)	32,662	27,697	4,965 (1,320–8,610)
Neutropenia TCoC, mean	30,382	24,952	5,430 (-31–10,891)	26,588	22,571	4,017 (-1,288–9,322)	32,975	31,051	1,924 (-4,629–8,477)
Thrombocytopenia TCoC, mean	30,566	22,535	8,031 (3,895–12,166)	31,434	22,958	8,476 (2,962–13,991)	33,863	30,751	3,113 (-3,310–9,536)

Costs of generic vs. branded nab-paclitaxel (nP) during first-line (1L) treatment of metastatic pancreatic adenocarcinoma (mPDAC) with gemcitabine + nab-paclitaxel (GnP) in a real-world commercial database.

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Background: In 2013, the FDA approved GnP for 1L treatment of mPDAC. The total cost of care (TCoC) to payers during treatment is driven by the cost of nP, which first faced competition from generics in April 2022. We examined cost per generic vs. branded nP claim and cost per patient (pt) before and after generic entry. **Methods:** This retrospective observational study utilized Optum Market Clarity claims + EHR data. Inclusion criteria were: adult pts diagnosed with mPDAC between January 1, 2015 and May 31, 2023; initiated 1L GnP (index date) -14 to +90 days from diagnosis; 6 months pre-index enrollment. Claims were grouped into branded vs. generic based on NDC. For per-pt-per-month (PPPM) analyses, pts were required to have ≥30 days post-index enrollment and no nP claims missing NDCs. Pts were grouped into 1) branded pts with all nP claims for the branded NDC, and 2) generic pts with all claims for generic NDCs. TCoC, inpatient, outpatient, and chemotherapy drug cost to payers were calculated. The full study period and the period of generic entry beginning April 1, 2022 were examined. **Results:** Costs for branded vs. generic nP claims and GnP pts for 1L treatment initiated April 1, 2022 and after are shown (Table). The mean cost of generic claims was affected by outliers and was more than twice that of branded. The median nP claim cost, less affected by outliers, was still $2,408 greater for generic vs. branded. Costs for branded nP claims were similar before and after generic entry (not shown). 20 1L GnP pts had only branded nP claims, 24 had only generic, and 156 had mixed or were missing ≥1 NDC. PPPM mean TCoC was approximately twice as high for generic vs. branded. Medians were more similar, with generic median $2,037 higher vs. branded. Differences were driven by drug cost; other inpatient and outpatient costs were similar between the two groups (not shown). Costs of branded GnP pts before and after generic entry were similar (not shown). **Conclusions:** Publicly available prices for branded nP are $1,673 vs. $1,460 for generic per 100mg. In real-world data, we found generic nP costs per claim and PPPM tended to be higher than branded. While GnP is accepted by payers and clinicians as a standard 1L regimen for mPDAC, its real-world drug cost is at the level of branded therapy, despite generic availability. Future research should evaluate this finding as more post-generic-entry data becomes available. Research Sponsor: Ipsen Biopharmaceuticals, Inc.

Branded vs generic costs.

	Statistic	Branded	Generic
Claims	**N**	**343**	**308**
Cost per claim	Mean (SD)	4,721 (2,197)	11,761 (40,559)
	Median (IQR)	3,482 (2,945–6,583)	5,890 (3,482–6,964)
Patients	**N**	**20**	**24**
TCoC, PPPM	Mean (SD)	25,410 (9,531)	52,129 (121,166)
	Median (IQR)	22,276 (17,911–32,180)	24,313 (17,231–36,774)
GnP drug cost PPPM	Mean (SD)	11,056 (4,334)	35,910 (118,205)
	Median (IQR)	10,162 (7,658–13,508)	10,554 (7,162–16,797)

Real-world use of supportive medications to prevent and manage key adverse events during first-line (1L) treatment of metastatic pancreatic adenocarcinoma (mPDAC) with FOLFIRINOX (FFX), FFX without 5FU bolus, and gemcitabine+nab-paclitaxel (GnP).

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Background: 1L treatment of mPDAC with FFX or GnP typically requires supportive care to alleviate toxic effects. Two of the most common adverse events are neutropenia, which may be treated with granulocyte colony-stimulating factor (G-CSF), and diarrhea, often treated with atropine and/or loperamide. Medications may be administered as prophylaxis (PPx) or to manage symptoms as they occur. Other methods to improve tolerability include modifying regimens with lower doses or omitted components, such as removing the 5FU bolus from FFX (FFXnb). This study examines PPx vs non-PPx G-CSF, atropine, and loperamide during 1L treatment with FFX, FFXnb, and GnP. **Methods:** This retrospective observational study utilized Optum Market Clarity claims + EHR linked data. Inclusion criteria were: adult patients (pts) diagnosed with mPDAC between January 1, 2015 and May 31, 2023; initiated treatment (index date) with 1L FFX, FFXnb, or GnP -14 to +90 days from diagnosis; no evidence of other cancers in 6 months pre-index continuous enrollment; at least one EHR record post-index. G-CSF PPx was defined as administration/fill within 7 days of the start of a treatment, inclusive. PPx with atropine or loperamide was defined as administration/fill the day before or day of a treatment; all other use was categorized "non-PPx". Pts may appear in both PPx and non-PPx categories. **Results:** Rates of overall use, PPx, and non-PPx for each medication of interest are shown (Table). The overall G-CSF use rate was >3.5 times higher for FFX and FFXnb vs GnP. While ≥70% of FFX/FFXnb pts received PPx, non-PPx still occurred in more than 1 in 8 pts. One-fifth of FFX and FFXnb pts received atropine at any time, vs. just 6% of GnP pts. Atropine PPx was less common than non-PPx for FFX/FFXnb and was rare in GnP pts. Similar to the other medications, loperamide was observed more for FFX and FFXnb vs. GnP, but was the least common of the medications, used by <10% of pts in any regimen. PPx and non-PPx use were similar. Use of the three medications was similar between FFX and FFXnb. **Conclusions:** In a RW cohort of 1L mPDAC pts, G-CSF and antidiarrheal use was more common in FFX and FFXnb than GnP pts. Most FFX and FFXnb pts received PPx G-CSF, reflecting clinician efforts to manage the high risk of neutropenia with these regimens. Non-PPx G-CSF still occurred in 14–18% of pts. FFXnb had similar rates of concomitant medications as FFX, indicating that removing the 5FU bolus may have minor impact on the need for some supportive medications. Research Sponsor: Ipsen Biopharmaceuticals, Inc.

Concomitant medication use.

Medication		1L FFX (n=511)	1L FFXnb (n=583)	1L GnP (n=958)
G-CSF	% treated overall	77%	71%	20%
	% PPx	76%	70%	17%
	% non-PPx	18%	14%	9%
Atropine	% treated overall	20%	20%	6%
	% PPx	11%	9%	2%
	% non-PPx	15%	16%	5%
Loperamide	% treated overall	7%	9%	3%
	% PPx	4%	5%	1%
	% non-PPx	4%	6%	2%

Real-world microscopic residual disease (MRD) monitoring with circulating tumor DNA (ctDNA) in pancreatic adenocarcinoma (PDAC).

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Background: ctDNA is a powerful tool that can detect MRD with limited but emerging data in patients (pts) with PDAC. We describe the clinical outcomes of pts with PDAC who underwent real-world MRD testing at our institutions. **Methods:** We retrospectively analyzed pts at 2 institutions with ≥1 tumor-informed ctDNA test (Natera, Inc) ordered for MRD testing post-operation (post-op) between June 2020-August 2024. Descriptive statistics were used to characterize clinicopathologic factors, ctDNA positivity (ctDNA+), and relapse rates (RR). Median relapse-free survival (mRFS) and overall survival (mOS) were calculated using Kaplan-Meier methods, comparisons between groups by log-rank test, and hazard ratios (HR) by Cox proportional hazard models. Associations between clinicopathologic factors and outcomes were assessed by Fisher's exact and Chi-square tests. **Results:** Of the 54 pts who underwent MRD testing, 48 pts (88.9%) had successful tests. Failures were due to insufficient tissue. Patients with successful tests were staged as follows: 31.2% stage 1, 50.0% stage 2, and 18.8% stage 3, with 12.5% having R1/2 resections and 46.8% being node-positive. With a median follow-up of 21.0 months (mo), 22 (45.8%) pts had relapsed and 22 (45.8%) were anytime-ctDNA+. Sensitivity and specificity of ctDNA for relapse was 77.3% and 86.4%, respectively. Between anytime- and never-ctDNA+ pts, RR were 77.0% vs. 23.1%, mRFS 14.2 mo vs. not reached (NR) (HR 4.9, 95% CI 1.9-12.9), and mOS 31.7 mo vs. NR (HR 3.3, 95% CI 1.0-10.9), respectively. For 27 pts tested in the MRD window (2-12 weeks post-op), mRFS was 6.6 vs. 25.0 mo (HR 3.1, 95% CI 1.0-9.4) and mOS was 25.5 mo vs. NR (HR 2.5, 95% CI 0.6-10.2) in ctDNA+ vs. negative pts. ctDNA+ preceded radiographic relapse in 54.5% of patients with a median lead time of 166 days (IQR 70-332). In anytime-ctDNA+ pts, there were non-statistically significant trends for improved RFS and OS based on ctDNA decrease and clearance (table). No clinicopathologic factors predicted ctDNA+. Post-op ctDNA+ ($p < 0.001$) and CA 19-9 ($p=0.03$) were the only factors that had a significant association with RFS. **Conclusions:** Post-op ctDNA is a strong prognostic marker for survival in PDAC, predicting relapse about five months before imaging. ctDNA+ status is independent of clinicopathologic factors and the strongest predictor of recurrence. These findings highlight a window where novel therapies could be tested to eradicate MRD. Longer follow up and prospective studies are needed to validate its predictive role with therapies in the MRD setting to improve outcomes in high-risk pts. Research Sponsor: None.

		RFS events/ total	mRFS (mo)	HR (95% CI)	p	OS events/ total	mOS (mo)	HR (95% CI)	p
ctDNA clearance	Yes	6/9	17.0	0.61 (0.2-1.7)	0.35	2/9	NR	0.37 (0.08-1.8)	0.20
	No	11/13	10.3			8/13	31.7		
ctDNA MTM/mL reduction	≥50%	8/11	15.5	0.93 (0.3-2.9)	0.90	3/11	NR	0.54 (0.1-2.4)	0.42
	<50%	6/8	13.6			4/8	31.7		

Changes in peri-operative chemotherapy and radiation use among patients with resected pancreatic ductal adenocarcinoma (PDAC) between 2004-2021 in the US SEER dataset.

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Background: PDAC peri-operative management changed over 20 years. Randomized data supports adjuvant and multiagent chemotherapy (CT), while peri-operative chemoradiotherapy (RT) was used selectively in the US. We examined recent trends in treatment with outcomes in surgically resected PDAC patients. **Methods:** Using the Surveillance, Epidemiology, and End Results (SEER) 17 dataset, we identified surgically resected primary M0/MX PDAC cases from 2004-2021 over 3-year periods (P1-P6). Neoadjuvant (NCT), adjuvant (ACT), or multiagent (MACT) CT scheduling was included. χ^2 analysis assessed cohort characteristics over time and multinomial logistic regression assessed odds of CT allocation adjusted by cohort characteristics. Multivariable Cox proportional hazards model assessed treatment-related overall survival (OS) adjusted for cohort characteristics. **Results:** Among 55,293 cases of M0/MX PDAC, 21,904 (40%) completed surgery. Median age was 66 years; stages were T1/T2/T3/T4 (8.0/25/59/6.0%) and N0/N+ (38/54%). RT use declined (P1:P6, 41%->18% p<0.001) while CT use increased (P2:P6, 63%->79%, p<0.001). NACT (P2:P6, 4.1%->23% p<0.001) and MACT (P2:P6, 1.6%->20%, p<0.001) use increased while ACT decreased (P2:P6, 58%->36%, p<0.001). CT is likely to be allocated to young patients (<65yo) with Stage II-III tumors located at the pancreatic head (OR>1.00, p<0.001). Median OS increased over time (P1:P6, 18mo->32mo, p<0.001). Multivariable analysis shows that Adjuvant RT (HR 0.84 [0.81-0.87], p<0.001) and all CT (NCT: HR 0.71 [0.66-0.77], p<0.001; ACT: HR 0.63 [0.60-0.66], p<0.001; MACT: HR 0.65 [0.60-0.71], p<0.001) improved OS. **Conclusions:** This study shows decreased RT and increased CT adoption from 2004-2021. Median OS improved over the years, potentially due to peri-operative advances during this study period. Research Sponsor: None.

Trends in treatment use over 3-year periods.

	RT N (%)	CT N (%)	Neo CT N (%)	MACT N(%)
P1: 2004-2006 N=2661	1087 (41)	0 (0)[a]	0 (0)[a]	0 (0)[a]
P2: 2007-2009 N=3185	1185 (37)	2021 (63)	129 (4.1)	50 (1.6)
P3: 2010-2012 N=3536	1226 (35)	2460 (70)	267 (7.6)	138 (3.9)
P4: 2013-2015 N=3947	1210 (31)	2880 (73)	434 (11)	244 (6.2)
P5: 2016-2018 N=4224	971 (23)	3256 (77)	676 (16)	491 (12)
P6: 2019-2021 N=4351	787 (18)	3430 (79)	1003 (23)	857 (20)

[a]No cohorts reported chemotherapy use during period 1.
RT: Radiation Therapy; CT: Chemotherapy; NCT: Neoadjuvant Chemotherapy; MACT: Multiagent Chemotherapy.

Investigating communication barriers impacting postoperative decision regret in patients with pancreatic ductal adenocarcinoma.

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Background: Patient communication barriers are associated with worse quality of care and clinical outcomes. However there are limited data exploring their impact on shared decision-making (SDM) in the preoperative setting, especially for pancreatic ductal adenocarcinoma (PDAC). This study investigates the impact of patient communication barriers on patient-physician communication surrounding the decision to pursue pancreatectomy. **Methods:** This cross-sectional study investigates the impact of communication barriers on the degree of decisional regret (DR) in patients with PDAC who have undergone curative-intent resection at least 6 months prior to study recruitment. Patients completed validated surveys assessing communication preferences, SDM participation, and DR. Health literacy was assessed by the BRIEF Health Literacy survey. Groups were stratified by presence or absence of DR, assessed by the Decisional Regret Scale (DRS). Characteristics and survey scores were compared by chi-square and independent t-test. Given significant findings, results of this interim analysis are reported below. **Results:** 45 patients met inclusion criteria and completed all questionnaires. 19 (42.2%) patients expressed regret about their decision to pursue surgery with 5 patients expressing moderate to severe regret (DRS $\geq$25). Baseline characteristics were similar between groups. There were no significant differences in neoadjuvant or adjuvant treatments, or operative characteristics. Both groups reported high participation in preoperative SDM (average score 34.47 $\pm$ 9.67 in DR group vs 36.08 $\pm$ 8.88, P=0.567). Despite similar levels of education between groups, those expressing regret had a lower level of health literacy (14.68 $\pm$ 4.10 vs 17.35 $\pm$ 2.07, P=0.016). Fewer patients also identified English as their primary language in the DR group (78.9% vs 100%, P=0.026). While most patients preferred an active role in the final decision making process, fewer patients in the DR group reported this happening in actuality (κ= 0.635 vs κ=0.934). **Conclusions:** Almost half of patients expressed some regret pursuing surgery for PDAC. Despite similar educational backgrounds, a higher proportion of those expressing regret did not identify English as their primary language and had lower health literacy. While there was high participation in SDM, there was greater discordance between preferred and actual roles in the final decision making process. This identifies an important disparity to address in future preoperative discussions. Research Sponsor: None.

699 **Poster Session**

Assessing impact of postoperative outcomes and quality of life on decisional regret after resection of pancreatic ductal adenocarcinoma.

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Background: Surgery is potentially curative for patients with localized pancreatic ductal adenocarcinoma (PDAC). Given the high rate of complications, decline in quality of life (QoL) postoperatively, and high rates of recurrence, clear preoperative communication and expectation setting is critical. This study investigates clinical outcomes that affect postoperative decisional regret (DR) in patients with PDAC. **Methods:** This mixed-methods study investigates the impact of clinical outcomes and QoL on the degree of DR in patients with PDAC who have undergone curative-intent resection at least 6 months prior to study recruitment. Patients completed validated surveys assessing DR and QoL by the European Organization for Research and Treatment of Cancer (EORTC). Groups are stratified by presence or absence of DR, assessed by the Decisional Regret Scale (DRS). Characteristics and survey scores were compared by chi-square, independent t-test, and median test. Narrative responses were thematically analyzed. Given significant findings, results of this interim analysis are reported below. **Results:** 45 patients met inclusion criteria and completed all questionnaires. 19 (42.2%) patients expressed regret about their decision to pursue curative-intent surgery, with 5 patients expressing moderate to severe regret (DRS $\geq$ 25). There was no significant difference in median postoperative timing of survey completion between groups (56 months in DR group vs 34 months, P=0.465). Baseline characteristics, medical treatments, and operative approaches were similar. Both groups had similar recurrence rates at time of survey completion. While 30 and 90 day readmission rates were similar, a greater proportion of the DR group had 30 day complications (42.1% vs 15.4%, P=0.045). Those expressing regret reported poorer physical functional status (mean score 79.30 $\pm$ 18.84 vs 92.31 $\pm$ 10.27, P=0.011) and greater impact on social activities (mean score 67.54 $\pm$ 33.55 vs 84.67 $\pm$ 19.19, P=0.038). Thematic analysis of narrative responses revealed that a majority of patients expressed a strong desire to have a greater focus on potential QoL changes during preoperative discussions. **Conclusions:** Patients expressing postoperative DR experienced more 30 day complications, and still report lasting effects on their current physical status and social activities. This emphasizes the importance of thorough preoperative counseling on surgical risks and potential changes in postoperative QoL so that patients may make the best informed decision. Research Sponsor: None.

700 **Poster Session**

Comparison of demographics, mortality outcomes and resource utilization in early-onset versus later-onset pancreatic cancer: An analysis of the US National Inpatient Sample (NIS).

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Background: Pancreatic cancer is associated with higher mortality, with 5-year relative survival of 12%. Early-onset pancreatic cancer (EOPC) is relatively uncommon and is seen in 5-12% of the cases. Recent studies indicate a rising trend in the incidence of EOPC. In this study, we categorized EOPC for ages less than 50 and later-onset pancreatic cancer (LOPC) for ages above or equal to 50 years. The aim of the study is to understand difference in demographics, mortality, and resource utilization between the two sub-categories of patients admitted to acute care hospitals in the United States. **Methods:** This is a retrospective cohort study of adult patients hospitalized with pancreatic cancer as the primary diagnosis at acute care hospitals across the United States in 2021, using the National Inpatient Sample (NIS) database. International Classification of Diseases, 10th Revision (ICD-10) codes were utilized to identify patients with a primary diagnosis of pancreatic cancer. The primary outcome is comparing demographics and in-hospital mortality between EOPC and LOPC. The secondary outcomes are length of stay (LOS) and total hospitalization charges. We performed multivariate logistic regression analysis to assess in-hospital mortality, LOS, and total hospitalization charges after adjusting for potential confounders such as sex, race, Charlson comorbidity index, hospital teaching status, and hospital region. **Results:** Our study included 38,654 patients with pancreatic cancer, of which 2,242 were EOPC and 36,412 were LOPC. The majority of the baseline characteristics were similar in both groups, except that EOPC has a higher Hispanic population (16.06% versus 8.77 %, $p < 0.001$), higher Medicaid as the payer source (28.28 % versus 7.32%, $p < 0.001$), and a more significant number of patients admitted to an urban teaching hospital (89.2 % versus 85.2%, $p < 0.05$) compared to LOPC subpopulation. No statistically significant difference in mortality was found between the two sub-categories (odds ratio 0.97, 95 % CI 0.60-1.5). Charlson index, hospital location, hospital bed size, and hospital region are independent predictors of mortality. There was no statistically significant difference between the two sub-categories for mean length of stay and total hospitalization charges. **Conclusions:** No significant difference was observed between EOPC and LOPC regarding in-hospital mortality, total length of stay, and total hospitalization charges. Our study highlights that EOPC disproportionately affects the Hispanic population and also a greater number of EOPC patients being admitted to urban teaching hospitals. Future research is needed to understand the disproportionate effect on the Hispanic population in the EOPC group and continued research in the domain, as the incidence of early-onset pancreatic cancer is expected to rise. Research Sponsor: None.

Understanding rates of attrition and barriers to surgical resection in patients undergoing neoadjuvant therapy for localized pancreatic cancer.

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Background: Treatment paradigms are shifting toward neoadjuvant therapy (NAT) for resectable (R) and borderline resectable (BR) pancreatic cancer (PC) based on the rationale of treating occult metastatic disease, potentially downstaging patients (pts) to resectability if anatomically unresectable and, most importantly, selection of biology and pts most likely to benefit from potentially morbid surgery. While the benefits of NAT for PC have been well described, what remains underreported are the true attrition rates and underlying reasons for failure to proceed to surgical resection among pts receiving NAT. Herein, we present true intent-to-treat (ITT) attrition data of pts undergoing NAT for localized PC (LPC). **Methods:** Pts with LPC and who underwent ≥ 1 cycle NAT from January 2013-2022 were included. Clinicopathologic data were compared between those who did and did not proceed to pancreatectomy. Multivariable models identified factors associated with odds of attrition. Overall survival (OS) was estimated using the Kaplan-Meier method. **Results:** 427 LPC pts received NAT, including 57 (13%) with R, 133 (31%) BR, and 237 (56%) locally advanced (LA) disease. 64% had head tumors with median tumor size of 3.3 cm (IQR 2.7-4.2) and median CA 19-9 of 267 U/mL (IQR: 58-766). 60% of pts received FOLFIRINOX and 28% Gemcitabine-Abraxane for a median of 4.2 months (IQR 2.5-5.7). During NAT, 22% switched regimens and 30% received radiation. Following NAT, 207 (48%) pts underwent attempted resection, and 182 pts (43%) had successful pancreatectomy. The rate of attrition among all LPC pts was 52%, of which 20% had disease progression, 18% remained inoperable, 11% had physiologic decompensation, 2% were lost to follow-up, and 1% declined surgery. Rates of attrition were: 23% for R, 44% for BR, and 73% for LA ($p<0.01$). Higher stage was associated with increased odds of attrition (OR 9.7, 95% CI 4.8 – 19.5 for LA and OR 3.0, 95% CI 1.4 – 6.1 for BR compared to R). The most common reasons for attrition were: metastatic progression (5% in R, 18% in BR, 26% in LA), the primary tumor was deemed inoperable (2% in R, 12% BR, 32% LA), or physiologic decompensation on NAT (7% R, 11% BR, or 12% LA). Pts undergoing pancreatectomy had a median OS of 30 vs 16 months for those not ($p<0.01$). ITT median OS inversely correlated with stage – R: 30, BR: 21 and LA: 18 months ($p<0.01$). There was no difference in OS for those who received NAT and resection – R: 31, BR: 27, and LA: 34 months ($p=0.66$). **Conclusions:** At a tertiary center with 87% BR and LA pts, 43% of pts proceeded to pancreatectomy following NAT including 27% with LA tumors. Attrition of pts remains a significant issue even in the setting of R, highlighting the importance of NAT in pt selection. Pts who receive NAT and pancreatectomy experience similar OS, regardless of stage, further supporting the NAT approach in choosing patients most likely to benefit from resection.
Research Sponsor: U.S. National Institutes of Health; T32CA211034.

Evaluation of the impact of lymph node ratio on survival in patients with pancreatic cancer undergoing neoadjuvant therapy.

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Background: Pancreatic cancer continues to have a dismal prognosis despite improvements in surgical care. Resection status and lymph node status continues to be key factors impacting prognosis. There remains significant debate on the utility of increasing nodal harvest in patients undergoing surgery, while others report an improvement in survival. We sought to evaluate the impact of lymph node positivity to lymph nodes resected in patients with pancreatic cancer undergoing resection. **Methods:** Utilizing the National Cancer Database, we identified patients diagnosed with pancreatic cancer who underwent neoadjuvant chemotherapy (NCT) or neoadjuvant chemoradiation (NCR) followed by resection. We then stratified by nodal status and calculated a LN+/LNR (lymph node resected) ratio. Pearson's Chi-square test was used to compare categorical variables. Survival analysis was estimated by the Kaplan-Meier method and associated log-rank test. Propensity score matching (PSM) was completed for variables of age, sex, tumor size, LN removed, LN positive, pathogenic T- stage and N-stage, grade, 30- and 90-day mortality, R0 resection, facility volume, response, adjuvant therapy, and LNR. Significance was considered at p-value $\leq$0.05. **Results:** We identified 4117 patients with a median age of 65 (25-90). There were 2323 patients who underwent NCT and 1794 who underwent NCR. The median LN resected in the NCT was 18 vs 15 in the NCR, $p<0.001$. However, there were more LN+ in the NCT vs NCR, 2.1 vs 0.9, $p<0.001$. The R0 resection rates were lower in the NCT vs NCR 81% vs 86%, $p<0.001$. Additionally, complete response rates were higher amongst the NCR, 9.2% vs 4.7%, $p<0.001$. Lymph node ratios were consistently higher in the NCT cohorts among all ratio strata, $p<0.001$. Patients with pN+ disease had a median and 5-year survival of 26.4mo and 17% compared to their pN- counterparts at 38.8mo and 36%, $p<0.001$. Patients treated with NCT demonstrating a LNR of 0.01-0.1 had median and 5-year survival of 32.8 and 27%, LNR 0.11-0.4 (25.5mo and 12%), and LNR>0.4 (19.1mo and 10%), $p<0.001$. Patients treated with NCR demonstrating a LNR of 0.01-0.1 had median and 5-year survival of 29.1 and 21%, LNR 0.11-0.4 (25.8mo and 17%), and LNR>0.4 (21.3mo and 5%), $p<0.001$. PSM confirmed a higher LNR in the NCT cohort compared to NCR, $p<0.001$, however survival was also longer in these patients among the LNR 0.01-0.1, and 0.11-0.4, $p<0.001$. **Conclusions:** Lymph node status in patients with pancreatic cancer continues to be a significant prognostic indicator of survival. Moreover, lymph node ratio may be a more accurate predicter of survival in patients undergoing neoadjuvant chemotherapy and neoadjuvant chemoradiation. Research Sponsor: None.

703 **Poster Session**

Perioperative versus adjuvant therapy in resectable pancreas cancer.

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Background: Pancreatic ductal adenocarcinoma (PDAC) is the third-leading cause of cancer death in the United States with a 5-year survival rate of 13%. Recurrence rates are high following resection and optimization of neoadjuvant and adjuvant therapy remains under investigation, with the hypothesis that neoadjuvant/perioperative therapy may convert borderline resectable tumors to resectable and prevent progression of micro-metastases in the postoperative setting. Our study evaluates patient outcomes and models predictive and prognostic markers to assist with clinical decision-making in consideration of neoadjuvant therapy for resectable and borderline resectable pancreatic cancers. **Methods:** We performed a single-institution retrospective analysis of patients who underwent pancreatic resection between January 1, 2016 and December 31, 2020 (n=532). We excluded patients with unresectable disease or a non-PDAC diagnosis. Analysis of the remaining patients (n=175) was performed using automated and manual chart review, basic statistical comparisons with T-tests or Chi-squared tests, Kaplan-Meier Survival analyses, and Cox Proportional Hazard Testing using intention to treat analysis. **Results:** There were no significant differences ($p>0.05$) the presenting demographics between the adjuvant (AJ, n=97) and neoadjuvant/perioperative (NP, n=76) groups. Patients were more likely to have a history of pancreatitis, a risk factor for PDAC development, in the NP group (p=0.0153). Patients in the AJ group included patients who ultimately did not receive the recommended adjuvant therapy for various reasons (n=25) and had overall worse performance status (p=0.004) and received less chemotherapy overall compared to the NP cohort (p=0.0003). Adjuvant therapy was more likely to be gemcitabine with capecitabine (n=37, p=0.0007) in the AJ group and gemcitabine with nab-paclitaxel (n=11, p=0.0018) in the NP group. In the neoadjuvant setting, 62% of patients received FOLFIRINOX and 28% received gemcitabine with nab-paclitaxel. Patients in the NP group were more likely to receive radiation (p=0016), including 6 patients who received neoadjuvant radiation. Median overall survival (33.9 [28.8-43.8] versus 38.2 [28.2-47.8] months) and progression free survival (21.1 [18.1-34.2] versus 16.6 [11.7-23.6] months) were not significantly different between the AJ and NP groups, respectively. **Conclusions:** These results demonstrate similar presenting characteristics between patients with resectable PDAC treated in a period when general practice shifted from adjuvant therapy to a neoadjuvant/perioperative approach. Although patients treated neoadjuvantly received more chemotherapy overall and the chemotherapy regimens differed between the groups, patients had similar outcomes in the two groups. Research Sponsor: None.

Outcomes of surveillance in patients with intraductal papillary mucinous neoplasms (IPMNs) who tested negative for high-risk mutations (HRMs) in next-generation sequencing (NGS).

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Background: HRMs in NGS have demonstrated high sensitivity and specificity for predicting advanced neoplasia in patients with IPMNs. However, the outcomes of surveillance in patients who tested negative for these HRMs have not been investigated in prior studies. This study aimed to identify the predictors of progression in patients with negative NGS results for HRMs. **Methods:** We conducted a retrospective review of prospectively collected data from patients who underwent NGS testing between 2016 and 2023 in our pancreatic cancer prevention program. Patients without IPMNs (i.e., negative for KRAS/GNAS), with HRMs, and those without at least one follow-up imaging post-NGS were excluded from the study. Progression was defined as the development of new or additional worrisome features or high-risk stigmata (per Kyoto guidelines), or advanced neoplasia (HGD/PDAC) following NGS testing. A Cox proportional hazards regression model was used to identify predictors of progression. **Results:** A total of 543 patients underwent NGS testing during the study period, of which 210 patients met the inclusion criteria. The median follow-up duration was 22 months (IQR = 12-36). Of these, 72 patients developed progression following NGS testing. A total of 11 patients underwent surgical resection, and 5 of them had HGD. Factors predictive of progression included a history of smoking (HR = 1.74; 95% CI = 1.06 - 2.86), larger cyst size (HR = 1.09; 95% CI = 1.03 - 1.16), and the presence of a dilated main pancreatic duct (MPD) (HR = 2.56; 95% CI = 1.5 - 4.4) prior to NGS testing. The median time to progression for patients with worrisome features was 33 months, compared to 59 months for those without worrisome features at baseline. Similarly, the median time to progression for patients with cysts $\geq$ 2 cm was 38 months, compared to 57 months for those with cysts $<$ 2 cm. **Conclusions:** Patients who test negative for high-risk mutations in NGS are not exempt from the risk of progression. Any smoking history and presence of any worrisome features before NGS testing showed to be independent predictors of disease progression. For patients with smoking history, worrisome features or cysts $\geq$ 2 cm, frequent surveillance should be recommended following NGS testing.
Research Sponsor: None.

Results of Cox proportional hazards regression model for identifying the predictors of progression.

Variables	Hazards ratio (HR)	95% Confidence Interval	P-value
Age	0.99	0.96 – 1.01	0.257
Male sex	1.02	0.62 – 1.7	0.93
White race	1.07	0.63 – 0.83	0.8
Family history of pancreatic cancer	0.6	0.26 – 1.36	0.22
Personal history of other cancers	0.95	0.56 – 1.62	0.86
Diabetes	1.22	0.72 – 2.05	0.46
Any history of smoking	1.74	1.06 – 2.86	**0.03***
Pre-NGS cyst size (cm)	1.09	1.03 – 1.16	**0.006***
Pre-NGS dilated main pancreatic duct (≥5mm)	2.56	1.5 – 4.4	**0.001***
Pre-NGS presence of mural nodules	0.32	0.04 – 2.37	0.26

*Significant p-value.

Trajectories of premorbid healthcare contact days in veterans with pancreatic cancer.

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Background: Early detection of pancreatic cancer is a key clinical priority. Although prior work has evaluated machine learning models to identify at-risk patients for pancreatic cancer screening, this is limited by complexity and cost. "Contact days''— days spent receiving healthcare outside the home for any reason or length of time—have previously been used to measure the ''time toxicity'' of cancer care post-diagnosis. We sought to characterize trajectories of premorbid contact days, alongside standardized risk scores and healthcare costs to explore their use as potential markers for early pancreatic cancer diagnosis. **Methods:** We identified decedents with pancreatic adenocarcinoma diagnosed between 2012 and 2022 who had at least one primary care visit in the Veterans Affairs (VA) system in each of the two years prior to diagnosis using the VA Clinical Cancer Registry (VACCR) and the Corporate Data Warehouse (CDW). We grouped patients into two cohorts: (1) surgical (receipt of curative intent surgery), and (2) non-surgical and collected sociodemographic and clinical characteristics. During a 2-year premorbid period we calculated monthly contact days (24 data points) and plotted them as percentages (e.g., 3 contact days in a month=10%) and visualized their trajectory using cubic smoothing splines. We plotted trajectories for each sociodemographic and clinical subgroup and visually evaluated differences. We similarly plotted monthly Care Assessment Need (CAN) scores, a validated VA risk score identifying patients at high risk of hospitalization/death, and monthly VA healthcare costs. **Results:** We included 8,654 patients (649 surgical - median age 69 years, 74% white, 78% stage 1-2 and 8,005 non-surgical - median age 71 years, 72% white, 21% stage 1-2). For both groups, trajectories of contact days were similar. Baseline contact days (-24 to -12 months) were 7-8% (≈2 days/month). Contact days started to rise on average 9 months prior to diagnosis at an approximate rate of 1% per month, reaching 15-20% at diagnosis (≈5-6 days/month). In the surgical group, patients with a Charlson comorbidity score of ≥2, urban residence, and Black race had higher baseline contact days. Those <65 years and rural residence had a steeper slope of rise in contact days. CAN score trajectories closely mirrored contact day trajectories in both groups. Cost trajectories were similar in both groups (≈$1000/month at baseline), rose gradually to ≈$2000/month by -3 months, and then rose sharply to >$6000/month 2 months before diagnosis. **Conclusions:** In this decade-long national study of Veterans diagnosed with pancreatic cancer, we found that contact days and CAN scores, but not costs, start to rise 9 months prior to diagnosis. These data provide the first evidence supporting their further evaluation in pancreatic cancer risk models which may aid early diagnosis. The similar trajectories in patients who go on to receive curative-intent surgery or not may imply they may be markers for pancreas cancer diagnosis but perhaps not outcomes. Research Sponsor: Pancreatic Cancer Action Network.

Second-line therapy with Nal-IRI/5-FU/FA after failure of gemcitabine/nab-paclitaxel in advanced pancreatic cancer (PC): Predictive role of 1st-line therapy (PREDICT) and impact of quality of life (QoL).

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Background: Metastatic PC has the lowest survival rate of all malignant diseases and is the fourth most common cancer. QoL remains an unresolved issue in PC for patients (pts) after treatment failure of 1st-line CTx (TTF1). Nanoliposomal irinotecan (Nal-IRI) with 5-FU/folinic acid (FA) increased survival of PC pts from 4.2 months (mo) to 6.1 mo as compared to 5-FU/FA alone, with a rate of toxicities that may impact QoL. PREDICT is an open label, single arm, multicenter Phase IIIb trial (NCT03468335) and examined 2nd-line Nal/IRI/5-FU/FA for PC. Here we report on QoL during 2nd-line Nal-IRI after failure of 1st-line gemcitabine/nab-Paclitaxel (gem/nab-Pac). **Methods:** In this prospective trial, 151 patients with locally advanced or metastatic PC were treated with biweekly Nal-IRI/5-FU/FA (70 mg/m^2, 2400 mg/m^2, 400 mg/m^2) after failure of gem/nab-pac (TTF1). Primary end point (EP) was the time to treatment failure of 2nd-line therapy (TTF2) and has been reported elsewhere. Secondary EP were QoL and Health related quality of life (HR-QoL) which were evaluated using questionnaires (EORTC QLQC30, QLQ-PAN26, EQ-5D-5L). Evaluation of time to definitive deterioration of QoL (TDD) during 2nd-line therapy, defined as the time from screening/baseline until loss of ≥10 points in the EORTC QLQ-C30 was compared to baseline. **Results:** QoL analyses were performed with all QoL evaluable subjects set (QAS). The QAS included 143 pts, of which 48 pts were included in TTF1 high (TTF of 1st-line ≥213 d) and 49 pts in the TTF1 low (TTF ≤119 d) cohort, with 79 (54.1%) female and 67 (45.9%) male pts. Mean age was 68.2±8.9 years. Overall, a low global health status / QoL mean score was observed at baseline for TTF1 high and low cohorts (about 48 points), which slightly improved during the study for TTF1 high (3.5±23.7) and slightly worsened for TTF1 low cohort (-0.7±22.5). Substantial improvements were observed for both cohorts in emotional functioning scale at certain study time points (TTF1 high: up to 9.1±25.7, TTF1 low: up to 6.6±17.0). A relatively high mean score was observed for cognitive function scale for both cohorts at baseline (TTF1 high: 70.1±30.2; TTF1 low: 72.5±21.4). Pts in the TTF1 low vs. high cohort showed more pronounced worsening of cognitive function. Median TDD was numerically longer (about one month) for TTF1 high (5,3 mo) compared to TTF1 low cohort (3,9 mo). The probability to maintain QoL after six months was similar between the two cohorts. **Conclusions:** QoL showed similar global and HR-QoL scores with only slight changes during the study. Median TDD was numerically longer for high TTF1 pts, but similar after six months between pts still on 2nd line Nal/IRI/5-FU regardless of duration of TTF in 1st-line. Clinical trial information: NCT03468335. Research Sponsor: None.

Disease-free survival as a surrogate endpoint for overall survival in adults with early-stage pancreatic cancer: A trial-level surrogacy analysis.

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Background: In early-stage cancer, overall survival (OS) is the gold-standard endpoint for assessing therapeutic efficacy. However, long follow-up periods make OS challenging and costly to measure in clinical trials. Disease-free survival (DFS), which captures the time to recurrence or death, has been proposed as a surrogate endpoint for OS in early-stage pancreatic cancer (PC). This surrogacy relationship could allow for a more rapid assessment of treatment benefits, but its validity requires robust analysis across clinical trials. Understanding the trial-level correlation between DFS and OS is crucial to optimizing trial design and grant accelerated drug approval. **Methods:** We conducted a systematic search of Embase, PubMed, and Cochrane databases through August 2024, identifying randomized controlled trials (RCTs) that evaluated adjuvant or neoadjuvant systemic therapies in adult patients with PC. Included studies reported both OS and DFS outcomes. Hazard ratios for DFS (HR_DFS) were analyzed as surrogate endpoints for OS (HR_OS). Statistical analyses utilized linear regression models, weighted by trial sample sizes, to determine the strength of the correlation. Surrogacy strength was categorized by the coefficient of determination (R^2), with $R^2 \geq 0.7$ indicating a strong correlation, 0.69–0.5 a moderate correlation, and values < 0.5 indicating weak correlation. **Results:** Out of 3,048 studies screened, 30 RCTs met the inclusion criteria, comprising 5,275 patients in adjuvant therapy trials and 557 patients in neoadjuvant therapy trials. The median follow-up duration ranged from 16 to 53.5 months, with 12 phase II trials included. Overall, DFS showed strong surrogacy for OS in adjuvant trials (R^2 = 0.70; 95% CI, 0.50–0.91). Subgroup analysis indicated strong correlation in phase III trials (R^2 = 0.71; 95% CI, 0.46–0.96) and moderate correlation in phase II trials (R^2 = 0.67; 95% CI, 0.21–1.00). The strongest surrogacy was observed in studies incorporating adjuvant radiation therapy (R^2 = 0.81; 95% CI, 0.52–1.00). In the analysis of neoadjuvant trials, HR-DFS also presented strong correlation with OS (R^2 = 0.90; 95% CI, 0.56–1.00). **Conclusions:** There is a strong trial-level correlation between DFS and OS in both adjuvant and neoadjuvant therapy trials for early-stage pancreatic cancer, particularly in phase III studies and trials incorporating adjuvant radiation therapy. However, the strength of this surrogacy varies depending on the treatment setting and trial design. Research Sponsor: None.

709 **Poster Session**

A mixed methods evaluation of the Canopy Cancer Collective's implementation of network-wide screening initiative in 2023.

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Background: The Canopy Cancer Collective, a Learning Health Network (LHN) comprised of 14 pancreatic cancer centers across the U.S., set ambitious goals in 2023 to improve screening rates across four key areas: clinical trial screening (from a baseline of 30% to 100%), assessment of enzyme replacement (37% to 100%), goals of care discussions (57% to 100%), and palliative care screening (30% to 100%). This study aimed to identify the barriers and facilitators influencing the implementation of these screening aims, with the goal of developing informed, actionable recommendations for future implementation efforts. **Methods:** Throughout 2023, data were gathered from monthly assessments completed by site coordinators and quality improvement coaches at each of the 14 centers. These assessments captured insights into planning, implementation, and challenges related to the screening aims. Additionally, a comprehensive site assessment conducted in October 2023 was analyzed. We employed content analysis to identify the key factors influencing the implementation process, with percentage improvements in screening aims documented. **Results:** A total of 128 assessments were reviewed. The most common barriers identified were insufficient personnel and time for screening implementation (n=7), delayed responses and lack of engagement from supporting teams (n=6), unanticipated delays and logistical barriers to the procedures (n=6), recruitment challenges (n=3), and changes in the staff or management (n=5). Facilitators that supported implementation included: identification and testing of patient-centered strategies for implementation of screening (n=11), team expansion (n=8), consistent data collection and sharing through the LHN database (n=7), launching of other initiatives tailored to institutional needs (n=5), and collaboration between teams within or across institutions (n=4). Among the barriers to form ideal interprofessional composition, resource shortages (n=7), provider shortages (n=6), and coordination difficulties (n=5) were included. By January 2024, the network's screening rates had improved significantly: 92% for clinical trial, 87% for enzyme need and palliative care each, and 97% for goals of care. **Conclusions:** While common challenges such as staffing limitations and logistical barriers posed obstacles, facilitators like patient-centered strategies, team growth, and improved collaboration were key to the successful implementation of the screening aims. These findings offer actionable insights for enhancing future implementation efforts, with a focus on addressing resource limitations and fostering team engagement. Moving forward, we plan to assess the outcomes of these screening initiatives to examine their impact on patient care and evaluate the perceptions of site on utilization of implementation strategies to promote its adoption. Research Sponsor: None.

Impact of neoadjuvant radiation therapy modalities on post-surgical lymphopenia in operable pancreatic cancer.

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Background: Recent advances in therapeutic vaccines show promise for *KRAS*-mutated pancreatic cancer (PC), with ongoing clinical trials exploring their potential further. Lymphocytes are essential in generating vaccine-induced immune response. Patients who develop lymphopenia following neoadjuvant radiation therapy (RT) are frequently excluded from these vaccine trials. We aimed to assess how different modalities of neoadjuvant RT impact the severity and incidence of lymphopenia. **Methods:** We identified patients with PC who received neoadjuvant therapy including RT followed by surgery at our institution between March 2009 and June 2024. Patients were stratified by the type of RT they received: conventional long-course chemo-RT (LCRT) with 50.4 Gy over 28 fractions (Fr), short-course chemo-RT (SCRT) with 36 Gy over 15 Fr, and stereotactic body RT (SBRT) with 30 Gy over 5 Fr. The absolute lymphocyte count (ALC, 10^3 cells/μL) was compared at various timepoints: baseline, during RT, 6-weeks after surgery (post-op), and every 3-month interval up to 1 year post-op. Moderate lymphopenia was defined as ALC $<$1.0 and severe lymphopenia as ALC $<$0.5. **Results:** Among the 627 patients included in the study, 522 had baseline pre-RT ALC data available for analysis. There was no difference in baseline ALC among the different RT groups (Table). During RT, patients receiving LCRT experienced a higher incidence of severe lymphopenia (86.1%; median ALC (mALC) nadir 0.28) compared to SCRT (41.9%; mALC nadir 0.50) or SBRT (20.0%; mALC nadir 0.72, p$<$0001). No difference in the mALC or rates of lymphopenia was observed beyond 6 weeks post-op. Notably, 53% of the patients experienced lymphopenia for as long as 1 year after surgery. **Conclusions:** LCRT resulted in more severe lymphopenia compared to SCRT or SBRT during RT, but this difference in severity diminished 6 weeks after surgery. Alternatives in treatment sequencing for operable pancreatic cancer can have profound effects on patterns of disease recurrence. When applying multiple treatments in series, toxicities such as lymphopenia may impact receipt of therapy. Techniques to mitigate such toxicities, while maximizing response, will be important to consider. Research Sponsor: None.

Absolute lymphocyte count (ALC, 10^3 cells/μL) at different time point.

Patients with baseline ALC	LCRT (n=454)	SCRT (n=34)	SBRT (n=34)	P-value
Baseline ALC, median (IQR)	1.48 (0.74)	1.27 (0.94)	1.51 (0.74)	0.74
6-week lymphopenia, n (%)				
No	173 (37)	13/27 (48)	18/35 (51)	0.10
Moderate	219 (46)	11/27 (41)	16/35 (46)	
Severe	79 (17)	3/27 (11)	1/35 (3)	
1-year lymphopenia, n (%)				
No	101/217 (47)	3/8 (37)	9/13 (69)	0.054
Moderate	90/217 (41)	5/8 (63)	1/13 (8)	
Severe	26/217 (12)	0/8 (0)	3/13 (23)	

A novel 2-2 5-5 model for stratifying liver metastasis burden in patients with pancreatic cancer.

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Background: In metastatic pancreatic cancer (MPC), hepatic metastasis is uniformly counted as just 'one organ metastasis' regardless of its metastatic tumor burden. This study aims to develop a novel stratification system for liver metastasis based on tumor burden, and to evaluate its prognostic significance in MPC patients. **Methods:** Patients with MPC treated at Seoul National University Bundang Hospital from 2011 to 2021 were included as the training cohort. Patients with liver-only metastasis were selected for analysis. Liver metastasis burden was categorized by the number and maximum diameter of metastases as measured from diagnostic CT images. Optimal cutoffs for metastasis number and size were determined using Cox proportional hazards regression, adjusted for chemotherapy regimen. An independent cohort from Seoul St. Mary's Hospital and Hallym University Sacred Heart Hospital served as the validation group. The 'H category classification' suggested by the Japanese Society for Cancer of the Colon and Rectum (JSCCR) was also referenced. **Results:** A total of 1,257 patients were included, with 1,084 in the training cohort and 173 in the validation cohort. In the training cohort, data from 311 patients with liver-only metastasis were analyzed to derive the Two-two five-five ('2-2 5-5') criteria, classifying metastasis into Oligo-metastasis (≤2 metastases and ≤2 cm), Meso-metastasis (non-oligo and non-mega), and Mega-metastasis (≥5 metastases or ≥5 cm). Median overall survival (OS) significantly differed between the groups: Oligo (11.4 months), Meso (9.2 months, HR 1.66, 95% CI 1.14–2.43), Mega (5.6 months, HR 2.63, 95% CI 1.94–3.58) in the training cohort; Oligo (5.8 months) vs Mega (1.81 months, HR 1.94, 95% CI 1.16–3.22) in the validation cohort. **Conclusions:** This 2-2 5-5 model effectively stratifies MPC patients with hepatic metastasis into three prognostic groups, demonstrating significant survival differences in both training and validation cohorts. This could rationalize the independent therapeutic strategies by extent of hepatic metastasis. Research Sponsor: None.

Demographic characteristics of patients classified according to "two-two five-five" criteria.

	Oligo-metastasis (n = 88)	Meso-metastasis (n = 59)	Mega-metastasis (n = 164)	Total (n = 311)	P-value*
Age, median (range)	66.0 (33.2-91.0)	67.1(37.2-86.9)	68.7 (40.1-94.6)	67.9 (33.2-94.6)	0.587
Male, n (%)	43 (48)	36 (61)	91 (55)	170 (54)	0.330
Tumor location(Head), n(%)	47 (53)	27 (45)	60 (36)	134 (43)	<0.05
Number of liver metastasis, median (range)	1 (1-2)	3 (1-4)	12 (2-14)	5 (1-14)	<0.05
Maximal size of liver metastasis, cm (IQR)	10.6 (8.1-13.1)	15.9 (10.1-24.2)	21.6 (16.8-33.7)	17.2 (10.9-25.2)	<0.05
Average size of liver metastasis, cm (IQR)	10.3 (8.0-12.1)	13.5 (7.8-19.1)	13.2 (10.5-19.2)	11.9 (9.2-17.1)	<0.05
CA19-9 (U/mL)	462 (103-1,334)	560 (89-1,480)	1,170 (100-9,500)	745 (100-3,700)	<0.05

Treatment preferences in advanced pancreatic ductal adenocarcinoma (PDAC) from qualitative patient (PT), caregiver (CG), and oncologist (OC) interviews.

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Background: Most treatment regimens for patients with PDAC focus on extending survival while managing symptoms and maintaining quality of life (QoL). Here we report insights from PT, CG, and OC interviews on PDAC treatment preferences. **Methods:** A targeted literature review of 7 quantitative/preference and 12 qualitative/patient-reported outcome studies was used to develop semi-structured interview guides. Single 60-minute interviews of US PTs with stage 4 PDAC, CGs, and OCs with ≥5 years' experience treating PDAC were undertaken virtually by staff from March to April 2024. Interviews had 5 sections: background/clinical experience (OC only); key treatment benefits and risks to avoid (ranked by importance); treatment convenience; and 2 hypothetical treatment preference questions. Participants completed a sociodemographic and clinical questionnaire. Interviews were recorded with permission, transcribed, and analyzed with ATLAS.ti v8+. **Results:** Ten PTs, CGs, and OCs completed interviews (N=30 total), including 4 PT/CG dyads. Mean PT age was 56.8 (standard deviation [SD] 12.15) years; 6 were men and 4 did not have CGs. Mean CG age was 51.1 (SD 14.2) years; 9 were women and 7 cared for their partners. OCs had mean 18.5 (SD 6.9) years' clinical experience and 10 were men. Top-ranked treatment benefits were increased overall survival (OS) (6 PTs [60%], 8 CGs [80%], 7 OCs [70%]); improved QoL (6 PTs [60%], 5 CGs [50%], 7 OCs [70%]); stopping disease progression (5 PTs [50%], 3 CGs [30%], 7 OCs [70%]), and increased life expectancy (7 PTs [70%], 5 CGs [50%], 3 OCs [30%]). PTs' top 3-ranked side effects to avoid included fatigue (n=4 [40%]), nausea (n=4 [40%]), and vomiting (n=3 [30%]). CGs ranked trouble remembering (n=4 [40%]), neutropenia (n=4 [40%]), and neuropathy (n=3 [30%]) in the top 3, while OCs ranked vomiting (n=5 [50%]), neuropathy [n=5 [50%]), and neutropenia (n=4 [40%]) highest. Most PTs/CGs were indifferent to ≥1 infusion/visit and infusion duration. In the first hypothetical treatment preference question, the treatment with higher efficacy but higher side effect risk and longer infusion time was preferred by 80% of PTs/CGs and 70% of OCs. In the second hypothetical treatment question, treatments had the same efficacy; 90% of PTs/CGs/OCs preferred the treatment with shorter infusion time and lower risk of gastrointestinal side effects and hypersensitivity reactions, despite higher risk of severe hypokalemia and severe neutropenia. **Conclusions:** OS and QoL were ranked as similarly important by participants, suggesting that extending survival is insufficient without acceptable QoL. Avoiding side effects was important to all participants, though PTs, CGs, and OCs ranked specific risks with differing importance. More studies are needed to quantify tradeoffs between treatment efficacy and side effects in the care of patients with advanced PDAC. Research Sponsor: Astellas Pharma Inc.

Effect of ketorolac on serum GDF-15 and IL-8 in patients with pancreatic cancer with cachexia.

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Background: Cancer cachexia deteriorates the quality of life, chemotherapy tolerance and survivability of pancreatic cancer (PC) patients. Although a prevalent condition affecting PC patients, there are no approved therapies for cachexia and its mechanism of action is poorly understood. Elevated growth differentiation factor 15 (GDF-15), a stress-induced cytokine, has been associated with the pathogenesis of cachexia. Similarly, prior studies have displayed positive correlations between the proinflammatory factor interleukin-8 (IL-8) and PC cachexia. This abstract focuses on the exploratory endpoints (serum GDF-15 and IL-8 levels) of a feasibility trial on ketorolac as a treatment for PC cachexia. **Methods:** Patients took ketorolac 10 mg PO QID for 5 consecutive days, with weights measured on Day 1 (D1) and Day 6 (D6). Serum cytokines were drawn on D1 and D6. IL-8 was measured using a HDF15 assay (Eve Technologies). GDF-15 was measured using a human R-plex GDF-15 assay. Activity (e.g., number of steps) was measured via Fitbit. Differences in weight, activity, GDF-15, and IL-8 were analyzed using two-sample paired t-tests. **Results:** Twenty-nine patients were enrolled, of which 19 were evaluable. Median age was 70, 47.4% (9/19) were female, and 94.7% (18/19) were adherent with ketorolac. Statistically significant increases in both weight per baseline BMI and activity were observed from D1 to D6 (0.0716 ± 0.1079 kg/BMI, $p=0.012$; 1423 ± 2477 steps, $p=0.0221$). Of the 19 evaluable patients, 16 patients provided complete D1 and D6 pairs of serum cytokines. From D1 to D6, the mean GDF-15 level decreased by 277.5 pg/mL ($p=0.529$) and mean IL-8 decreased by 65.13 pg/mL ($p=0.103$). In all patients who gained weight ($n=14$), a trend toward reduced IL-8 from D1 to D6 was observed (79.57 ± 39.57 pg/mL, $p=0.065$). Similarly, for patients who gained $\geq$ 1-kg from D1 to D6 ($n=10$), a trend toward reduced GDF-15 was observed (951.2 ± 482.3 pg/mL, $p=0.080$). Patients with increased activity from D1 to D6 ($n=12$) had mean reductions of 157.7 pg/mL ($p=0.643$) and 70.17 pg/mL ($p=0.179$) in GDF-15 and IL-8, respectively. **Conclusions:** Given the growing literature suggesting associations between inflammatory cytokines and cancer cachexia, it is critical to investigate treatments that may possibly alter cytokine levels. Although our findings suggest that ketorolac aids in weight gain and increased activity, larger sample sizes are needed to determine whether these differences are related to reductions of specific serum cytokines. However, given reductions that trended towards significance for both GDF-15 and IL-8 serum levels within subsets of patients that gained weight on ketorolac, further studies investigating these cytokines are warranted to hopefully provide insight on the mechanistic action of ketorolac in cachexia. Clinical trial information: NCT05336266. Research Sponsor: Yinuoke, Ltd.

The value and indications for surgery in pancreatic cancer with synchronous liver metastases.

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Background: Over half of pancreatic cancer patients present with advanced disease and distant metastases, primarily in the liver, at initial diagnosis. Although systemic combination therapies have improved outcomes, the role of surgery remains contentious. This study seeks to pinpoint prognostic factors for patients with synchronous liver metastatic pancreatic cancer (sPDACLM) who receive effective treatments, and to identify patients likely to benefit from surgery. **Methods:** This is a retrospective cohort study. Included patients received first-line chemotherapy (AG/mFOLFIRINOX), with efficacy assessed every 4-6 cycles through imaging and biochemical evaluations. Imaging assessments were conducted according to RECIST 1.1 criteria, while biochemical evaluation focused on the reduction of CA19-9 (calculating cutoff values). Patients with good responses were considered for curative surgery, while those with minor responses decided between surgical treatment or continued maintenance therapy after being fully informed. **Results:** From May 2017 to February 2024, a total of 48 patients with sPDACLM who met the inclusion criteria were recruited for this study. The prognosis of patients with resectable or borderline resectable primary tumors was significantly superior to unresectable cases (18 months vs 11 months, p = 0.0011). Multivariate analysis revealed independent favorable prognostic factors: surgery (HR 0.22; p = 0.026), partial response (PR) per RECIST 1.1 criteria (HR 0.09, p = 0.040), and tumor marker reduction >85% (HR 27.99; p = 0.005). Achieving conversion effectiveness, as defined by imaging and biochemical criteria, was associated with a significantly improved prognosis compared to cases where conversion was ineffective (28 months vs. 12 months, p = 0.0086). Notably, effective surgery patients had a median survival of 49 months and a 5-year survival rate of 33%. Conversely, for patients who did not respond to systemic therapy, surgery did not confer any significant impact on overall survival; however, the corresponding survival curve indicated a trend towards shorter survival compared to those who did not undergo surgery. **Conclusions:** Patients with sPDACLM fulfill both imaging and biochemical criteria, exhibit improved prognoses; additionally, surgery has been shown to significantly prolong their median overall survival times. Research indicates that accurate screening for cases suitable for conversion surgery can significantly improve the survival rates of sPDACLM. Research Sponsor: National key clinical specialty construction project of General Surgery Department (Hepatobiliary Surgery Department); 2021-QTL-004.

Overall survival time and survival rates in patients with synchronous liver metastatic pancreatic cancer undergoing conversion therapy.

	OS	1-year survival rate	2-year survival rate	3-year survival rate	5-year survival rate
Effective group					
Surgery	49 months	100%	88%	66%	33%
Non-surgery	18 months	80%	20%	-	-
Non-effective group					
Surgery	11 months	33%	-	-	-
Non-surgery	12 months	36%	-	-	-

OS, overall survival time.

Nivolumab plus ipilimumab (N+I) in patients (pts) with pancreatic cancer (PC) with *BRCA1/2* mutation (mut): Results from the Targeted Agent and Profiling Utilization Registry (TAPUR) study.

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Background: TAPUR is a phase II basket study evaluating antitumor activity of commercially available targeted agents in pts with advanced cancers with genomic alterations. Results from a cohort of pts with PC with *BRCA1/2* mut treated with N+I are reported. **Methods:** Eligible pts had measurable disease, ECOG performance status (PS) 0-2, adequate organ function, and no standard treatment (tx) options or prior immune checkpoint inhibitor tx. PD-L1 expression testing was not required. Genomic testing was performed in CLIA-certified, CAP-accredited site-selected labs. Most genomic tests did not distinguish between germline or somatic mut. Pts received I at 3 mg/kg every 3 weeks (wks) for 4 doses with N at 1 mg/kg IV every 3 wks for 4 doses. N alone was then continued at 240 mg every 2 wks or 480 mg every 4 wks until disease progression. Primary endpoint was disease control (DC) per investigator defined as complete (CR) or partial (PR) response per RECIST v. 1.1, or stable disease of at least 16 wks duration (SD16+). CR is based on radiographic assessment. CA 19-9 levels were not collected. Simon 2-stage design tested null DC rate of 15% vs. 35% (power = 0.85; α = 0.10). If ≥2 of 10 pts in stage 1 have DC, 18 more pts are enrolled; otherwise, the cohort is closed. If ≥7 of 28 pts have DC, the null DC rate is rejected. Secondary endpoints were objective response (OR), progression-free survival (PFS), overall survival (OS), duration of response (DOR) and SD, and safety. DOR is defined as time from pt's first documented OR to progressive disease. **Results:** 32 pts with PC with *BRCA1* (n=8), *BRCA2* (n=22) or both *BRCA1/2* (n=2) were enrolled between October 2017 and March 2023. 4 pts were not evaluable for efficacy. Pts had adenocarcinoma (n=29), acinar cell carcinoma (n=2), and poorly differentiated carcinoma (n=1). Table shows demographics and outcomes. 1 CR, 3 PR and 4 SD16+ were observed for a DC rate of 31% (90% CI, 18 to 100) and OR rate of 14% (95% CI, 4 to 33). The pt with CR had a *BRCA1* mut, the pts with PR had *BRCA1* (n=1) and *BRCA2* (n=2) muts, and the pts with SD16+ had *BRCA1* (n=2) and *BRCA2* (n=2) muts. The null DC rate was rejected (p=0.04). The pt with CR remains on tx with a DOR of 311 wks as of May 2024. 15 pts received prior tx with olaparib and 1 of those pts had SD16+ on this study. 17 pts had ≥1 drug-related grade 3-4 adverse event (AE) or serious AE. All were consistent with N+I labels except generalized muscle weakness and lymphopenia. **Conclusions:** N+I shows anti-tumor activity in pts with PC with *BRCA1/2* mut and warrants further study. Clinical trial information: NCT02693535. Research Sponsor: Bristol-Myers Squibb; Astrazeneca, Bayer, Boehringer Ingelheim, Eli Lilly and Company, Genentech, Merck, Pfizer, Seagen, now a wholly owned subsidiary of Pfizer Inc., Taiho Oncology.

Demographics (N=32) and efficacy outcomes (n=28).

Median (Med) age, years (range)		**66 (37-80)**
ECOG PS, %	0-1	31 (97)
	2	1 (3)
Prior systemic regimens, %	0-2	10 (31)
	≥3	22 (69)
DC rate, % (OR and SD16+) (90% CI)		31 (18, 100)
OR rate, % (95% CI)		14 (4, 33)
Med PFS, wks (95% CI)		9 (7, 16)
Med OS, wks (95% CI)		34 (14, 46)

Effect of dose adjustments on overall survival (OS) in patients with metastatic pancreatic ductal adenocarcinoma (mPDAC) treated with NALIRIFOX: A post hoc analysis of NAPOLI 3.

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Background: NALIRIFOX is approved by the US Food & Drug Administration and European Medicines Agency for the first-line treatment of adults (no upper age limit) with mPDAC. Approval was based on the results of NAPOLI 3 (NCT04083235), in which NALIRIFOX demonstrated significant improvements in OS (primary outcome) and progression-free survival compared with nab-paclitaxel plus gemcitabine. In this exploratory post-hoc analysis of NAPOLI 3, we examined the impact of NALIRIFOX dose adjustments on OS in patients treated in North America. **Methods:** Patients (N = 770) with confirmed untreated mPDAC were randomized (1:1) to receive liposomal irinotecan 50 mg/m^2 + oxaliplatin 60 mg/m^2 + leucovorin 400 mg/m^2 + 5-fluorouracil 2400 mg/m^2 (NALIRIFOX) on days 1 and 15 of a 28-day cycle or nab-paclitaxel 125 mg/m^2 and gemcitabine 1000 mg/m^2 on days 1, 8 and 15 of a 28-day cycle. In this analysis, liposomal irinotecan dose reductions, delays, and exposure were examined descriptively in patients who received NALIRIFOX at centers in North America. OS was evaluated using Kaplan–Meier methods; no statistical tests were performed. **Results:** Overall, 120 patients were randomized to receive NALIRIFOX at centers in North America (intention-to-treat [ITT] population) and 112 received treatment (safety population). Dose adjustments (dose reductions and dose delays) occurred in a higher proportion of patients presenting with longer OS. Patients with the longest duration of liposomal irinotecan exposure and highest cumulative dose had the longest OS. **Conclusions:** This finding suggests that tolerability-guided dose modification of liposomal irinotecan does not adversely affect efficacy outcomes and suggests a path forward to further optimize the OS of patients with mPDAC receiving NALIRIFOX. Clinical trial information: NCT04083235. Research Sponsor: Ipsen.

Impact of *UGT1A1*28* polymorphism on tolerability in patients with metastatic pancreatic ductal adenocarcinoma (mPDAC) treated with NALIRIFOX in NAPOLI 3.

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Background: SN-38, the active metabolite of irinotecan, is cleared by the liver enzyme uridine diphosphate glucuronosyltransferase 1A1 (UGT1A1) and biliary excretion. *UGT1A1*28* (genotype 7/7) homozygosity is associated with reduced activity of UGT1A1 and a recommendation for dose adjustment of non-liposomal irinotecan, a component of the FOLFIRINOX regimen. In this post-hoc analysis, we analyzed the impact of *UGT1A1*28* homozygosity on the incidence and profile of treatment-emergent adverse events (TEAEs) among patients enrolled in NAPOLI 3 (NCT04083235). **Methods:** Patients (N = 770) with confirmed untreated mPDAC were randomized (1:1) to receive liposomal irinotecan 50 mg/m^2 + oxaliplatin 60 mg/m^2 + leucovorin 400 mg/m^2 + 5-fluorouracil 2400 mg/m^2 (NALIRIFOX) on days 1 and 15 of a 28-day cycle or nab-paclitaxel 125 mg/m^2 and gemcitabine 1000 mg/m^2 (Gem+NabP) on days 1, 8 and 15 of a 28-day cycle. *UGT1A1*28* status was evaluated in all patients before enrollment. All patients, regardless of *UGT1A1*28* status, received the full starting dose of liposomal irinotecan and followed the same dose reduction rules. This exploratory analysis of TEAEs by *UGT1A1*28* status was descriptive; statistical analyses were not performed. **Results:** Overall, 83 patients (11.0%) were homozygous for *UGT1A1*28*, among whom the incidence of TEAEs was similar to patients with non-homozygous status (Table). The most frequently reported TEAEs (≥ 40%) among the homozygous group (vs the overall NAPOLI 3 population) were diarrhea (59.0% vs 70.5%), nausea (56.4% vs 59.5%), vomiting (46.2% vs 39.7%) and anemia (41.0% vs 26.2%) in the NALIRIFOX arm and nausea (45.5% vs 42.7%), fatigue (45.5% vs 37.7%), diarrhea (45.5% vs 36.7%) and anemia (40.9% vs 40.4%) in the Gem+NabP arm. **Conclusions:** The incidence of TEAEs, including TEAEs leading to death, was similar between patients with homozygous and non-homozygous *UGT1A1*28* status, in both treatment arms. The profile of TEAEs was consistent with that of the overall NAPOLI 3 population. *UGT1A1*28* status did not substantially influence the safety profile or subsequent need for dose reductions of liposomal irinotecan in NAPOLI 3. Clinical trial information: NCT04083235. Research Sponsor: Ipsen.

	*UGT1A1*28* homozygous		*UGT1A1*28* non-homozygous	
TEAE, n (%)	**NALIRIFOX (n = 39)**	**Gem+NabP (n = 44)**	**NALIRIFOX (n = 328)**	**Gem+NabP (n = 331)**
Any	39 (100.0)	44 (100.0)	327 (99.7)	328 (99.1)
Related to any drug	39 (100.0)	41 (93.2)	310 (94.5)	308 (93.1)
Grade ≥ 3	31 (79.5)	34 (77.3)	288 (87.8)	288 (87.0)
Serious	24 (61.5)	24 (54.5)	176 (53.7)	168 (50.8)
Leading to treatment discontinuation	15 (38.5)	10 (22.7)	103 (31.4)	100 (30.2)
Leading to dose reduction of any drug	23 (59.0)	14 (31.8)	182 (55.5)	174 (52.6)
Leading to interruption of any drug	1 (2.6)	1 (2.3)	15 (4.6)	3 (0.9)
Leading to death	2 (5.1)	4 (9.1)	20 (6.1)	19 (5.7)

Optimal preoperative treatment strategy for conversion surgery in unresectable locally advanced pancreatic cancer: A project study by the Japanese Society of Hepato-Biliary-Pancreatic Surgery.

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Background: Conversion surgery (CS) for unresectable locally advanced pancreatic cancer (UR-LAPC) has been widely accepted when feasible. However, optimal preoperative treatment strategies remain unclear. To determine the most effective preoperative approach for UR-LAPC in the current treatment landscape, we conducted a project study in the Japanese Society of Hepato-Biliary-Pancreatic Surgery. **Methods:** We analyzed 465 UR-LAPC patients who underwent CS following chemotherapy with FOLFIRINOX (FFX), gemcitabine plus nab-paclitaxel (GnP), or modified regimens between 2015 and 2020 across 84 Japanese board-certified training institutions. Overall survival (OS) was the primary endpoint. We used the Kaplan-Meier method for estimating survival analyses, the Cox proportional hazards model for multivariate analyses, and Maximally Selected Rank Statistics to determine the optimal preoperative treatment duration. **Results:** Median OS was 43.8 months, with a 5-year survival rate of 37.2%. The optimal preoperative treatment duration was 6.1 months. Patients receiving >6 months of preoperative treatment (n=350) showed significantly better median OS and recurrence-free survival (RFS) compared to ≤6 months (n=115) (50.4 vs 29.7 months and 15.6 vs 9.1 months, respectively; both $P<0.001$). FFX-based regimens (n=116) showed better OS and RFS than GnP-based regimens (n=349) (65.0 vs 36.5 months and 19.3 vs 12.4 months, respectively; both $P<0.01$). Multivariate analysis identified preoperative treatment >6 months (HR 0.53, 95% CI 0.407-0.697, $P<0.001$) and initial FFX-based regimens (HR 0.63, 95% CI 0.464-0.843, $P=0.002$) as independent prognostic factors. Additional favorable factors included normal CA19-9 and CEA levels, PNI>45 before CS, adjuvant chemotherapy, R0 resection, and Evans grade III/IV. **Conclusions:** This large-scale retrospective study has demonstrated that preoperative treatment duration >6 months and the use of FFX-based regimens as initial treatment were associated with improved survival outcomes in UR-LAPC patients undergoing CS. These findings provide critical information for considering CS in UR-LAPC patients, especially in current clinical practice. Research Sponsor: None.

Intra-arterial gemcitabine versus intravenous gemcitabine: Pharmacokinetic sub-study of the TIGeR-PaC phase 3 clinical trial.

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Background: Localized, dual balloon catheter-mediated, intra-arterial delivery of gemcitabine (IAG) targeted to tumors/tissue can provide higher local drug potency (1). Furthermore, IAG may result in a decreased systemic drug concentration and associated side effects. In patients with locally advanced pancreatic cancer (LAPC), this approach is currently being tested in the TIGeR-PaC Phase 3 clinical trial. Herein we report the results of a 19-patient pharmacokinetics (PK) sub-study analysis within TIGeR-PaC. **Methods:** PK analyses were performed for a total of 19 participants across 6 TIGeR-PaC study sites; 11 participants received IAG with the RenovoCath dual balloon catheter at 1000 mg/m^2 over 20 minutes and 8 participants received intravenous gemcitabine (IVG) (5 at 1000 mg/m^2, 2 at 800 mg/m^2, 1 at 500 mg/m^2) over 30 minutes. IAG target treatment arteries were either the superior mesenteric artery (SMA) (n=5) or branches of the celiac axis (n=6). At T = -5, 10, 15, 20, 30, 40, 60, and 90 minutes from the onset of infusion, 2 mL of blood was collected in heparinized tubing containing 25 mcg/mL of tetrahydrouridine. Plasma from each sample was frozen and shipped to a reference lab for gemcitabine assays. Maximum plasma drug concentration (C_{max}) and the area under the drug plasma concentration curve (AUC) based on the terminal phase were compared between the two groups; for these dose-dependent parameters, only participants who received 1000 mg/m^2 gemcitabine (IAG, N=11; IVG, N=5) were included in the comparison analysis. **Results:** As shown in the table, AUC was significantly lower with IAG (4.99) compared to IVG (9.97) (P = 0.019). Peak plasma gemcitabine concentrations were also lower with IAG (12.9 mcg/mL) vs. IVG (14.6 mcg/mL) despite a 50% higher drug concentration during IAG vs. IVG (1000 mg/m^2 infused over 20 minutes vs. 30 minutes, respectively). There was no difference in plasma levels between IAG treatment sites (SMA vs. celiac axis). **Conclusions:** In this analysis, localized, dual-balloon catheter-mediated IAG resulted in decreased systemic levels of gemcitabine compared to IVG. Thus, in addition to providing increased local potency, the IAG approach may also be beneficial in decreasing gemcitabine-related systemic side effects. 1. Farsad K, et al. 2024. *JVIR* 35:1043-48 e3. Clinical trial information: NCT03257033. Research Sponsor: RenovoRx.

Effects of treatment mode on mean pharmacokinetic parameters for gemcitabine.

PK parameter	IAG Group (N=11)	IVG Group (N=5)
C_{max} (mcg/mL), Mean (SE)	12.9 (±2.4)	14.6 (±1.2)
AUC (hr · mcg/mL), Mean (SE)	4.99 (±0.96)	9.97 (±1.9)

AUC = area under the curve; C_{max} = maximum serum concentration; IAG = intra-arterial gemcitabine; IVG = intravenous gemcitabine; PK = pharmacokinetic; SE = standard error.

Influence of time to first therapy (TTFT) variable on survival outcomes of neo-adjuvant chemotherapy versus upfront surgery approach in resectable pancreatic ductal adenocarcinoma.

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Background: Evidence demonstrating overall survival (OS) benefit of neoadjuvant chemotherapy (NAC) followed by surgical resection over upfront surgical resection for resectable pancreatic ductal adenocarcinoma (PDAC) has been mixed. The time to first therapy (TTFT) variable has not been studied as a contributing factor. **Methods:** A nationwide retrospective analysis using the National Cancer Database to evaluate patients with clinical stage T1 and T2 PDACs from 2010 to 2020. Cox proportional hazards model was used to evaluate the impact of NAC followed by definitive surgery compared to upfront surgery on OS with and without TTFT. **Results:** Total of 43,196 patients were included – 9,880 patients with clinical stage T1 and 33,316 patients with T2 PDACs. There were increasing trends in the NAC approach from 2.9% in 2010 to over 25% by 2020 and decreasing trends in the upfront surgery approach from 69.35% in 2010 to 31.84% by 2020. There were significant differences in TTFT according to the treatment choice with upfront surgery group having a significantly shorter TTFT – proportion of those receiving first treatment within the 1st week was 24.32% in the upfront surgery compared to 4.22% in the NAC group. In the adjusted cox regression without the TTFT variable, there was a 25% higher rate of death in the upfront surgery compared to the NAC group (HR 1.25, 95% CI 1.19 – 1.30). When the adjusted regression was performed with addition of a TTFT interaction term, there was survival disadvantage of upfront surgery approach in patients whose TTFT occurred after 1 week, but not in those with TTFT occurring in less than 1 week (HR 1.01, 95% CI 0.86 - 1.17). **Conclusions:** Our study emphasizes the importance of incorporating TTFT variable when comparing NAC versus upfront surgery approach in PDAC. Future studies comparing NAC to upfront surgery in resectable PDAC should consider incorporating the TTFT variable. Research Sponsor: None.

Real-world evidence of nanoliposomal irinotecan and fluorouracil with folinic acid in patients with unresectable or recurrent pancreatic cancer: Final results of a multicenter observational study.

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Background: Nanoliposomal irinotecan (NAL-IRI) and fluorouracil with folinic acid (NFF) is the standard regimen after gemcitabine-based therapy for unresectable or recurrent pancreatic cancer (urPC). However, we have almost no prospective data on its efficacy and safety in the real-world, so we conducted this study to investigate them both retrospectively and prospectively (NAPOLEON-2 study). We previously reported the retrospective data in ASCO-GI 2023, and here we report the final results of the prospective part. **Methods:** We prospectively collected data of urPC patients treated with NFF who had received at least one previous chemotherapy line in 17 hospitals in Japan from June 2021 to October 2023. The primary endpoint was overall survival (OS). Secondary endpoints were overall response rate (ORR), disease control rate (DCR), progression-free survival (PFS), dose intensity (DI) and adverse events (AEs). In addition, OS and PFS among the therapeutic lines of NFF were also analyzed. **Results:** A total of 150 urPC patients were prospectively enrolled. The median follow-up period was 7.2 months (95% confidence interval [CI], 6.3–8.8); median age, 72 years (range, 45–85), with 81 female patients (54%). The Eastern Cooperative Oncology Group performance status was 0/1/2/3 in 46/93/10/1 patients, respectively. Sixteen patients (11%) had locally advanced disease; 134 (89%) had metastatic disease; 77 (51%) had liver metastasis; and 47 (31%) had peritoneal metastasis. All patients previously received gemcitabine-based therapy. NFF was administered as 2nd/3rd/4th-or-later-line therapy to 87/53/10 patients, respectively. The median OS was 7.8 months (95% CI, 6.6–9.2); median PFS, 3.7 months (95% CI, 2.8–4.9); overall response rate, 11%; and disease control rate, 56%. The relative dose intensity was 72.7% with NAL-IRI and 79.4% with fluorouracil. The initial dose of NAL-IRI was reduced in 84 patients (56%), mainly owing to decreased organ function (17%), followed by age (12%), worsened performance status (10%), or UGT1A1 examination status (5%). Dosage reduction of NAL-IRI during treatment (independent of the initial dose reduction) was performed in 74 patients (49%), mainly owing to neutropenia (22%) and anorexia (11%). Frequent Grade 3/4 adverse events were neutropenia (27%), anorexia (19%), and leukopenia (16%). Grade 5 peritoneal infection was observed in only one patient. The median OS and PFS for NFF in the 2nd-line group, compared with the 3rd-or-later-line group, were 7.4 vs 7.8 months (hazard ratio [HR], 0.97; 95% CI, 0.68–1.39; p=0.88) and 3.3 vs 4.2 months (HR, 1.04; 95% CI, 0.74–1.45; p=0.84), respectively. **Conclusions:** NFF had appropriate efficacy and manageable toxicity profiles, consistent with our previous report. NFF can be a candidate for 2nd-or-later-line regimens in the real-world. Clinical trial information: UMIN000043939. Research Sponsor: None.

Safety, efficacy, and on-treatment circulating tumor DNA (ctDNA) changes from a phase 1 study of RMC-6236, a RAS(ON) multi-selective, tri-complex inhibitor, in patients with RAS mutant pancreatic ductal adenocarcinoma (PDAC).

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Background: PDAC is the third leading cause of cancer mortality, with limited treatment options. Even with multi-agent chemotherapy, patients with second-line (2L) PDAC have a median progression-free survival (PFS) of ≈2–3.5 months and median overall survival (OS) of ≈6–7 months. RAS mutations occur in >90% of patients with PDAC, mainly KRAS G12X (G12X = non-synonymous mutations in KRAS codon 12 [G12]). RMC-6236 is an oral, RAS(ON), multi-selective, tri-complex inhibitor of the active GTP-bound state of both mutant and wild-type RAS. In a Phase 1 study, RMC-6236 demonstrated efficacy and manageable safety in patients with PDAC harboring KRAS G12X or other RAS mutations (NCT05379985). Data on ctDNA reduction with targeted therapy in PDAC are sparse. We report safety, updated efficacy, and exploratory analyses of early ctDNA reduction with clinically active doses of RMC-6236 in patients with RAS mutant PDAC. **Methods:** Escalating doses of RMC-6236 were administered orally to patients with previously treated RAS mutant PDAC. Additional patients were enrolled for dose optimization and expansion. Patients receiving clinically active doses (160–300 mg QD) of RMC-6236 ≥14 weeks before the July 23, 2024 data cutoff were included. Plasma samples were collected for ctDNA analysis of change in RAS mutant variant allele fraction at baseline (BL; cycle 1, day 1 [C1D1]), and on treatment (C2D1 or C3D1). **Results:** As of July 23, 2024, 127 patients with RAS mutant PDAC had received RMC-6236 160–300 mg QD. The most common (≥10% of patients) any-grade treatment-related adverse events were rash (91%), diarrhea (48%), nausea (43%), vomiting (31%), stomatitis (31%), fatigue (20%), paronychia (13%), mucosal inflammation (13%), decreased appetite (11%), and peripheral edema (10%). The Table shows objective response rate (ORR), PFS, and OS with 2L RMC-6236. Paired plasma samples were tested in 106/127 patients. Of these, 68 patients with RAS mutant allele ctDNA at BL were evaluable for ctDNA response (Table). **Conclusions:** RMC-6236 showed a manageable safety profile and encouraging efficacy in patients with previously treated RAS mutant PDAC, and early and deep reductions in RAS mutant ctDNA. RASolute 302, a global, randomized, Phase 3 trial evaluating RMC-6236 as 2L treatment vs chemotherapy in patients with metastatic PDAC, is ongoing. Clinical trial information: NCT05379985. Research Sponsor: None.

	KRAS G12X	RAS mutant
Efficacy with 2L RMC-6236		
	(n=42)	**(n=57)**
ORR, % (95% CI)	29 (16–45)	25 (14–38)
[confirmed + pending confirmation]		
Median PFS, months (95% CI)	8.5 (5.3–11.7)	7.6 (5.9–11.1)
Median OS, months (95% CI)	14.5 (8.8–not evaluable)	14.5 (8.8–not evaluable)
ctDNA Response with 2L+ RMC-6236		
	(n=56)	**(n=68)**
>50% decrease from BL, n (%)	53 (95)	63 (93)
100% decrease from BL, n (%)	28 (50)	32 (47)

Evaluating the relative treatment efficacy of CD40 agonist mitazalimab in combination with mFOLFIRINOX in patients with metastatic pancreatic ductal adenocarcinoma (mPDAC) using unanchored indirect treatment comparisons (ITCs).

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Background: Mitazalimab, a human CD40 agonistic IgG1antibody has demonstrated encouraging results in combination with mFOLFIRINOX (mFFX) in the phase 2 single arm trial OPTIMIZE-1 in patients with mPDAC. We conducted unanchored ITCs to assess the relative efficacy of mitazalimab + mFFX vs FOLFIRINOX (FFX) / mFFX and NALIRIFOX (NFX), the most effective therapies to date. **Methods:** Using data from OPTIMIZE-1 and published data from five RCTs identified via a literature review, matching-adjusted indirect comparisons (MAIC) and simulated treatment comparisons (STC), two commonly used ITC methods for health technology assessment (1) were conducted to balance effect modifiers and prognostic factors between trials. Analysis outcomes were overall survival (OS), objective response rate (ORR), and progression free survival (PFS). Reconstructed survival data from FFX, mFFX, and NFX studies were pooled together for the comparison with FFX/mFFX (Pool #1) and FFX/mFFX/NFX (Pool #2) pooled studies. **Results:** In the STC analysis mitazalimab + mFFX showed a significantly higher OS versus Pool #1 [hazard ratio (HR): 0.64; 95% CI: 0.46 – 0.87] and #2 [HR: 0.65; 95% CI: 0.47 – 0.87], corresponding to an improvement in median OS by 3.4 and 3.3 months respectively. The OS HRs of study-level pairwise comparisons ranged from 0.57 (95% CI: 0.39 – 0.80) to 0.85 (95% CI: 0.53 – 1.39) with only one non-significant comparison. MAICs showed similar trends, although the comparison with Pool #1 was close to statistical significance (HR: 0.68; 95% CI: 0.45 – 1.02). Results of the PFS and ORR comparisons are shown in the table below. **Conclusions:** Literature based ITCs show a significantly better efficacy of mitazalimab + mFFX versus the current available therapies in patients with mPDAC. These trends will have to be confirmed in a randomized Phase 3 study. 1. Phillippo DM et al, 2019. Clinical trial information: NCT02829099. Research Sponsor: Alligator Bioscience.

ITC results – HR (for OS and PFS) and odds ratios (for ORR) and their 95% CIs.

	STC			MAIC		
Comparator study	**OS**	**PFS**	**ORR**	**OS**	**PFS**	**ORR**
PRODIGE 4 (FFX)	**0.57 (0.39 – 0.80)**	**0.63 (0.44 – 0.88)**	1.62 (0.87 – 2.99)	**0.60 (0.40 – 0.89)**	**0.56 (0.37 – 0.82)**	1.55 (0.84 – 2.87)
PANOPTIMOX (FFX)	**0.58 (0.39 – 0.84)**	**0.68 (0.47 – 0.98)**	1.17 (0.59 – 2.30)	**0.60 (0.38 – 0.94)**	0.67 (0.44 – 1.02)	1.04 (0.52 – 2.05)
AVENGER500 (FFX)	**0.71 (0.50 – 0.98)**	1.15 (0.82 – 1.60)	1.35 (0.75 – 2.42)	0.69 (0.46 – 1.03)	1.11 (0.76 – 1.62)	1.30 (0.72 – 2.34)
SWOGS1313 (mFFX)	0.85 (0.53 – 1.39)	0.89 (0.57 – 1.36)	0.90 (0.43 – 1.89)	0.91 (0.55 – 1.50)	0.86 (0.55 – 1.33)	0.88 (0.42 – 1.86)
Pool #1	**0.64 (0.46 – 0.87)**	0.81 (0.59 – 1.10)	1.33 (0.77 – 2.32)	0.68 (0.45 – 1.02)	0.81 (0.56 – 1.18)	1.20 (0.69 – 2.09)
NAPOLI-3 (NFX)	**0.68 (0.49 – 0.93)**	0.85 (0.61 – 1.17)	0.97 (0.55 – 1.71)	**0.67 (0.46 - 0.97)**	0.83 (0.58 – 1.20)	0.92 (0.52 – 1.62)
Pool #2	**0.65 (0.47 – 0.87)**	0.82 (0.60 – 1.11)	1.17 (0.68 – 2.02)	**0.68 (0.47 – 0.99)**	0.84 (0.59 – 1.19)	1.11 (0.64 – 1.91)

Preliminary safety, antitumor activity, and circulating tumor DNA (ctDNA) changes with RMC-9805, an oral, RAS(ON) G12D-selective tri-complex inhibitor in patients with KRAS G12D pancreatic ductal adenocarcinoma (PDAC) from a phase 1 study in advanced solid tumors.

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Background: RMC-9805 is a potent, oral, RAS(ON) G12D-selective, covalent, tri-complex inhibitor targeting the active, GTP-bound state of oncogenic RAS G12D isoforms. Despite low response rates to current standard of care (SOC), no RAS-targeted therapy is approved for G12D PDAC. **Methods:** In this Phase 1 study (NCT06040541), patients with previously treated, advanced KRAS G12D solid tumors received escalating RMC-9805 doses (150-1200 mg once daily [QD] or 300-600 mg twice daily [BID]). Treatment cycles were every 3 wks. Antitumor activity was assessed every 6 wks the first 24 wks then every 9 wks. Additional patients were enrolled at RMC-9805 doses that cleared the dose-limiting toxicity (DLT) evaluation to further characterize pharmacokinetics, safety, antitumor activity, and biomarkers. Plasma for ctDNA analysis was collected at baseline (BL; cycle 1, day 1 [C1D1] or screening), and on treatment (OT; C2D1 or C3D1). **Results:** As of Sept 2, 2024, 179 patients with KRAS G12D solid tumors (most having PDAC; n=104 [58%]) received 5 escalating dose levels of RMC-9805 (150-1200 mg daily). No DLTs or Grade 4 or 5 treatment-related adverse events (TRAEs) were reported, and the maximum tolerated dose (MTD) was not reached. Among patients who received a candidate recommended phase 2 dose (RP2D) of 1200 mg daily (1200 mg QD [n=60] or 600 mg BID [n=39]), the most common (≥10% of patients) TRAEs were nausea (27%), diarrhea (20%), vomiting (15%), and rash (10%), all of which were Grade 1 or 2 severity. One Grade 3 TRAE (ALT elevation) was observed in a patient with PDAC and a history of ALT elevation, biliary stenting, and liver metastasis. No patients treated with 1200 mg daily discontinued treatment due to TRAEs and 4% of patients had dose reductions due to TRAEs. In patients with PDAC receiving RMC-9805 daily at 1200 mg QD (n=20) or 600 mg BID (n=20) who enrolled at least 14 wks prior to data cutoff, the objective response rate (confirmed response or pending confirmation) was 30%, and the disease control rate was 80%. Of 28 patients who had KRAS G12D mutations detected in ctDNA at BL and were evaluable for OT assessment, 86% had an OT decrease >50% from BL of detectable KRAS G12D mutations, and 39% had a 100% OT clearance. Updated data will be presented. **Conclusions:** Oral RMC-9805 showed encouraging initial antitumor activity with early and deep reductions in KRAS G12D ctDNA in patients with KRAS G12D PDAC. Tolerability was favorable relative to SOC chemotherapy for PDAC and manageable. This overall safety profile and antitumor activity support continued evaluation as monotherapy in patients with KRAS G12D PDAC, and in combination with chemotherapy and targeted therapies, including the RAS(ON) multi-selective inhibitor RMC-6236. Clinical trial information: NCT06040541. Research Sponsor: Revolution Medicines.

MEK inhibitor-based therapy in metastatic pancreatic adenocarcinoma with *KRAS* G12R alteration.

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Background: Pancreatic ductal adenocarcinoma (PDAC) is largely driven by oncogenic alterations in the *KRAS*, *TP53*, *CDKN2A/B*, and *SMAD4* genes. *KRAS* G12R accounts for nearly 20% of *KRAS* alterations in PDAC, is biologically unique, and offers the potential for improved response to MEK inhibitor (MEK-inh)-based therapy. As part of the MCW-Master-PREDICT (NCT05802069) observational study, we evaluated the efficacy of MEK-inh-based therapy matched to the unique molecular alterations in the tumor of each patient as recommended by the molecular tumor board. **Methods:** From March 2022 to December 2023, 8 patients with metastatic PDAC were treated with MEK-inh-based therapy. Molecular profiling was performed by Tempus (648 genes) in 7 patients, and FoundationOne (324 genes) in 1 patient. Description of molecular alterations, matched therapies, and efficacy of treatment are provided. **Results:** The median age of patients starting MEK inhibitor-based therapy was 69 years, with 62% of them being female. The molecular characteristics and matched therapies are summarized (Table). MEK-inh-based therapy was administered to 3 (37.5%) patients as a first or second-line treatment, and to 5 (62.5%) patients as a third to fifth-line treatment for metastatic PDAC. The overall survival and progression free survival (PFS) on MEK-inh-based therapy was 8.4 and 4.9 months, respectively. Four patients had PFS beyond 4 months (4.9, 5.5, 5.6 and 6.9 months). **Conclusions:** Individualized MEK-inh-based therapy demonstrated efficacy in late-line treatment of metastatic PDAC with *KRAS* G12R alterations. Using this approach earlier in the treatment course and in combination with RAS inhibitors or cytotoxic systemic therapies may further enhance outcomes. Research Sponsor: None.

Tumor alterations and matched therapies.

Gene altered	Number of patients (%)	Number of patients matched (%)	Drug family matched
KRAS G12R	8 (100)	8/8 (100)	MEK inhibitor
TP53	7 (87.5)	4/7 (57)	VEGF/VEGR inhibitor*
CDKN2A	4 (50)	4/4 (100)	CDK4/6 inhibitor**
SMAD4	2 (25)	2/2 (100)	MEK inhibitor + (EGFR or Pan-HER inhibitor)***

*PMID: 27466356; **PMID: 33472910; ***PMID: 36127339, PMID: 24625091.

MEK inhibitor-based therapy in metastatic pancreatic adenocarcinoma with *KRAS* G12D or *KRAS* G12V alteration.

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Background: Up to 95% of pancreatic ductal adenocarcinomas (PDAC) are driven by oncogenic alterations in the *KRAS* gene, with *KRAS* G12D (40%) and *KRAS* G12V (30%) mutations being the most common. However, in most cases, *KRAS* alterations are accompanied by alterations in other tumor suppressors such as *TP53*, *CDKN2A/B*, and *SMAD4*. Therefore, targeting *KRAS* alone may not be sufficient for effective treatment of PDAC . As part of the MCW-Master-PREDICT (NCT05802069) observational study, we evaluated the effectiveness of MEK inhibitor (MEK-inh)-based therapy, tailored to the unique molecular alterations of each patient's tumor, in treating metastatic PDAC (mPDAC) with *KRAS* G12D or *KRAS* G12V alterations. **Methods:** From April 2022 to December 2023, 19 patients (*KRAS* G12D=10 (53%), *KRAS* G12V=9 (47%)) were treated with MEK-inh-based therapy. Comprehensive genomic profiling was performed using Tempus xT (648 genes) in 15 patients (79%), FoundationOne (324 genes) in 4 patients (21%), and STRATA (429 genes) in 1 patient. The genomic characteristics, matched therapies, and treatment outcomes are provided (Table). **Results:** Of the 19 patients, 11 (58%) were female. The median age at the initiation of MEK-inh-based therapy was 67 years. Six (32%) patients received MEK-inh-based therapy as first or second-line treatment, while 13 (68%) patients were treated in the third to fifth-line. Among the 19 patients, the median overall survival (mOS) and median progression-free survival (mPFS) from the initiation of MEK-inh-based therapy were 5 and 2 months, respectively. The mOS from the time of MEK-inh-based therapy was 5.1 months for *KRAS* G12D, and 4.7 months for *KRAS* G12V (P=0.87). The mPFS from the time of MEK-inh-based therapy was 2.3 months for *KRAS* G12D and 1.4 months for *KRAS* G12V (P=0.12). Among all 19 patients, 4 (21%) patients had mPFS above 4 months (*KRAS* G12V=1, *KRAS* G12D=3; 4.1, 4.3, 4.7, 9.9 months). **Conclusions:** The efficacy of MEK-inh-based therapy was limited in the late-line treatment of mPDAC for patients with *KRAS* G12D and *KRAS* G12V mutations. The potential role of MEK inhibitors in earlier lines of therapy, and their combination with *KRAS* inhibitors, requires further prospective evaluation. Research Sponsor: None.

Tumor alterations and matched therapies.

Altered gene	KRAS G12V (N=9)	KRAS G12D (N=10)	Patients matched (N=19)	Type of matched therapy
KRAS G12D/V	9 (100%)	10 (100%)	19 (100%)	MEK inhibitor
TP53	7 (78%)	8 (80%)	8 (42%)	VEGF/VEGR inhibitor*
CDKN2A/B	8 (89%)	5 (50%)	9 (47%)	CDK4/6 inhibitor**
SMAD4	3 (33%)	4 (40%)	4 (21%)	MEK inhibitor + (EGFR or pan-HER inhibitor) ***

*PMID: 27466356; **PMID: 33472910; ***PMID: 36127339, PMID: 24625091.
N= number of patients.

Phase II study of niraparib and dostarlimab for the treatment of germline or somatic homologous recombination repair mutated (HRD) metastatic pancreatic cancer.

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Background: PARP inhibition (PARPi) is a standard maintenance therapy for patients (pts) with germline *BRCA1/2* mutations in pancreatic cancer. Combination of PARPi therapy and immunotherapy has potential to increase efficacy and may offer benefit beyond germline *BRCA1/2*. **Methods:** We performed a multisite single-arm phase 2 study using the highly potent PARPi niraparib with anti-PD1 drug dostarlimab in pts with germline or somatic HRD mutations in *BRCA1/2, PALB2, RAD51C/D,* or *BARD1*. Pts were required to have metastasis, progressed on ≥1 regimen, and prior platinum-exposure unless refused/intolerant. Pts were treated with niraparib 200 mg orally once daily continuously. Dostarlimab 500 mg was administered IV every 3 weeks for 4 cycles, then 1000 mg IV every 6 weeks. The primary endpoint was disease control rate (DCR) at 12 weeks (DCR12) using iRECIST criteria. At least 8 pts with DCR12 in the first 19 eligible pts (41%) would be considered promising for further trials. **Results:** 22 pts were enrolled from Dec 2020 to Oct 2023. Two pts withdrew prior to receiving therapy, and 2 were later found to be ineligible, leaving 18 eligible pts for final analyses. Median age was 63.0 yrs, 39% male, 94% non-Hispanic white, 1 black (6%). 17 pts were platinum exposed, 7 were platinum-refractory, and 5 had previously progressed on PARPi maintenance. Mutation distribution was: *BRCA2* 13/18 (72%), *BRCA1* 4/18 (22%), *BARD1* 2/18 (11%). One pt had both *BRCA1* (somatic) and *BRCA2* (germline) mutations. Germline mutations were identified in 14/18 (78%). DCR12 was achieved in 5/18 (27.8%), so the primary endpoint (>41%) was not met. Pts received a median 2 cycles (range 1-8) of niraparib + dostarlimab. No clinical factors were significantly associated with improved DCR12 (all $p > 0.05$) in subgroup analyses, but we did observe the following statistically nonsignificant associations. Pts who were platinum-refractory had a lower DCR12 compared to those not (14.3% vs. 36.4%). Counterintuitively, PARPi exposed/refractory pts had a higher DCR12 than non-exposed/non-refractory (50 vs 17% exposed/nonexposed and 40% vs 23% refractory/nonrefractory). Pts with germline mutations had a higher DCR12 compared to those without (38.5% vs. 0%). Non-*BRCA2* mutations had a higher DCR12 than those with *BRCA2* mutations (50% vs.17%), with highest DCR12 rates among *BRCA1* patients (67%). Tolerability was in line with similar combinations, with 60% non-hematologic Gr3 or higher, including fatigue (15%), AST increase (10%), nausea/vomiting (10%), sepsis (10%). Two grade 5 events were not attributed to treatment. **Conclusions:** PARPi plus anti-PD1 checkpoint inhibition was not sufficiently active as later line therapy in metastatic pancreatic cancer for the majority of patients with HRD mutations. Additional molecular analyses to elucidate predictive markers of sensitivity/resistance are ongoing. Clinical trial information: NCT04493060. Research Sponsor: GSK.

AGITG ASCEND: Randomised, double-blind phase II study of certepetide or placebo added to gemcitabine plus nab-paclitaxel in patients with untreated metastatic pancreatic ductal adenocarcinoma: Initial results.

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Background: Gemcitabine (GEM) plus nab-paclitaxel (NAB-PAC) is a standard first-line chemotherapy regimen for advanced/metastatic pancreatic ductal adenocarcinoma (PDAC), with median Progression Free Survival (mPFS) and Overall Survival (mOS) of 5.5 & 8.7 in the MPACT trial and 5.6 & 9.2 months (mo) in the NAPOLI 3 trial, respectively. Certepetide (formerly LSTA1 or CEND-1) is a novel cyclic peptide that improves targeted penetration of co-administered drugs into tumor and stroma, leading to potentially increased anti-neoplastic activity. ASCEND is a randomized phase II trial designed to investigate the impact of adding certepetide to GEM/NAB-PAC. (NCT05042128). **Methods:** Eligible participants (pts) with histologically confirmed advanced PDAC and ECOG 0-1 were randomized 2:1 to receive GEM/NAB-PAC plus certepetide (3.2 mg/kg) or placebo (PLA) on days 1, 8, & 15 of a 28-day cycle. Stratification was by age (<65/≥65 years), ECOG (0/1), presence of liver metastasis (Y/N) and trial site. The primary objective was to determine the effect of adding certepetide to GEM/NAB-PAC on PFS. Objective tumor response rate (OTRR), safety and OS were secondary objectives. This non-comparative phase II design targeted a PFS increase of 17% at 6 mo from 47% to 63%. A sample size of 65 patients in the certepetide arm was expected to have 80% power, with 95% confidence to exclude an uninteresting 6-mo PFS rate of 47%. **Results:** 95 pts (66 certepetide, 29 PLA) were enrolled between May 2022 to December 2023. 6-mo PFS in the certepetide and PLA groups was 49.0% (95% CI 36.4%, 60.5%) and 40.8 (95% CI 22.6%, 58.3%) respectively, with mPFS of 5.5 mo in both groups. mOS was 12.42 mo for the certepetide and 9.72 mo for the PLA. An OTRR of 38.3% (certepetide) and 26.9% (PLA) was observed. Of note, 4 complete responses were observed in the certepetide group vs. 0 in PLA group. In subjects with ECOG 0, 6 mo PFS was 68.1% (95% CI 48.7%, 81.4%) in the certepetide compared to 36.4% (95% CI 11.2%, 62.7%) in the PLA. Grade ≥3 toxicities were similar in both groups at 16% (certepetide) and 15% (PLA). **Conclusions:** The addition of certepetide is safe and despite showing no improvement in 6-mo PFS, a possible signal of benefit in OS and OTRR, including the 4 complete responses was observed, warranting further investigation. A further cohort of the ASCEND study evaluating the addition of a second dose of certepetide is ongoing. Clinical trial information: NCT05042128. Research Sponsor: Lisata Therapeutics, Inc.

Preliminary result of a phase Ib study: Efficacy and safety of FG-M108 plus gemcitabine/nab-paclitaxel as first-line (1L) treatment in patients with Claudin18.2-positive locally advanced unresectable or metastatic (LA/m) pancreatic cancer.

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Background: FG-M108, an ADCC-enhanced anti-CLDN18.2 monoclonal antibody, showed significant efficacy in 1L treatment of gastric cancer. Herein, we report the safety and efficacy results of FG-M108 in 1L treatment of pancreatic cancer (cohort C2&D2). **Methods:** In this open-label, multicenter phase I/II study patients received FG-M108 (cohort C2: 300mg/m^2 or cohort D2: 600mg/m^2 Q3W) plus gemcitabine (1000mg/m^2, d1, d8, Q3W) and nab-paclitaxel (125mg/m^2, d1, d8, Q3W). Eligible patients were those with CLDN18.2 positive (IHC 1+/2+/3+≥10%) previously untreated LA/m pancreatic cancer. The primary endpoint were the incidence of adverse events (AEs) and preliminary clinical efficacy (ORR, DCR, DOR, PFS, and OS). **Results:** As of Sep 5 2024, 50 patients were enrolled and received FG-M108+gemcitabine/nab-paclitaxel treatment (39 patients in cohort C2, 11 patients in cohort D2). The median age was 61 (range 30-72). 47 (94%) patients were with CLDN18.2 moderate-high expression (IHC 2+/3+≥40%). Out of 50 patients, 44 patients had at least one tumor assessment after baseline and included in the efficacy analysis set. The results of ORR and DCR in cohort C2 were 32.4% and 100.0%, while that in cohort D2 were 30.0% and 90.0%, respectively. In cohort C2, the median PFS and median DOR were 6.8 months and 9.8 months, respectively. In the subgroup patients with CLDN18.2 moderate-high expression, the median PFS and median DOR were 7.6 months and 9.8 months, respectively. The median OS has not been reached. FG-M108 related AEs were observed in all patients, with 39 patients (78%) having Grade 3/4 AEs and no FG-M108 related fatal AE was observed. The most common FG-M108 related AEs in cohort C2 & D2 were anemia (56.2% vs 63.6%), nausea (56.4% vs 36.4%), vomiting (48.7% vs 45.5%), and hypoalbuminemia (46.2% vs 54.5%). **Conclusions:** The combined therapy of FG-M108 plus chemotherapy as 1L treatment for patients with CLDN18.2 positive pancreatic cancer was well tolerated with encouraging survival especially in patients with CLDN18.2 moderate-high expression, deserving further investigation. Clinical trial information: NCT04894825. Research Sponsor: None.

GOBLET study: Results of the safety run-in for first-line metastatic pancreatic ductal adenocarcinoma (PDAC) patients treated with pelareorep + modified FOLFIRINOX +/- atezolizumab.

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Background: Immunotherapies are effective in only the small subset of metastatic PDAC patients with MSI-H or dMMR tumors. Pelareorep (pela) is a non-genetically modified, intravenously administered reovirus that selectively infects cancer cells. It stimulates the expansion of pre-existing tumor-infiltrating lymphocyte (TIL) clones, and it makes tumors visible to the immune system by upregulating interferon-induced gene expression including PD-L1, CXCL9, CXCL10, and CXCL11. PD-L1 upregulation also provides a basis for potential synergy between pela and immune checkpoint inhibitors. Pela combined with gemcitabine/nab-paclitaxel/atezolizumab showed promising tumor responses in mPDAC. Here, we are assessing the safety and preliminary efficacy of pela + mFOLFIRINOX +/- atezolizumab in a new cohort in the ongoing GOBLET study. **Methods:** GOBLET is a phase 1/2, open-label, Simon two-stage study in patients with advanced or metastatic GI cancers. In this new GOBLET cohort (Cohort 5), patients with newly diagnosed mPDAC are randomized to receive either pela + mFOLFIRINOX or pela + mFOLFIRINOX + atezolizumab. The primary endpoints are safety and objective response rate (ORR). In Stage 1, 15 evaluable patients will be enrolled in each arm. Both arms include a 3-6 patient safety run-in that must be successfully completed prior to opening the study to full enrollment. If pre-specified ORR success criteria are met, one or both arms may be advanced to Stage 2 during which 17 additional evaluable patients/per arm will be enrolled. **Results:** The 3-patient safety run-in for both arms of Cohorts 5 (6 patients total) have been enrolled, and all patients have completed the required 1-month evaluation period. Cohort 5 safety data have been reviewed by the independent Data Safety Monitoring Board (DSMB). The DSMB identified no safety signals attributable to the combination of pela and mFOLFIRINOX +/- atezolizumab and recommended that enrollment into these arms of Cohort 5 continue without modification. Reported adverse events are consistent with the known toxicities of the components of the treatment regimen. Tumor response results are pending. **Conclusions:** The results of the GOBLET Cohort 5 safety run-in indicate that pela can be given safely in combination with mFOLFIRINOX +/- atezolizumab to newly diagnosed mPDAC patients. Safety and efficacy of these combination therapies will continue to be monitored (Eudra-CT: 2020-003996-16). Clinical trial information: 2020-003996-16. Research Sponsor: None.

Clinical outcomes of adjuvant chemoradiotherapy in pancreatic adenocarcinoma with R1 resection.

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Background: The benefit of chemoradiotherapy (CRT) following adjuvant chemotherapy in pancreatic cancer is unclear, though recent findings suggest some disease-free survival benefit over chemotherapy alone. We aimed to assess overall (OS) and progression-free survival (PFS) and patterns of disease recurrence following adjuvant chemoradiotherapy (CRT) in pancreatic adenocarcinoma patients with R1 resection (defined as <1mm). **Methods:** We retrospectively collected data of patients with pancreatic adenocarcinoma who underwent adjuvant CRT (50.4-54Gy) following surgery with R1 resection between 2007-2023 in a single centre in the UK. Clinical data was collected from electronic medical records and information on radiotherapy was collected from treatment planning systems (TPS). Toxicity during and after CRT was graded with Common Terminology Criteria for Adverse Events (CTCAE). OS and PFS were measured with Kaplan-Meier curves from the time of surgery to death or disease progression or last follow up. In-field and locoregional recurrence were determined from restaging scans compared against radiotherapy field in TPS. **Results:** Among 33 patients included, 27 (81.8%) died within median follow-up 39 (14.8-111.4) months. All patients received adjuvant chemotherapy prior to CRT. Median OS was 42 (21.6-52.7) months and median PFS was 18.2 (15.5-28.4) months. In all but one patient, the R1 involved the posterior resection margin. Most patients (84.8%) had documented disease progression. In the majority the site of first progression was distant metastatic disease (51.5%), whilst six (18.2%) had local recurrence within the radiotherapy field. Four (12.1%) relapsed locoregionally. No statistically significance difference in OS or PFS was seen with presence of perineural or lymphovascular invasion, though analysis was limited by low number of patients. The majority of patients (72%) were node positive on pathology. Two (6.1%) patients had grade 3-4 toxicity during CRT (neutropenic and non-neutropenic sepsis). **Conclusions:** Our study supported acceptable safety following CRT as an adjuvant treatment in pancreatic cancer patients with R1 resection on surgery. Despite R1 resection, we observed survival outcomes with adjuvant chemotherapy and CRT somewhat above previously reported data. However the benefit of CRT after adjuvant chemotherapy remains unclear. Further evaluation is warranted to clarify which subgroups of patients benefit.
Research Sponsor: None.

Clinical outcomes and molecular characteristics of patients with locoregional ampullary carcinoma.

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Background: Ampullary carcinoma (AC) is a rare malignancy arising from the ampulla of Vater which has a more favorable prognosis compared to other pancreatic malignancies. Current guidelines favor surgical resection followed by adjuvant therapy for average risk patients (pts), however the type of regimen as well as other perioperative treatment modalities are still being explored. The objective of this retrospective study aims to provide insight into the management of locoregional disease. **Methods:** Pathology records from Mayo Clinic (AZ, FL, MN) denoting AC between 2010 to 2024 were searched using Mayo Data Explorer and selected for retrospective review. Pt demographics, treatment courses, and next generation sequencing (NGS) data were collected. Statistical analysis was conducted using SAS version 9.04. **Results:** A total of 137 pts (62% male, n = 85) were identified with median age 66 years old. Histologic subtypes were 54% pancreatobiliary (44/81), 40% intestinal (32/81), 6% mixed (5/81), with 56 pts unknown. Stages at diagnosis were 81% resectable (111/137), 11% locally advanced (15/137), and 8% metastatic (11/137). Nearly all (93%, n = 103/111) pts with localized disease had resection with a 97% R0 resection rate (100/103). Majority (68%, n = 75/111) of resectable pts had perioperative chemotherapy with 5-FU or gemcitabine-based regimens (24%, n = 18/75 neoadjuvant vs. 76%, n = 57/75 adjuvant). Perioperative chemoradiation was given in 17% (19/111) of pts. At median follow up of 30.5 months, recurrence rates were 40% (41/103), primarily to the liver (18/41). Median recurrence free survival (RFS) was 29.3 months (95% CI 24.7 – NE). Median overall survival (OS) has not been reached. The 5-year RFS and OS rates were 0.37 (95% CI 0.25 - 0.55) and 0.69 (95% CI 0.58 - 0.82), respectively. Univariate analysis showed no difference in OS with the addition of neoadjuvant or adjuvant chemotherapy or chemoradiation. Somatic NGS testing showed pathogenic mutations in 90% of pts (63/70): 57% KRAS (36/63; 44% G12D, 19% G12V, 8% G12C, 8% G12D, 19% others); 13% homologous recombination (HR) (8/63); 6% *ERRB2*/Her2 amplification (4/63), and 5% mismatch repair (MMR) (3/63). **Conclusions:** AC has a favorable prognosis in resectable pts. However, high recurrence rates indicate the need for better systemic therapies and improved selection of pts who would benefit from these treatments. Future research should focus on refining perioperative strategies to enhance outcomes and reduce recurrence. Research Sponsor: None.

733 Poster Session

Cost-effectiveness of NALIRIFOX compared to other first-line treatments for metastatic pancreatic cancer.

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Background: NALIRIFOX was recently approved for first-line treatment of metastatic pancreatic cancer, where combination chemotherapy remains the main treatment modality. We conducted a cost-effectiveness analysis of NALIRIFOX compared to both FOLFIRINOX and gemcitabine/protein-bound paclitaxel (GP). **Methods:** A partitioned survival model with three health states (progression-free, post-progression and death) was constructed to model the disease course. Using published data from NAPOLI-3, parametric survival curves were generated for NALIRIFOX, from which survival curves for FOLFIRINOX and GP were extrapolated using pooled efficacy data reported by a network meta-analysis of clinical trials. Model inputs were retrieved from the literature or other publicly available resources. Specifically, drug acquisition costs were from average wholesale prices in the Red Book while administration, disease monitoring and supportive care costs were from the CMS Physician Fee Schedule. Utility values to estimate quality-adjusted life years (QALY) were from the CALGB80303 trial. Adverse effects that were considered included anemia, neutropenia, febrile neutropenia, thrombocytopenia, diarrhea, peripheral neuropathy, nausea and vomiting, and fatigue. Adverse effect rates were based on data pooled from clinical trials while costs and disutility were obtained from a secondary analysis of claims data and a clinician survey, respectively. The time horizon was 5 years and utilized a third-party payer perspective. Both costs ($USD 2024) and outcomes were discounted at 3% annually. A probabilistic sensitivity analysis was conducted by varying model parameters based on 95% confidence intervals if known or 10% in both directions otherwise. **Results:** The overall cost associated with NALIRIFOX was $129,314, followed by GP ($104,593) and FOLFIRINOX ($41,528). Among the 3 regimens, NALIRIFOX incurred the highest treatment-related cost ($105,533) but the lowest adverse effects management cost ($16,541). On average, patients on NALIRIFOX gained a total of 1.14 life years (LY) or after adjusting for utility, 0.86 QALY. FOLFIRINOX led to 1.09 LY or 0.82 QALY while GP resulted in 0.99 LY or 0.75 QALY. Due to its higher cost and lower survival benefit, GP was dominated by FOLFIRINOX. Compared to FOLFIRINOX, the incremental cost-effectiveness ratio (ICER) of NALIRIFOX was $1,952,062/QALY. In our sensitivity analysis, the probability of NALIRIFOX being cost-effective was 0% at all conventional ICER thresholds up to $200,000/QALY. **Conclusions:** Due to the high drug acquisition cost, the ICER of NALIRIFOX far exceeds currently acceptable thresholds for economic value and NALIRIFOX is considered not cost-effective. This is despite that in our model among first line treatments for metastatic pancreatic cancer, NALIRIFOX was associated with reduced costs for managing adverse effects. Research Sponsor: None.

The effect of renin-angiotensin system inhibitors on patients with pancreatic cancer receiving systemic treatment: A retrospective cohort study from TriNetX US collaborative networks.

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Background: This study aimed to assess the impact of renin-angiotensin system inhibitors, including angiotensin receptor blockers (ARBs) and angiotensin-converting enzyme inhibitors (ACEIs), on patients with pancreatic cancer undergoing systemic chemotherapy. **Methods:** We conducted a global, retrospective collaborative network analysis using TriNetX data from January 1, 2014, to December 31, 2023, to explore this question. The study captured data on pancreatic cancer diagnoses, comorbidities, medications, and survival outcomes. Propensity score matching (PSM) was used to create 1:1 matched groups. Hazard ratios (HR) with 95% confidence intervals (95% CI) were calculated to assess survival outcomes, while the Kaplan–Meier method was employed to evaluate survival probabilities. Subgroup analyses were performed based on sex, age, surgery, race, and comorbidities. **Results:** The study included 1,861 patients who were on ARBs/ACEIs and 4,875 patients who were not on these medications at baseline. After 1:1 propensity score matching, both groups were well-balanced with 1,627 patients each. Compared to the non-ARBs/ACEIs group, patients receiving ARBs/ACEIs demonstrated significantly better overall survival (HR: 0.846, 95% CI: 0.753–0.949). Subgroup analyses revealed significantly improved survival outcomes in patients on ARBs/ACEIs, including those without surgery (HR: 0.846, 95% CI: 0.749–0.956), males (HR: 0.762, 95% CI: 0.640–0.908), patients aged 65 years and older (HR: 0.801, 95% CI: 0.709–0.906), and White patients (HR: 0.806, 95% CI: 0.704–0.924). Notably, similar survival benefits were observed in patients with comorbidities such as diabetes, ischemic heart disease, dyslipidemia, and hypertension. **Conclusions:** In the largest TriNetX matched cohort study on pancreatic cancer, the use of ARBs/ACEIs was associated with a reduced risk of overall mortality. However, prospective studies are needed to further evaluate the effects of concomitant ARB/ACEI use in patients with pancreatic cancer undergoing systemic treatment. Research Sponsor: None.

Neoadjuvant therapy in borderline resectable and resectable pancreatic cancer.

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Background: To improve clinical outcomes in borderline resectable and resectable pancreatic cancer patients, preoperative chemotherapy with or without radiation can be administered. According to NCCN, there is limited evidence to recommend a specific neoadjuvant chemotherapy regimen. The Dutch randomized phase III PREOPANC trial showed that preoperative chemoradiotherapy led to improved disease free survival. We compared overall survival (OS) between two chemotherapy regimens; mFOLFIRINOX (mFOL) and Gemcitabine/nabPaclitaxel (Gem/nabP). **Methods:** We performed a retrospective analysis of biopsy-proven pancreatic cancer patients from 1/1/2019 to 2/1/2024. We included borderline resectable and resectable pancreatic cancer patients who received neoadjuvant chemotherapy and then underwent pancreatic resection. We divided them into two cohorts: those receiving mFOL and those receiving Gem/nabP. Kaplan-Meier and multivariate Cox regression analyses were performed to analyze OS and rates of R_0 resection amongst the two groups. A p-value < 0.05 was deemed statistically significant. **Results:** From January 2019 to January 2024, 85 pancreatic cancer patients received preoperative chemotherapy prior to pancreatic resection. 50 patients received mFOL and 35 patients received Gem/nabP. The mean overall survival (OS) for the mFOL group was 1,137 days, while for the Gem/nabP group it was 876 days. The hazard ratio was estimated as 1.692 (confidence interval 0.901- 3.177, p-value 0.10). After matching for pathological stage and grade, preoperative radiation, postoperative chemotherapy, and presence of perineural and lymphovascular invasion, the hazard ratio was 1.981 (p-value 0.065). The R_0 resection rate was 92.1% for mFOL and 82.2% for Gem/nabP. The odds ratio for having negative margins with mFOL compared to Gem/nabP was 2.005 with a p-value of 0.37. **Conclusions:** Preoperative chemotherapy with mFOL did not demonstrate improved OS compared to Gem/nabP. The probability of achieving negative margins was also similar amongst the two groups. The SWOG S1505 which is a randomized phase II trial comparing perioperative chemotherapy with mFOL versus Gem/nabP in resectable pancreatic cancer patients has also shown similar OS and progression free survival in both groups. Research Sponsor: None.

XPO1 inhibition as a clinically viable strategy to enhance durability of response to $KRAS^{G12D}$ inhibitor in pancreatic ductal adenocarcinoma.

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Background: The OFF State (GDP bound) $KRASG^{12D}$ inhibitor MRTX1133 is currently being evaluated in pancreatic ductal adenocarcinoma (PDAC) patients. Nevertheless, as with other OFF state $KRAS^{G12C}$ inhibitors sotorasib and adagrasib, the $KRAS^{G12D}$ inhibitor may yield only a modest increase in disease-free survival due to emergence of drug resistance. This underscores the need to identify strategies that can enhance the efficacy of KRAS inhibitors in PDAC. Nuclear protein transport plays an important role in several pro-tumorigenic pathways and has been shown to be a therapeutic vulnerability in KRAS mutant cancers. XPO1 is the major nuclear exporter and plays a pivotal role in shuttling various critical tumor suppressors, genome surveillance proteins, and transcription factors out of the nucleus and is frequently overexpressed in various cancers, including PDAC. In this study, we employed a $KRAS^{G12D}$ inhibitor resistant model and evaluated its sensitivity to nuclear exporter protein inhibitor. **Methods:** We examined the cytotoxic and molecular effects of $KRAS^{G12D}$ inhibitor MRTX1133 in combination with nuclear transport inhibitor eltanexor in $KRAS^{G12D}$ inhibitor resistant models. The preclinical antitumor efficacy of the combination was evaluated in $KRAS^{G12D}$ mutant PDAC cell derived xenograft (CDX) subcutaneous and orthotopic models. **Results:** Eltanexor treatment reversed MRTX1133 resistance *in vitro* yielding suppressed proliferation of $KRAS^{G12D}$ mutant 2D and 3D cultures of PDAC cell lines and patient derived primary tumors cells. High throughput P-kinome analysis showed suppression of a larger repertoire of MAPK substrates in combination treatment compared to single agent group. Eltanexor and MRTX1133 combination led to reduction in protein expression of KRAS downstream effectors and cell cycle markers. Furthermore, a combined administration of eltanexor and MRTX1133 at sub-optimal doses resulted in remarkable tumor growth inhibition in multiple subcutaneous and orthotopic $KRAS^{G12D}$ mutant mice models. Moreover, eltanexor as a maintenance therapy also prevented tumor relapse indicating durability of response in PDAC. **Conclusions:** This is the first study demonstrating that nuclear transport inhibitors can overcome $KRAS^{G12D}$ inhibitor MRTX1133 resistance in PDAC cells. This study provides a rationale for combining MRTX1133 with eltanexor for the treatment of PDAC patients with $KRAS^{G12D}$ mutant tumors. Research Sponsor: U.S. National Institutes of Health; NIH R37CA215427.

Predictive factors for the efficacy of nal-IRI + 5FU/LV therapy in patients previously treated with conventional irinotecan.

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Background: There were some reports that nal-IRI + 5FU/LV (NAPOLI) therapy was effective for patients with unresectable pancreatic cancer who had a history of treatment containing conventional irinotecan (IRI) though the others did not. We aimed to investigate the factors that contribute to the effectiveness of NAPOLI therapy for patients who have already received IRI-containing treatment. **Methods:** We retrospectively enrolled the consecutive patients who had previously treated with IRI-containing treatment and started NAPOLI therapy at our hospital between June 2020 and December 2022. We compared the patient backgrounds of patients who achieved CR, PR, or SD according to RECISTv1.1 (effective group) and those who did not (ineffective group). Continuous variables were divided into two groups by the median of the entire population, and factors contributing to the effectiveness of NAPOLI therapy were determined by logistic regression analysis. **Results:** The subjective were 52 patients. The reason for discontinuation of IRI-containing treatment was refractory and intolerant in 45 and 7 patients, respectively. The median age was 65 years. Twenty-three patients (44%) had ECOG PS of 0, and 41 patients (79%) had metastatic disease. The median albumin, CRP, CEA and CA 19-9 was 3.8 g/dL, 0.3 mg/dL, 8.5 ng/mL and 2082 U/mL, respectively. Twenty-one patients (40%) showed SD and they were allocated in the effective group whereas 31 patients were in the ineffective group. Of the 21 patients in the Effective Group, 16 patients(76%) were refractory, while of the 31 patients in the Ineffective Group, 29 patients (94%) were refractory. The factors contributing to disease control were PS of 0 (odds ratio, 5.79; 95% confidence interval, 1.34–25.0), CA 19-9 <1000 U/mL (odds ratio, 5.53; 95% confidence interval, 1.26–24.3), and Alb >3.8 g/dL (odds ratio, 5.19; 95% confidence interval, 1.14–23.7). Age, sex, disease stage, serum level of CRP, and CEA did not affect the effectiveness of NAPOLI therapy. **Conclusions:** Even in patients who have already received IRI treatment, the NAPOLI regimen is expected to be effective in patients with PS of 0, Alb >3.8 g/dL, and CA 19-9 <1000 U/mL. Research Sponsor: None.

Chemotherapy selection for advanced or metastatic pancreatic cancer using machine learning models trained with multi-center datasets.

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Background: Several factors must be considered when selecting the chemotherapeutic regimen for advanced or metastatic pancreatic cancer. The decision between FOLFIRINOX and Gemcitabine/Nab-paclitaxel (GnP) as the first-line therapy is challenging, as survival is influenced by the efficacy and toxicity profiles of these treatments, along with individual patient characteristics and therapeutic vulnerabilities. This study aims to facilitate the selection of an appropriate first-line regimen for advanced or metastatic pancreatic cancer by employing machine learning (ML) methods trained on multi-center datasets. **Methods:** Our study is based on a cohort of 191 patients who underwent systemic chemotherapy for advanced or metastatic pancreatic cancer at the Gangneung Asan Hospital and Asan Medical Center in South Korea between 2014 and 2023. The dataset consisted of 17 types of demographic and clinical characteristics, as well as time-course of response and survival outcomes. We divided the cohort into training and test groups (4:1) in a stratified manner. ML models predicting overall survival (OS) were trained on the former group using the CatBoost method. The models utilizing a minimal subset of variables were selected with respect to ROC-AUC during 5-fold cross validation. A patient was classified as having high risk to a therapy when the predicted probability of death following treatment initiation was above the threshold value as determined during training of the respective model. **Results:** The median age of the entire cohort was 62 years, with 64% being male. The ML models achieved ROC-AUCs of 0.81 and 0.82 when predicting early death within 12 months following the initial administration of FOLFIRINOX or GnP, respectively. The predictive accuracies of the models were 0.77 and 0.80 for the unseen datasets from the two treatment groups, respectively. The median OS of the high- versus low-risk groups of FOLFIRINOX predicted by the ML models were significantly different (9 vs 15 months, $P < 0.0001$), with a hazard ratio (HR) of 2.8. Similarly, the median OS of the high- and low-risk groups of GnP were significantly different (9 vs 18 months, $P < 0.01$, HR of 2.5). We observed that 27% of the patients predicted as high risk to FOLFIRINOX therapy were predicted as low risk to GnP and vice versa for 32% of the patients predicted as high risk to GnP therapy. **Conclusions:** We developed ML models to compute the probability of early death following FOLFIRINOX or GnP therapy from routinely collected data of the patients with advanced or metastatic pancreatic cancer. After confirming robust predictive performance with multi-center datasets, we believe that these models may assist clinicians in selecting a first-line regimens, potentially enhancing personalized treatment strategies for this challenging disease.
Research Sponsor: Medical Research Promotion Program through the Gangneung Asan Hospital, funded by the Asan Foundation.

Surufatinib combined with TAS-102 as a late-line therapy in patients with metastatic pancreatic cancer (mPDAC): Updated results of a single-arm, phase II trial.

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Background: Patients (pts) with mPDAC recurrent after FOLFIRINOX and AG chemotherapies have few treatment options. Previous reports of NCT05481463 showed promising clinical benefits of surufatinib (a small-molecule inhibitor of VEGFR1-3, FGFR1 and CSF-1R) plus TAS-102 (combination of trifluridine and tipiracil hydrochloride) in recurrent mPDAC pts. Here, we report the updated results. **Methods:** This is a single-arm, phase II trial (NCT05481463). Eligible pts with histologically confirmed mPDAC, who had progressed after ≥ 2 lines of previous therapies were treated with surufatinib (250mg, qd, po, q4w) and TAS-102 (35 mg/m^2, bid, po, days 1–5 and 8–12, q4w). The primary endpoint was progression-free survival (PFS). Key secondary endpoint was overall survival (OS). Other secondary endpoints included objective response rate (ORR), disease control rate (DCR), and safety. **Results:** Up to Sep 18, 2024, 20 pts were enrolled (median age 58 years, male 70.0%, TNM stage IV 100.0%, ECOG PS 1 100.0%, liver metastasis 50.0%, peritoneum metastasis 50.0%). Median baseline CA19-9 level of the 20 pts was 996.7u/ml, with a range from 2u/ml to 16612u/ml. 40.0% of pts had received 3L prior therapies before enrollment and 70.0% with primary site on pancreatic body and tail. All pts had received at least one post-baseline tumor assessment (evaluable), among which ORR was 15.0% (3 PRs), DCR was 30.0% (3 PRs, 3SDs). 35.0% pts had stable or shrinking tumors. Median PFS was 2.35m (95% CI 1.91-3.94). Median OS was 6.44m (95% CI 3.81-NR). In subgroup analysis, significantly longer PFS&OS were observed in pts without liver metastasis (mPFS: 3.42m vs 2.01m, p=0.049; mOS: 9.92m vs 3.70m, p=0.026), or with ≤ 2 metastatic organs (mPFS: 3.94m vs 1.91m, p=0.005; mOS: 10.09m vs 3.81m, p=0.003). Additionally, OS was significantly prolonged in pts with baseline CA19-9 level ≤ 996.7u/ml, in contrast with those with baseline CA19-9 level more than 996.7u/ml (9.92m vs 3.70m, p=0.008). All pts experienced treatment emergent adverse events (TEAEs). Most common (≥ 10%) TEAEs (any grade) were anemia (59.1%), neutropenia (54.6%), leukopenia (50.0%), and lymphopenia (45.5%). Grade ≥ 3 TEAEs included neutropenia (31.8%), lymphopenia (13.6%), and anemia (9.1%). **Conclusions:** Surufatinib plus TAS-102 showed encouraging survival benefits and promising anti-tumor activity with acceptable toxicity in the late-line treatment of refractory mPDAC, especially in patients without liver metastasis, with ≤ 2 metastatic organs, or with baseline CA19-9 level ≤ 996.7u/ml. The results warrant further investigations in a large cohort. Clinical trial information: NCT05481463. Research Sponsor: None.

Chidamide plus envafolimab combined with S-1 as second-line treatment in advanced and metastatic pancreatic cancer (P-henomS/SCOG-P002): A single-arm, exploratory, multicenter, phase 2 trial—Interim report.

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Background: Pancreatic ductal adenocarcinoma (PDAC) is one of the most lethal cancers with poor prognosis worldwide. Chidamide, a subtype-selective histone deacetylase (HDAC) inhibitor, has significant anti-tumor effects and extensive synergistic effects with immunotherapy. Envafolimab is a light-chain deficient PD-L1 antibody. Here we conducted a single-arm, multicenter, prospective phase II clinical study (ChiCTR2200058431) to evaluate the efficacy and safety of Chidamide plus Envafolimab combined with S-1 as second-line treatment in advanced and metastatic pancreatic cancer. **Methods:** Patients with metastatic PDAC receive Envafolimab (400 mg, on day 1), Chidamide (20 mg orally twice weekly, on days 0, 3, 7, and 10), and S-1 (40-60 mg according to body surface area, orally twice daily from day 1 to 14) every 3 weeks until disease progression, unacceptable toxicity, or patient refusal. The primary end points are safety and ORR. The secondary endpoints were PFS, OS, DCR, QoL and nutrition score. **Results:** Recruitment completed with 16 patients as of September 2023 and the data cut-off for analysis was August 2024, 13 were evaluable. Median age was 66 (range 59 - 80), with 5 males and 11 females. After a median follow-up of 8.5 months, the ORR and DCR by RECIST v1.1 were 30.77% and 76.92%, respectively. Median PFS was 5.83 months (95%CI 2.592-9.068) and median OS was not reached. No new safety signals were observed. Grade ≥3 treatment-related adverse events (TRAEs) occurred in 31.25% of patients. The most common TRAEs were anemia (12.5%), decreased platelet count (6.25%) and neutropenia (12.5%). No treatment-related deaths occurred. **Conclusions:** Preliminary data suggest that Chidamide and Envafolimab in combination with S-1 may be an effective second-line treatment with a manageable safety profile for patients with PDAC. Clinical trial information: ChiCTR2200058431. Research Sponsor: None.

Efficacy evaluation.

Efficacy Evaluation	All (N = 13) n(%)
Best efficacy evaluation	
Partial Response (PR)	4 (30.77%)
Stable Disease (SD)	6 (46.15%)
Progressive Disease (PD)	3 (23.08%)
Objective response rate (ORR)	30.77%
95%CI	9.09-61.43
Disease control rate (DCR)	76.92%
95%CI	46.19-94.96
mPFS (95%CI)	5.83 (2.592-9.068)
mOS (95%CI)	- (NR)

741 Poster Session

Interim analysis of feasibility, safety, and tumor control in two first-in-human trials of a novel alpha-emitting radionuclide for pancreatic adenocarcinoma.

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Background: This study aims to report the initial feasibility, safety, and tumor response of delivering Diffusing Alpha-Emitter Radiation Therapy (Alpha DaRT) to the pancreas. This innovative approach involves the interstitial implantation of a novel alpha-emitting radiation source via endoscopic ultrasound (EUS) for the treatment of pancreatic cancer. **Methods:** The study enrolled patients treated with Alpha DaRT to the pancreas across two protocols (PANC and ALL). The primary objective was to assess the safety of Alpha DaRT delivery to the target lesion via EUS, while the secondary objective was to evaluate initial tumor response using the Response Evaluation Criteria in Solid Tumors version 1.1 at 1 and 3 months post-Alpha DaRT source insertion. Toxicity was evaluated according to the Common Terminology Criteria for Adverse Events version 4.03. The PANC protocol included patients with biopsy-proven adenocarcinoma, either unresectable, recurrent, or metastatic, who were unfit for standard treatment. The ALL protocol allowed for any malignancy with a targetable lesion and permitted concurrent systemic therapy. For this analysis, only patients from the ALL protocol with pancreatic adenocarcinoma treated to the primary tumor were included. Pre-procedure planning involved CT simulation and EUS, and EUS-guided Alpha DaRT placement was confirmed by CT. The median follow-up time was 91 days. **Results:** A total of 12 patients (9 male, 3 female) median age 73 (range: 59-85) were enrolled (4 in PANC, 8 in ALL). Four patients had localized, unresectable disease and eight had metastatic disease. Three patients had a history of prior radiotherapy. Successful placement of Alpha DaRT sources into the targetable pancreatic lesions was achieved in all cases (100%), as confirmed by post-treatment imaging. Grade 1-2 acute toxicities included fever in 2 of 12 patients (16.7%), fatigue/weakness in 2 of 12 patients (16.7%) and abdominal or back pain in 3 of 12 patients (25.0%). One patient was hospitalized for obstructive jaundice (Grade 3) possibly secondary to inflammation seven days post-implantation requiring percutaneous transhepatic cholangiography; all hepatic laboratory values returned back to normal post procedure. All adverse events resolved. No long-term toxicities ($>$ 3 months) have been seen to date. Tumor response data were available for 8 of 12 patients. Local control was 100% for all evaluable patients. Of these, 6 (75.0%) exhibited stable disease, while 2 (25.0%) showed a partial response ($>$ 30% tumor reduction). One patient had a complete metabolic response on PET-CT at both the primary pancreatic lesion and liver metastasis. **Conclusions:** Alpha DaRT delivered via endoscopy is feasible with a favorable safety profile and demonstrates promising tumor responses. Ongoing longer follow-up and larger studies are required to validate these findings. Clinical trial information: NCT05781555 and NCT05657743. Research Sponsor: None.

Safety and preliminary efficacy of ivaltinostat, an HDAC inhibitor, plus capecitabine in patients with pretreated metastatic pancreatic adenocarcinoma (mPDAC): Results of a phase Ib study.

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Background: Front-line chemotherapy for patients (pts) with mPDAC typically consists of mFOLFIRINOX or gemcitabine/nab-paclitaxel given until disease progression or intolerable toxicity. For pts with durable disease control on front-line treatment, maintenance therapy that can effectively delay disease progression while preserving quality of life with minimal cumulative toxicity is highly desirable. Ivaltinostat (ival) is a pan-histone deacetylase inhibitor that increases histone acetylation, suppresses cell proliferation, and promotes apoptosis. Preclinical data demonstrated synergy with 5-FU/capecitabine (cape) in mouse models. We report here results of the Phase Ib portion of a Ib/2 trial evaluating the safety and recommended phase 2 dose (RP2D) of this combination to use in a randomized study of ival + cape vs cape as maintenance therapy. **Methods:** Key eligibility criteria for phase 1b included unresectable or metastatic PDAC; ≥18 years old; ≥1 prior therapy; ECOG PS 0-1; and no known germline BRCA1/2 mutation. Three cohorts were enrolled, receiving ivaltinostat at 60, 125, or 250mg/m^2 iv (weekly on days 1 and 8) in combination with cape (1000mg/m^2 po BID on days 1-14) of a 21-day cycle. 1o objectives: safety, tolerability and RP2D; 2o objectives: pharmacokinetics (PK) of ival + cape. **Results:** A total of 28 patients were enrolled at the 60 (n=6), 125 (n=10) and 250 (n=12) mg/m^2 ival dose levels. Median prior lines of therapy: 2 (range 1-7). Median age: 67 years (range 50-83). Only one DLT (G4 thrombocytopenia) was observed at 250mg/m^2 dose level. Treatment-emergent AE profile was similar across dose levels, with common TEAEs including fatigue (n=15, 53.6%), nausea (n=14, 50.0%), palmar-plantar erythrodysesthesia syndrome (n=12, 42.9%), diarrhea (n=10, 35.7%), and constipation (n=9, 32.1%). G3 AEs included neutropenia (n=6, 21.4%), anemia (n=4, 14.3%), abdominal pain (n=3, 10.7%) and diarrhea (n=3, 10.7%). G4 AEs included 1 (3.6%) event of neutropenia and 1 (3.6%) of thrombocytopenia. No SAEs related to study drugs occurred in any of the 3 dose cohorts. PK analyses indicated dose proportional increases in exposure with increasing doses of ival. Histone H3K27 acetylation for ival showed the similar trend in optical density evaluation. Across all dose levels and prior treatments, median OS and PFS were 9.6 and 3.3 months, respectively. **Conclusions:** The combination of ival and cape was generally well tolerated in this heavily pretreated patient population. Ival 250 mg/m^2 is the dose selected for the ongoing randomized study of ival + cape vs cape in the maintenance setting for mPDAC pts post-FOLFIRINOX, which is expected to finish accrual in 2025. Clinical trial information: NCT05249101. Research Sponsor: CG Pharmaceuticals Inc.

A phase 1b dose escalation trial of gemcitabine and nab-paclitaxel in combination with lixumistat in patients with advanced pancreatic cancer.

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Background: Despite recent advances, there are limited effective therapeutic options for pancreatic ductal adenocarcinoma (PDAC). Conventional chemotherapy can be difficult to tolerate and offers modest survival benefits. Lixumistat is a small molecule biguanide that inhibits the first and rate-limiting step of the oxidative phosphorylation pathway, which is critical to the survival of PDAC tumor cells. A first-in-human dose escalation trial of Lixumistat in solid tumors demonstrated a favorable safety profile and a recommended phase 2 dose (RP2D) of 800 mg per day (QD) (1). **Methods:** A single center Phase 1b study is currently enrolling patients with treatment-naïve metastatic PDAC to evaluate the safety and tolerability of Lixumistat when combined with standard-of-care gemcitabine (Gem) plus nab- paclitaxel (NP) (NCT05497778). The initial escalation cohort received oral Lixumistat at 400 mg QD; the subsequent escalation cohort received oral Lixumistat at 800 mg QD. All patients also receive fixed doses of intravenous (IV) NP (125 mg/m^2) and IV Gem (1000 mg/m^2) on Days 1, 8, and 15 (28 day cycle). Here we report the preliminary results from the escalation phase of the trial. **Results:** Thirteen patients have been treated: 7 at 400 mg and 6 at 800 mg. There were no grade 4-5 toxicities. Two patients experienced a DLT at the 800 mg dose (grade 3 diarrhea and grade 3 fatigue); no DLTs were observed at the 400 mg dose. The most common Lixumistat related toxicities were Grade 1/2 nausea/vomiting (controlled with anti-emetics), skin rash, fatigue, and diarrhea. Grade 1-3 thrombocytopenia, neutropenia, and peripheral sensory neuropathy events were no more than expected with this combination and managed with Gem/NP dose reduction, omission of day 8, or therapy hold until resolved to Grade 1. Based on toxicities observed at 800 mg, 400 mg is the RP2D. Among 6 response-evaluable patients treated at the RP2D, 2 achieved an objective partial response (ORR = 33%, 95% CI: 4.3%-77.3%); 4 patients had stable disease with tumor shrinkage observed in 3 of these patients (12%, 22% and 23% reduction); and the DCR was 100% (95% CI: 54.1%-100%). Notably, a patient from the 800 mg cohort who was dose reduced to 400 mg after DLT achieved a complete response in their target lesions. Current median follow-up time is 13.2 months, and 2 patients have died. Estimated median PFS is 12.9 months (95% CI: 3.65, NA) and estimated median OS is 18 months (95% CI: 6.42, NA). Four patients remain on active treatment at the time of analysis. **Conclusions:** When combined with Gem and NP; a dose of 400 mg of Lixumistat QD was demonstrated to be safe, tolerable and identified as the RP2D. This trial is currently enrolling patients at the RP2D in the expansion cohort. Updated data on safety and efficacy will be presented at the meeting. 1. Janku 2022. Clinical trial information: NCT05497778. Research Sponsor: This research is supported by The MD Anderson Cancer Center SPORE in Gastrointestinal Cancer Grant P50 CA221707; Immunomet.

Galeterone monotherapy in advanced pancreatic ductal adenocarcinoma: Results from a phase two trial.

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Background: Advanced pancreatic ductal adenocarcinoma (PDAC) is a devastating disease with limited treatment options. Dysregulated MAPK and PI3K signaling in PDAC activates the eukaryotic translational initiation complex, promoting cell growth, chemoresistance and metastatic spread (PMID 25593033). Galeterone, a novel steroidal anti-androgen, downregulates critical mediators of this complex including Mnk1/2 and phosphorylated eIF4E, thereby blocking epithelial-to-mesenchymal transition and NF-kB activity. In-vitro, galeterone exhibits anti-tumor activity both alone and in combination with gemcitabine and slows tumor growth in a MiaPaCa-2 xenograft murine model (PMID 28881737). We present results of the monotherapy arm from an ongoing phase two trial evaluating galeterone +/- gemcitabine for patients with advanced PDAC. **Methods:** Patients with locally advanced or metastatic PDAC, an ECOG performance status of 0-2 and adequate organ function, who progressed on two prior lines of systemic therapy were eligible. Galeterone was administered orally at 2550mg daily and gemcitabine intravenously at 1000mg/m^2 weekly for three weeks on 28-day cycle. The primary endpoint was radiographic response rate per RECIST v1.1 and secondary endpoints included progression-free survival (PFS), overall survival (OS) and safety. We used a Simon's optimal two-stage design with a null hypothesis of p_0=0.05 and one-sided alternative of p_1=0.20 with an early stopping rule for futility in each arm. Due to poor accrual and no signal of activity, the galeterone alone arm closed after enrollment of four patients. **Results:** Four patients were evaluable for the primary endpoint. Ages ranged from 43 to 68. Two patients were male and two were female. Two patients were white and two were black; one patient was Hispanic/Latino. All patients had ECOG PS of 1. No patients had a radiographic response; the best response was progressive disease (PD). PFS from cycle 1 day 1 ranged from 0.9 – 1.7 months and overall survival between 1.8 – 3.6 months. The only treatment-related adverse events (TRAEs) were grade 1 dizziness and grade 3 fatigue in one patient each. There were no treatment-related serious adverse events. No patients required dose modification or discontinued due to TRAEs. **Conclusions:** Galeterone showed no clinical activity, but with a favorable safety profile, as a single-agent in heavily pre-treated patients with advanced PDAC. Activation of the Mnk-eIF4E axis mediates chemoresistance in PDAC and galeterone may prove more efficacious with gemcitabine. This combination arm is currently open to enrollment. Clinical trial information: NCT04098081. Research Sponsor: None.

Clinical outcomes with galeterone monotherapy in PDAC.

Pt #	Age	Prior lines of Therapy	Time on Treatment (days)	Best Response	PFS (months)	OS (months)
1	68	4	46	PD	1.5	3.6
2	59	2	54	PD	1.7	3.8
3	55	2	47	PD	1.4	2
4	43	2	35	PD	0.9	1.8

SIL-204 siRNA encapsulated in extended release microparticles for the treatment of localized cancer that harbors a KRAS G12x or G13D mutation.

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Background: In a Phase 2 trial, treatment of non-resectable locally advanced pancreatic cancer (LAPC) patients harboring the KRAS G12D or G12V mutations, with SiG12DLoder combined with chemotherapy resulted in superior median overall survival (OS) of 9.3 months over chemotherapy (not statistically significant) and Objective Response Rate of 55% (35% over the chemotherapy control), when administered once every 3 months (Clinical trial # NCT01676259, ESMO 2023, abstract FPN: 1626). SIL-204 which is a modified version of siRNA siG12D is composed of a 21-base sense and 23-base antisense strands, both chemically modified and linked to a lipid. SIL-204 is encapsulated in biodegradable PLGA microparticles (MP), for direct release and injection into KRAS mutated solid tumors using an endoscopic ultrasound procedure. The strategy of preventing the synthesis of KRAS mutated protein may have an advantage over other approaches targeting KRAS function using small molecules. **Methods:** Methods are presented in results section. **Results:** SIL-204 has an increase stability over siG12D.To enhance stability, base modifications and phosphorothiate bonds at cleavage sites were implemented. SIL-204 exhibited a half-life of more than 48 hours in human serum. SIL-204 has robust KRAS silencing. When tested in Hepa1-6 cells transfected with individual human KRAS mutations, SIL-204 effectively silenced the various G12x mutations found in pancreatic cancer (D,V,R,C) at sub-nanomolar concentration (IC50 range=0.19-0.59) as well as G13D (IC50 =0.37). Addition of a hydrophobic tail to a prototype of SIL-204, SIL-101, increased the silencing of KRAS. Using PANC-1 cells (KRAS G12D mutation), lipid conjugation of the siRNA doubled the siRNA's silencing efficiency in these cells. In addition, release of SIL-204 from the MP was tested in rats and indicates a prolong and sustained release over our intended treatment regimen SIL-204-MP was administered to tumors from human pancreatic tumor cell lines Capan-1 (KRAS G12V mutation and labelled with luciferase) grown in NSG mice and Panc-1 (grown in Athymic Nude mice). Bioluminescence imaging of tumor growth showed that SIL-204-MP significantly reduced Capan-1 tumor growth compared to vehicle controls, $p<0.0005$. Furthermore, histopathological analyses of tumor center slices from Capan-1 and Panc-1 models showed induction of significant tumor necrosis with SIL-204 compared to vehicle controls. **Conclusions:** The preclinical results of SIL-204-MP are promising and as the potential to further enhance the anti-tumor effect demonstrated by siG12DLoder in pancreatic patients. SIL-204 is now in toxicology studies as preparation for clinical trials in non-resectable pancreatic cancer. Research Sponsor: Silexion Therapeutics Corp.

Immunotherapy engagement in pancreatic adenocarcinoma: Provisional results of iLSTA study— Durvalumab, LSTA1 (certepetide), gemcitabine, and nab-paclitaxel for locally advanced pancreatic ductal adenocarcinoma.

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Background: Pancreatic ductal adenocarcinoma (PDAC) is characterised by a dense, extracellular matrix-rich stroma, which creates a physical barrier against drug penetration. Evidence suggests that the iRGD peptide, LSTA-1, can increase the penetration of anti-cancer drugs to tumours through selective targeting of tumour endothelial cell receptors. Additionally, LSTA-1 administration promotes CD8+ immune cell tumour infiltration, potentially enabling immunotherapy as a treatment modality. iLSTA examined safety, tolerability and effect of the combination. **Methods:** Participants diagnosed with locally advanced PDAC were divided into 3 cohorts in a 1:1:4 ratio. Cohort 1 (n=5) received gemcitabine (1000mg/m^2), nab-paclitaxel (125mg/m^2), placebo LSTA-1 (3.2mg/kg) and placebo durvalumab (750mg). Cohort 2 (n=5) received gemcitabine, nab-paclitaxel, active and placebo LSTA-1, and placebo durvalumab. Cohort 3 (n=20) received gemcitabine, nab-paclitaxel, LSTA-1 and durvalumab. Tissue biopsies via endoscopic ultrasound were obtained pre-treatment and between weeks 12 and 16 for analysis of tumour infiltrating lymphocytes (TILs). Patient tumour sizes were assessed every 8 weeks using radiological imaging according to RECIST v1.1, and serum CA19-9 levels were analysed monthly. **Results:** 6/17 patients showed significant RECIST partial response after 2 cycles of treatment (5 patients in cohort 3), with the remaining 11 patients exhibiting stable disease. After 4 cycles of treatment, 10/17 patients demonstrated partial response (9 patients in cohort 3). Of the remaining 7 patients, 6 demonstrated stable disease, and 1 patient (cohort 2) exhibited a RECIST complete response. 13/17 patients who completed 4 treatment cycles showed a decrease in CA19-9 levels. 6 patients demonstrated >90% reduction in CA19-9 (5 in cohort 3), with the remaining 7 patients showing a >50% reduction in CA19-9 levels (5 in cohort 3). 12 patients underwent repeat biopsies 12-16 weeks post-treatment to assess TILs. 10 patients showed immune cell infiltration (5% to 50% stroma infiltration), 2 had no tumour found (Cohort 3). The combination was safe with no unexpected toxicities. **Conclusions:** The preliminary results of this study suggest that the combination of gemcitabine and nab-paclitaxel with LSTA-1 and durvalumab is safe and potentially induces tumour infiltrating lymphocytes and RECIST response in locally advanced pancreatic adenocarcinomas. Clinical trial information: ACTRN12623000223639. Research Sponsor: WARPNINE.

Telotristat ethyl to promote weight stability in patients with advanced pancreatic cancer.

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Background: Cachexia a key feature of pancreatic ductal adenocarcinoma (PDAC), affects up to 80% of patients (pts), and is associated with worse outcomes. Serotonin, due to its tumorigenic potential and effect on gut function may contribute to weight loss in PDAC. Telotristat ethyl (TE) inhibits serotonin production and improves gut motility dysfunction. We developed a phase II trial to evaluate the impact of TE on the weight of patients with metastatic PDAC (mPDAC) receiving first-line chemotherapy. **Methods:** Treatment-naïve pts with mPDAC who had lost at least 10% of baseline weight (Group 1) received gemcitabine and nab-paclitaxel (GnP) in combination with TE (250mg administered orally three times a day). Pts with < 10% weight-loss were recruited in Group 2 and received chemotherapy without TE. Serum serotonin was collected monthly during the study. The primary endpoint was weight stability, evaluated as % weight change at 3 months compared to baseline, with a goal of < 5% decrease. A key secondary endpoint was change in serotonin levels at 3 months compared to baseline. The groups were analyzed independently. **Results:** We enrolled 22 pts, 14 in group 1 (planned 40) and 8 in group 2 (planned 40). The median age was 70 years (IQR 62-75), majority were male (n = 15; 68%) and 11 (50%) were Black. Mean weight at treatment start was 77.9 kg (SD 19.16) group 1, and 90.6kg (SD 24.71) group 2. Mean serotonin level at baseline was 165.07 ng/ml (SD 60.49) group 1, and 296.9 ng/ml (SD 166.65) group 2. Mean % weight change over 3 months in group 1 was -2% (90% CI -5.2 to 1.3, P < -5% = 0.0595). Mean change in group 2 was -5.6% (90% CI -9.4 to -1.7%, P < 5% = 0.53). At 3 months, the median change in serotonin level in group 1 was -37.8% (IQR -59.1 to -13, P = 0.064) and -42.6% (IQR-57.8 to -29.9, P = 0.016) for group 2. Adverse events (AE) were consistent with known AE for GnP and TE. The median overall survival was 10.7 months (95% CI 5.4-16.9) group 1 and 12.2 (95% 6.1 to 21.4) group 2, HR 1.41 (95% CI 0.57 to 3.52 P = 0.46). **Conclusions:** Among pts with mPDAC and cachexia, the addition of TE to chemotherapy led to weight stabilization. These pts experienced 2% weight loss on average during the study (the upper limit of the CI just crossed -5% at -5.2%). The average weight loss was 5.6% in those who did not receive TE suggesting a signal of activity. A larger study is warranted to evaluate the role of TE in the management of cachexia in mPDAC. Funded by Lexicon Pharmaceuticals (NCT03910387). Clinical trial information: NCT03910387. Research Sponsor: Lexicon Pharmaceuticals.

Implication of the novel microtubule targeting agent, AUS_001, in pancreatic cancer cellular signaling.

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Background: Pancreatic cancer (PanCa) is considered a major therapeutic challenge due to its poor prognosis, high mortality rate and resistance to most therapies, underlining the need for new treatment approaches. Our previous investigations revealed that the prenylated hydroxystilbene, AUS_001, inhibits β-tubulin polymerization via its unique binding to the colchicine site of tubulin. A cell-based profiling platform identified sensitivity of 12 out of 13 PanCa lines to AUS_001 with a median concentration causing 50% growth inhibition of 0.227μM. The goal of the current study was to elucidate the signaling activity and cellular responses following treatment with AUS_001 and explore the prospect of the latter in PanCa treatment. **Methods:** A series of functional and biochemical assays were performed to delineate the effect of AUS_001 on the growth, colony formation ability, invasive capacity and oxidative status of PanCa cells. Tumor growth was evaluated in PANC-1 implanted CB.17 SCID mice and MIA PaCa-2 implanted nude mice upon intraperitoneal administration of AUS_001 once weekly. Metabolic profiling was carried out using the Seahorse XF Analyzer. RNA silencing and adenovirus-mediated overexpression of the Dominant-Negative Mutant of c-Jun were employed to explore the link of c-Jun activation and AUS_001-induced cytotoxicity. **Results:** The potential of AUS_001 to disrupt PanCa cell functions was established *in vitro* and *in vivo*. Viability and compromised membrane integrity were multiplexed and resulted in half maximal effective concentration agreement indicating drug-induced cytotoxicity. AUS_001-treated cells underwent apoptosis followed by secondary necrosis, exerted diminished anchorage-independent growth and metastatic potential *in vitro*. Importantly, tumor growth was significantly reduced in PanCa murine xenograft models by AUS_001 as a monotherapy. Pretreatment with N-acetyl-l-cysteine did not alter cellular sensitivity to AUS_001, suggesting that the time-dependent induction of reactive oxygen species upon AUS_001 exposure is not the primary cause of cytotoxicity. Notably, the oxygen consumption rate, extracellular acidification rate and function of all mitochondrial complexes (I-IV) of live MIA PaCa_2 cells were significantly impaired in both glucose- and galactose-containing media 48h after AUS_001 administration. Integration of transcriptomic and proteomic data originating from AUS_001-challenged cells uncovered the dramatic induction of c-Jun and FosB stress responders. Further studies demonstrated that c-Jun transcriptional activation acts as a partial mediator of AUS_001-induced cytotoxicity. **Conclusions:** Collectively, our findings provide critical insights into the molecular events triggered by AUS_001 in PanCa cells and serve as supporting data for future exploration of AUS_001 as a novel PanCa therapeutic agent. Research Sponsor: Australis Pharmaceuticals, Inc.

749 Poster Session

Toxicology and clinical protocol development of DUO-207: An ultrasmall nanomedicine co-delivering gemcitabine and paclitaxel for the treatment of metastatic pancreatic cancer.

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Background: The long-term survival rates for metastatic pancreatic cancer have remained stubbornly low for decades. Most pancreatic cancers have dense, fibrous stroma tissue surrounding their tumors, which contributes to tumor growth and acts as a barrier to many therapies. Nanomedicines have been developed for use in oncology to reduce toxicity and improve efficacy, but a majority are too large (> 100nm) to effectively penetrate dense stroma. DUO-207 is a novel ultrasmall nanomedicine (~ 20nm) composed of copolymer-conjugated gemcitabine that encapsulates paclitaxel. Numerous preclinical efficacy models have demonstrated DUO-207's ability to penetrate and accumulate in tumor tissue. Here we describe results from toxicology studies performed in canines that assisted in the design of a phase 1b clinical trial using the continual reassessment method (CRM). **Methods:** A single cycle, bridging toxicology study of DUO-207 with a recovery period was performed in canines. Animal equivalent doses of nano-albumin-bound paclitaxel (nAb-PTX) and gemcitabine were administered as the standard of care (SoC) comparator. Blood was analyzed for hematological toxicity, biomarkers of major organ toxicity, and drug toxicokinetics. Toxicology results from DUO-207 were then used in a statistical model to explore plausible scenarios for clinical dose escalation. **Results:** DUO-207 treated canines demonstrated improved clinical observations compared to SoC. As expected, hematologic toxicities occurred in a dose dependent fashion in the DUO-207 group but no hepato- or nephrotoxicity was noted. Based on these data, a phase 1b CRM design has been developed with 90% power to detect the maximum tolerated dose in key scenarios with an average sample size of 11 patients per group. **Conclusions:** In summary, DUO-207 is a novel, ultrasmall nanomedicine that can co-deliver paclitaxel and gemcitabine in a single intravenous infusion to treat stroma-rich cancers. DUO-207 demonstrated dose dependent hematological toxicities but improved clinical observations compared to SoC and lacked signals of renal and hepatoxicity. Toxicokinetic data from canines was used in a statistical model to determine patient numbers for phase 1b dosing using the CRM. Duo Oncology will conduct its clinical trial in 2025. Research Sponsor: None.

Dose-escalated radiation for locally advanced unresectable pancreatic cancer.

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Background: Patients with unresectable locally advanced pancreatic cancer have limited therapeutic options. Historical treatments have included systemic chemotherapy or palliative radiation. New advanced radiotherapy treatment techniques have allowed dose escalated ablative radiotherapy to be delivered for patients with cancer limited to the pancreas. This study retrospectively reviews the long term outcomes of patients receiving dose escalated radiation for unresectable localized pancreatic cancer. **Methods:** Patients with locally advanced unresectable pancreatic underwent neoadjuvant chemotherapy with either gemcitabine or FOLFIRINOX based regimens. When maximum response was achieved with chemotherapy or this was no longer tolerated, patients were reassessed with imaging. Patients with localized unresectable disease then proceeded to definitive dose escalated radiation with either SBRT radiosurgery (40Gy in 5 fractions) or IMRT radiation (65-70Gy in 30-35 fractions) with concurrent 5FU. Surveillance of disease was performed with CT imaging and Ca19-9 biomarker assessment. **Results:** A total of 38 patients (21 male, 17 female) underwent definitive radiation. Radiation technique was determined by treating physician based on patient anatomy, with 7 patients having SBRT and 31 IMRT/5FU. Average patient age at diagnosis was 71. All patients underwent at least 6 months of follow up and mean follow up was 14 months. At time of latest follow up, 30 patients had no evidence of local progression and 12 patients had no evidence of progression at any site. The most common side effect at last follow up was grade 2/3 duodenal ulcer experienced by 7 patients. **Conclusions:** Advances in systemic chemotherapy have increased the population of patients with locally advanced unresectable pancreatic cancer who have persistent pancreatic disease after induction chemotherapy. Definitive dose escalated radiation is a new and effective technique to provide local control in this setting. This approach is well tolerated with manageable toxicity and effective durable local control. Patients continue to experience distant disease failure despite improving local control rates, though at a lower rate than historical controls. Control of primary pancreatic malignancy with dose escalated radiation can potentially improve quality of life and function for these patients for whom additional chemotherapy or surgery is not indicated. Research Sponsor: None.

Beyond Khorana score 2: Molecular correlates of venous thromboembolism in pancreatic cancer.

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Background: Pancreatic ductal adenocarcinoma (PDAC) is associated with a hypercoagulable state leading to thrombosis. Risk models such as the Khorana score automatically classify PDAC as intermediate-high risk, and recent guidelines recommend consideration of thromboprophylaxis. However, little is known about the molecular correlates of PDAC for venous thromboembolism (VTE). **Methods:** We examined clinical and genomic data from the prospective multi-institution COMPASS trial (NCT02750657), which enrolled patients with treatment-naïve metastatic PDAC who underwent a fresh tumor biopsy for real-time whole genome and transcriptome sequencing. Laser capture microdissection was performed. Patients underwent restaging scans at 8-week intervals. Detailed chart review was conducted to focus on VTE risk factors, timing of VTE diagnosis, and anticoagulation. We also compared clinical and molecular factors based on timing of VTE diagnosis. Overall survival was defined as time from VTE to death. **Results:** Of 268 patients enrolled in the COMPASS trial, 166 patients had detailed clinical data available regarding VTE status. 82 patients (49%) developed a VTE, where 16 (20%) had breakthrough clots. Baseline epidemiological variables were similar for those with and without VTE, including no differences between age, sex, BMI, and baseline CA 19-9. No patients were on routine prophylactic anticoagulation. SMAD4 mutations were more frequently seen in the VTE subgroup (57% vs. 40%, p= 0.04), but no differences were seen in other driver genes (KRAS, TP53, CDKN2A), Moffitt subtype, or homologous recombination deficiency status. Patients who developed an early VTE (within 3 months) had a higher baseline CA 19-9 (median 4015 vs. 891, p= 0.02), and none were port-associated. A higher incidence of KRAS wildtype cases (10%) were observed in the early vs. later VTE groups (p= 0.03), but no differences in KRAS allelic status. There were no differences in burden of SNVs, indels, or SVs. Early VTE occurred in 46% of all basal cases, which are typically more aggressive, and only in 37% of classical subtypes. Overall survival was shorter with early VTE (HR 1.74, p= 0.02). **Conclusions:** A higher incidence of SMAD4 alterations were observed among patients with metastatic PDAC who developed VTE. VTE diagnosed earlier were associated with shorter survival, suggesting that early thromboprophylaxis should be considered at the time of diagnosis. Clinical trial information: NCT02750657. Research Sponsor: None.

752 Poster Session

Preclinical development of the lipid nanoparticle–based mRNA vaccine with multiple natural T cell epitopes for pancreatic cancer.

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Background: Immunotherapy has recently emerged as a promising avenue in cancer treatment; however, its effectiveness on pancreatic ductal adenocarcinoma (PDAC) is hindered by the lack of effector T cells in the tumor microenvironment. One of the strategies to overcome the resistance of PDAC to immunotherapy involves identifying immunogenic tumor-associated and specific antigens for the development of personalized cancer vaccines. The mRNA platform has become an attractive platform for antigen presentation. The delivery of mRNA cancer vaccines via lipid nanoparticles (LNPs) enables intracellular delivery and controlled release of the payload, offering improved vaccine efficacy. However, mRNA vaccines presenting human T cell epitopes often lack a preclinical model for antitumor immune efficacy testing. **Methods:** A syngeneic mouse model of PDACs was established by implanting KPC tumor cells into the livers by hemispleen surgery. mRNA vaccine expressing five 25-mer peptide regions were synthesized. Two mouse CD8+ T cell epitopes were predicted. Their MHC binding was tested by the T2 binding assay. CD8+ T cells specific for these two epitopes were examined by ELISpot assays and were quantified by ImmunoSpot ELISpot Reader to identify changes in IFN-g secretion. **Results:** Using mass spectrometry analysis, we successfully identified natural HLA class I and class II-binding epitopes in the neoplastic tissues of human PDACs. For the design of mRNA vaccine, we chose five 25-mer peptide regions that each contains both class I and class II epitopes punctuated by AAY and GPGPG linkers and followed by an MITD sequence, which all aim to improve the efficiency of antigen presentation to dendritic cells. Using NetMHC, we predicted H2Kb or H2Db-binding epitopes LYPFPLAL and SMVQMTFL, within these five human peptide regions. The T2 binding assay showed a lack of evident binding of both peptides to the T2-Kb or T2-Db cells. However, the ELISpot assay showed that LYPFPLAL, but not SMVQMTFL, can induce the IFN-g expression from CD8+ T cells derived from splenocytes of the KPC tumor mice vaccinated by a peptide vaccine containing both peptides. Therefore, this result confirmed that LYPFPLAL is a mouse CD8+ T cell epitope and can be used to monitor antigen-specific CD8+ T cell response in response to the mRNA vaccine treatment in the syngeneic mouse model of PDACs. **Conclusions:** Antitumor efficacy of our above-designed mRNA vaccine can be tested in the syngeneic mouse hemispleen model of KPC tumors. Mouse LYPFPLAL epitope-specific CD8+ T cell response can serve as a surrogate marker for the antitumor efficacy of this mRNA vaccine in mouse models for PDAC. Research Sponsor: None.

Effects of tumor treating fields (TTFields) with FOLFIRINOX on BRCA WT pancreatic cancer cells.

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Background: Pancreatic cancer remains one of the most aggressive malignancies, frequently diagnosed at the locally advanced or metastatic stage. In the advanced setting, first-line chemotherapy options include gemcitabine with nab-paclitaxel or FOLFIRINOX, a combination regimen of fluorouracil, oxaliplatin, irinotecan and leucovorin. The FOLFIRINOX regimen is however associated with greater toxicity and is hence used only in patients with good performance status. BRCA-mutated tumors seem to be more sensitive to FOLFIRINOX due to the DNA-damaging, platinum-based component of this treatment regimen. Tumor Treating Fields (TTFields) are electric fields that disrupt cellular processes crucial for cancer cell viability that have shown efficacy in preclinical pancreatic cancer models; and, in various cancer types, have been shown to reduce expression of proteins from the BRCA-dependent DNA repair pathway. The current study examined the potential use of TTFields for sensitization of BRCA wild-type pancreatic cancer cells to treatment with FOLFIRINOX. **Methods:** Human pancreatic BRCA wild-type cancer cells BxPC3 and AsPC1 were treated with TTFields (150 kHz; 0.7 and 1 V/cm RMS, respectively), using the inovitro device. FOLFIRINOX was administered to the cells at increasing concentrations, with or without co-treatment with TTFields. After 72h of treatment, cell count, colony formation, and apoptosis were measured. For mechanistical insight, gene and protein expression were evaluated following 24, 48, or 72h of TTFields treatment. **Results:** TTFields application to the pancreatic cells together with FOLFIRINOX elevated the cytotoxic, clonogenic and apoptotic effects induced by FOLFIRINOX or TTFields alone. RNA sequencing data revealed that TTFields downregulated DNA damage repair, DNA replication, and cell cycle related processes. Specifically, real-time PCR and Western blot analysis demonstrated reduced expression of several central players in the BRCA DNA damage repair pathway. **Conclusions:** TTFields application together with FOLFIRINOX in pancreatic cancer cells lacking background BRCA mutations exhibits potential improvement relative to FOLFIRINOX alone, that may be rationalized based on downregulation of key DNA repair pathways. Future studies should explore the possibility of reducing FOLFIRINOX treatment dose, potentially alleviating FOLFIRINOX-related toxicity. Research Sponsor: Novocure Ltd.

$CD8^+$ T cell proximity analysis of the tumor microenvironment (TME) in patients with pancreatic ductal adenocarcinoma (PDAC) and associated clinical outcomes.

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Background: Immunosuppressive cells (cancer-associated fibroblasts [CAFs], tumor associated neutrophils [TANs], M2/M1 macrophages) can influence $CD8^+$ T cell infiltration in the PDAC TME. The spatial proximity of these cells to $CD8^+$ T cells and association with clinical outcomes is not well known. We hypothesize the cell distances between $CD8^+$ T cells and these phenotypes: CXC chemokine receptor 4 $(CXCR4)^+$ TANs, CAFs, focal adhesion kinase $(FAK)^+$ tumor cells, FAK^+ M2 and M1 macrophages will correlate with survival outcomes. **Methods:** Twenty-two stage I/II PDAC patients who consented for tissue collection were stratified into two groups: standard responders (n=11) with overall survival (OS) >2 years and poor responders (n=11) with OS ≤ 2 years. Multiplex IHC was performed on specimens using multiple biomarkers (CD8, CXCR4, CD66b, FAK, FAP, CD68, CSF1R, CD163, EPCAM) to identify cell phenotypes. Comprehensive spatial analysis with HALO software was used to measure mean cell distance between $CD8^+$ T cells and other cells. The mean cell-to-cell distance among evaluable specimens was determined by phenotype. Patients were divided into two groups with average cell distance below or above the mean by phenotype. OS of each subgroup was compared using Kaplan-Meier method. **Results:** Median OS for standard and poor responders was 47.6 months vs 13.1 months ($p<.0001$), respectively. There was a higher % of $CD8^+$ T cells closer (<50 μm) to $CXCR4^+$ TANs in standard responders (56% vs 28%, $p<.0001$). There were no significant differences in $CD8^+$ T cell distance to other phenotypes. Longer survival was seen with average cell distance below the mean when comparing $CD8^+$ T cells to FAK^+ tumor cells (1274 d vs 742 d, p=.306), to $CXCR4^+$ TANs (854 d vs 442 d, p=.947) and to FAP^+FAK^+ CAFs (890.5 d vs 672 d, p=.662). In contrast, longer survival was seen with a distance above the mean comparing $CD8^+$ T cells to M2 (622 d vs 1274 d, p=.732) and to M1 (613 d vs 1397 d, p=.517) macrophages. **Conclusions:** Our study shows clinical outcomes may be correlated with spatial distance among TME immune cells in PDAC. Longer survival was seen when $CD8^+$ T cells were closer to FAK^+ tumor cells, $CXCR4^+$ TANs, and CAFs and further away from M2 and M1 macrophages. Larger cohort analyses may better elucidate the prognostic and biological significance of cell distance among TME phenotypes, which may have potential implications for response to immunotherapy. Research Sponsor: None.

Machine learning–based liquid biomarker analysis from the NEONAX trial for response prediction to perioperative (PO) and adjuvant (A) gemcitabine/nab-paclitaxel in resectable PDAC (rPDAC).

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Background: The phase II NEONAX trial examined PO and A gemcitabine/nab-paclitaxel (G/nP) efficacy for rPDAC pts with a comprehensive biomarker program. This translational study aimed at identifying liquid biomarker for early prediction of G/nP success in each treatment group (PO and A) and for the combined group using machine learning (ML). **Methods:** Pts from both study arms were stratified into groups based on Short- or Long-DFS (n=80 pts total; PO n=42 pts; A n=38 pts). Blood plasma from selected G/nP naïve pts was analyzed by multiplexed ELISA (mELISA, 80-Plex ProcartaPlex Human Immune Response) and mass spectrometry (MS; feature list generation) and clinical data were acquired for each patient. For ML pts were divided into training (80%) and validation (20%) datasets. The Weka-based algorithm WrapperSubsetEval (WSE) with 8 different classifiers was used for feature selection. The best performing (highest accuracy, best ROC-AUC, minimal signature) classifier with the corresponding feature panel was selected by a 10x10-fold cross-validation (CV). Respective panels for all features combined as compared to clinical features and Ca19-9 blood levels were tested for performance via CV and bootstrap aggregating for training and validation datasets (10x with replacement). **Results:** The feature generation process generated 579 features from G/nP-naive pts (mELISA: 80; clinical data: 99; MS: 400). For response prediction of the whole rPDAC group to G/nP (S-/L-DFS), the ML-based flow identified a panel of 8 proteins from MS (SERPINA1, C1QB, KRT1, C4B, UBB, VCAM1, HPD, LYZ) and 3 proteins from mELISA (Galectin3, IL34, CCL4) combined with a logistic regression classifier. The predictive panel with the best performance for determining PO G/nP success was established using a kernel logistic regression classifier and included 2 proteins from mELISA (CXCL2, IL17A), 4 proteins from MS (C3, IGLV3-25, HLA-B, SHBG), and 2 clinical data (WHO grade, Na/Mg blood level ratio). The best performing biomarker for predicting success in A included 2 clinical data (tumor size at staging, hematocrit) and 1 protein from mELISA (IL22) and was established with a random forest classifier. All panels represented minimal feature signatures (rPDAC: 11; PO: 8; A; 3) and showed a high performance with ROC-AUC (training) > 0.90, ROC-AUC (CV) > 0.85, and ROC-AUC (validation) > 0.90. All panels described were superior compared to similar predictive signatures (with corresponding ML classifiers) for all clinical data or Ca19-9 blood levels. **Conclusions:** We show that minimal liquid biomarker signatures for early prediction of G/nP success in the NEONAX trial based on mELISA, MS and clinical data can be established by ML. The study shows the potential of ML for biomarker panel development and the value of mELISA/MS for generation of feature lists to use in ML. Clinical trial information: NCT02047513. Research Sponsor: Celgene/Bristol-Myers Squibb.

Impact of pancreatic tumour biomechanics on diagnosis and post-operative complications.

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Background: Around 600 people are diagnosed with pancreatic cancer in Ireland each year, with 90% of these having pancreatic ductal adenocarcinoma (PDAC). Five-year survival for PDAC is around 10%. PDAC upregulates stromal collagen formation which contributes to tissue stiffness. Stiffer pancreatic tumours have worse oncological outcomes due to faster progression and increased chemoresistance. Softer pancreata, conversely, are more likely to develop post-operative pancreatic fistulae (POPF), a significant post-pancreatectomy complication. While incorporated in risk scores, softness is currently determined by subjective intra-operative observation. **Methods:** Atomic force microscopy (AFM) is the gold standard technique used to measure biomechanics at a cellular length scale. We obtained fine needle biopsies (FNBs) of pancreatic tumours during diagnostic endoscopic ultrasound. This is the first time tissue biomechanics have been measured from pancreatic biopsies. Separately, we applied this technique to pancreatic tumours and adjacent normal tissue following resection. We also assessed collagen organisation in these pancreatic tumours to understand its contribution to the observed mechanical properties. **Results:** Of the patients with PDAC who underwent FNB, those presenting with metastatic disease had significantly stiffer tumours than those without, 1,490 Pa (SD 2,048.03 Pa) vs 250 Pa (SD 555.06 Pa) ($p<0.0001$). Neuroendocrine tumours exhibited larger variance compared to PDAC ($p < 0.0001$), SD 5,885.97 Pa vs SD 1,797.23 Pa. Regarding tumours which have undergone curative resection, tumour stroma is significantly stiffer than both cancer and benign epithelium. The collagen strands in this stroma are straighter, more numerous, and more packed than epithelial tissues. Pancreata which develop post-operative pancreatic fistulae (POPF) are softer than those which do not 96.42 Pa (SD 691.14 Pa) vs 343.99 Pa (SD 1,678.37 Pa). **Conclusions:** We describe a novel technique to measure the biomechanical properties of pancreatic biopsies. Using this technique, we now have the opportunity to measure these properties for the first time on all pancreatic tumours, including those presenting with unresectable disease. It is possible to distinguish between different types of tumour based on the biomechanical profile of biopsies. Previous ex vivo studies have linked stiffer tumours to a higher metastatic potential, and our results are the first to suggest that this may be true in a human population. In resected tumours, we confirm the anecdotal belief that softer pancreata are associated with POPF. Overall, this research demonstrates that tissue biomechanics has potential diagnostic and prognostic applications in pancreatic cancer. Research Sponsor: Irish Research Council; Breakthrough Cancer Research; St Vincent's Foundation.

757 Poster Session

Potential effect of the gut microbiome on tumour response to anti-cancer treatment in patients diagnosed with pancreatic ductal adenocarcinoma.

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Background: Chemotherapy is the standard of care for pancreatic ductal adenocarcinoma (PDAC), however, recent research has identified bacteria in the human gut microbiome that may influence patient response to chemotherapy. These bacteria are also present in faeces, and are thought to be representative of the bacteria in the pancreatic ductal system. Preliminary studies have suggested that patients with more advanced PDAC demonstrate lower levels of Firmicutes, and higher levels of Proteobacteria. Additionally, research has shown bacteria in the Firmicutes phylum possess several anti-cancer properties, which may improve patient treatment response, meanwhile, Gammaproteobacteria (Proteobacteria phylum), have been shown to attenuate gemcitabine efficacy and worsen treatment response. **Methods:** This prospective study involved 14 participants with locally advanced PDAC. Participants received combinations of Gemcitabine, Nab-Paclitaxel, and durvalumab (immunotherapy). Participant tumours were assessed every 8 weeks using radiological imaging according to RECIST v1.1, and serum CA19-9 levels were analysed monthly. Gut microbial profiling was completed on stool samples taken pre-treatment using PacBio HiFi full-length 16S rRNA sequencing. **Results:** 16S rRNA sequencing identified that of the 4 patients that died due to disease progression, 3 patients had a low Firmicutes abundance, and a high Proteobacteria abundance. The rother patient demonstrated low levels of Actinobacteriota and Bacteroidota, and a low bacterial diversity. One participant showed a significant increase in CA19-9 (1066.7%), and exhibited a low abundance of Actinobacteriota and Bacteroidota, a high abundance of Proteobacteria, and a low diversity. 5 participants demonstrated a significant treatment response (according to CA19-9 and RECIST analysis), with 4 patients showing a high Firmicutes abundance, high levels of Actinobacteriota or Bacteroidota, and low levels of Proteobacteria. The remaining patient demonstrated a low abundance of Firmicutes and Proteobacteria, yet, displayed the highest abundance of Actinobacteriota and Bacteroidota. **Conclusions:** This exploratory study suggests an association between microbial composition and treatment response, with Firmicutes potentially being associated with an improved treatment response, and Proteobacteria with a worsened response. We believe this study to be one of the first to explore correlation of microbiome with clinical treatment response in PDAC. Research Sponsor: WARPNINE.

Genomic analysis of paired primary and metastatic pancreatic ductal adenocarcinoma tumors.

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Background: The genomics of pancreatic ductal adenocarcinoma (PDAC) have been increasingly understood allowing for improvements in treatment, yet the clinical relevance of PDAC evolution is less well studied. The aim of this study is to evaluate genomic differences and clinical correlations between primary pancreatic tumors and paired metastases using a targeted sequencing panel. **Methods:** Patients who underwent surgical resection for PDAC with recurrent disease who had targeted genomic sequencing of both the primary tumor and a metastasis were included. Genomic alterations were determined using a panel of cancer-related genes (MSK-IMPACT). OncoKB was used to annotate driver alterations. MATH scores were used to quantify intratumoral heterogeneity. Comparison between paired samples was performed and association with clinical variables was assessed. **Results:** There were 55 patients included; 20% (n=11) received neoadjuvant therapy, 91% (n=50) had adjuvant therapy and 36% (n=20) had adjuvant radiation. The most common sites of metastasis sequenced were lung (36%, n=20), liver (35%, n=19) and local (13%, n=7). There was concordance between the first site of recurrence and site sequenced in 93% (n=51) of cases with a median time between resection and biopsy of recurrence of 25 months. The most commonly altered genes are *KRAS* (96%), *TP53* (83%), *CDKN2A* (40%), *SMAD4* (25%) and *RNF43* (13%). The majority of alterations are shared between the primary tumor and metastasis. Clonal mutations were defined as cancer cell fraction (CCF) >0.8. The majority of shared mutations are clonal; 52% of mutations are clonal in both the primary and metastasis, and clonality is preserved in 76% of shared mutations in metastases. The most commonly shared alterations are in driver genes and the majority are clonal- *TP53* (80%), *KRAS* (78%), *CDKN2A* (77%) and *SMAD4* (64%). The proportion of clonal mutations among drivers is higher than variants of unknown significance (p=0.005). There are 31 mutations private to the metastasis in 19 patients, 12 of which are clonal (39%). Mutations private to the metastasis are more common in patients who received adjuvant systemic therapy (p=0.035); however this was not seen when comparing patients who received adjuvant radiation to those who did not (p=0.38). MATH scores are higher in the metastasis as compared to the primary in all patients (p=0.0096), including those who received adjuvant treatment (p=0.002), indicating increased intratumoral heterogeneity in metastases. **Conclusions:** A targeted sequencing panel demonstrates that the majority of driver mutations are clonal and preserved between the primary tumor and metastasis. Additional private mutations and increased intratumoral heterogeneity are noted in metastasis, and our data suggests that this may be associated with exposure to systemic therapy. Research Sponsor: None.

759 Poster Session

MUC5AC levels as predictors of adjuvant therapy outcomes in pancreatic ductal adenocarcinoma: A study of post-surgical gemcitabine-based regimens.

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Background: The role of tissue mucin 5 AC (MUC5AC) in PDA is not fully understood. In our previous work we showed that mature MUC5AC (MM) detected in the extracellular (EC) space and in apical region has better prognostic value than perinuclear immature MUC5AC (IM). Our previous research demonstrated that mature MUC5AC (MM), present in the extracellular (EC) space and apical region, offers better prognostic value than perinuclear immature MUC5AC (IM). Preclinical studies suggest that MM may confer chemoresistance to gemcitabine (Gem), commonly used in adjuvant therapy (AT) either alone or with capecitabine (Cap) or nab-paclitaxel (NP) in patients not eligible for FOLFIRINOX. This study investigates MM's effect on outcomes in PDA patients undergoing post-surgical treatment with Gem-only, Gem-NP, and Gem-Cap. **Methods:** Formalin-fixed, paraffin-embedded tissue samples from resected PDA cases (January 2010 - June 2021) were sourced from the Ohio State University biorepository. Immunohistochemistry using monoclonal antibodies 45M1 (for MM) and CLH2 (for IM) was employed to examine cellular MM and IM expression, with H-scores calculated. Statistical analysis of progression-free survival (PFS) and overall survival (OS) was done using the log-rank test. **Results:** We had 49 patients available for analysis. The median age of the group was 66 years (range: 38 to 89) with majority Caucasians (90%) and 53% (n=26) females. AT distribution within this group was, 31 Gem-only, 7 Gem-NP, and 11 Gem and capecitabine (Cap). Median expression levels for MM and IM were both 150, and 82% (n=40) were EC-MM positive. Using the median MM H-score as a threshold (<150 vs. ≥ 150), outcomes were evaluated within each therapy subgroup (Table). **Conclusions:** This preliminary study suggests that PDA patients with low MM expression may benefit from Gem-Cap therapy. However, larger, prospective studies are needed for definitive conclusions. Research Sponsor: IRP, Pelotonia.

Comparing outcomes in the sub-groups.

Sub-group (n)	OS in days (p-value)	PFS in days (p-value)
MM H-score < 150 vs. ≥ 150		
Gem-only (11 vs. 20)	775 vs. 1006 (0.4)	444 vs. 413 (0.4)
Gem NP (4 vs. 3)	1409 vs. 551 (0.8)	391 vs. 418 (0.5)
Gem-Cap (4 vs. 7)	not evaluable (NE) vs. 766 (0.02)	NE vs. 415 (0.01)

760 **Poster Session**

Spatially resolved protein profiling in pancreatic adenosquamous carcinoma.

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Background: Pancreatic adenosquamous carcinoma (PASC) is a rare histopathological type of pancreatic cancer that is more aggressive than pancreatic ductal adenocarcinoma (PDAC). Although PASC has a unique tumor biology and poorer prognosis, there are no guidelines for differentially treating PASC patients, and its rarity makes it significantly understudied. We used Nanostring GeoMX digital spatial profiling (DSP) to provide a targeted proteomic characterization of key tumor-promoting proteins in PASC and PDAC. **Methods:** FFPE tissue from PASC (n=10 primary, n= 4 met) and PDAC (n=21 primary, n=5 met) cases were analyzed using the DSP platform. Sections were stained with an 82-protein cocktail of antibodies conjugated to UV-photocleavable DNA barcodes. A pan-cytokeratin morphology marker was used to select tumor regions of interest (ROI) and inferred surrounding stromal (i.e., pan-CK negative) ROIs. UV excitation of the defined ROIs was used to release the DNA barcodes and were quantified on the NanoString nCounter system. For PASC, squamous enriched ROI were inferred from corresponding p40 IHC. Corresponding clinical and mutational status for all cases were abstracted from medical records. **Results:** TP53 (PDAC, 80%; PASC, 83%) and KRAS (PDAC, 76%; PASC, 100%) were the most prevalent mutations.In comparing the tumor ROIs of PASC and PDAC (all cases), DSP-assessed global expression and activation of EGFR-MAPK signaling is significantly increased in PASC, whereas PI3K-AKT signaling is significantly decreased. Compared to PDAC, EGFR expression and phosphorylation (p-EGFR), as well as p-MEK1, are significantly increased in PASC. PASC showed significantly lower levels of PLCG1 and INPP4B, while the majority of other AKT signaling effectors were unchanged. For DNA damage response signaling, PASC showed significantly higher levels of p-ATM, p-CHK2, and p21, but not p-ATR. Likewise, expression of oncoproteins c-MYC and BCL6 were significantly increased in PASC (vs. PDAC). **Conclusions:** Given that the normal pancreas lacks squamous cells, their histological presence is a uniquely aberrant feature that can be used to guide deep spatially-resolved interrogation of complex cellular and molecular programs. Our preliminary findings show that regions of squamous cell differentiation within PASC have intrinsically higher levels of active EGFR signaling. With emerging RAS-targeted agents poised to alter the treatment landscape, high EGFR signaling in PASC suggest a possible compensatory resistance mechanism that may require additional targeting. Likewise, reliance on ATM signaling highlights this as preferential target for DDR-directed therapies. Building on these results with additional multiomic assays will inform strategies for improving the management of patients with PASC. Research Sponsor: Knight Cancer Institute; Kuni Foundation.

TP53 Y220C mutations in pancreatic ductal adenocarcinoma (PDAC): A genomic landscape study.

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Background: A novel targeted therapy designed to restore wild-type function to the p53 Y220C mutant protein has demonstrated promising activity in an early-phase trial (NCT04585750). While the *TP53* Y220C mutation is found in < 1% of all solid malignancies, its prevalence is enriched in certain cancers including PDAC. Here we report results of comprehensive genomic profiling (CGP) in a cohort of patients with advanced PDAC, with a focus on *TP53* Y220C mutations. **Methods:** A total of 27,377 cases of clinically advanced PDAC underwent hybrid capture-based CGP to assess all classes of genomic alterations (GAs). Cases were sequenced to a mean coverage depth of 650X. Microsatellite instability (MSI) status and tumor mutation burden (TMB) were determined using sequencing data; PD-L1 expression was measured by immunohistochemistry (Dako 22C3, tumor proportion score [TPS]). **Results:** *TP53* Y220C alterations were identified in 488 (1.8%) PDAC patients (Y220C+). In comparison to PDAC patients lacking the *TP53* Y220C alteration (Y220C-), Y220C+ patients were of similar age, but were more likely to be female (52.9% vs 46.8%; p = 0.0008) and harbor *KRAS* mutations (96.7% vs 92.7%; p = 0.007). Meanwhile, alterations in *ATM* were more frequent in the Y220C- group (4.0% vs 1.4%; p = 0.0008). Frequencies of other GAs were similar between Y220C+ and Y220C- groups, including alterations in the homologous recombination defect-associated genes *BRCA1/2* and *RAD21*, as well as *ERBB2*, *BRAF*, *MTAP* and *PIK3CA*. MSI-high status and elevated TMB (≥ 10 mutations/Mb) were extremely uncommon in both groups. PD-L1 expression was similar in the Y220C+ and Y220C- groups, with positive PD-L1 expression (TPS ≥ 1) observed in 31.5% and 34.9% of patients, respectively. **Conclusions:** The potentially targetable *TP53* Y220C mutation was identified in close to 2% of clinically advanced PDAC cases. Those with *TP53* Y220C mutations were more likely to be female and harbor *KRAS* mutations, and less likely to harbor *ATM* mutations. However, *TP53* Y220C mutations were not associated with other distinct genomic characteristics. These findings are consistent with previous reports of interactions between *TP53* and *KRAS* alterations in PDAC. Given the emergence of drugs aimed at restoring p53 function in cases with the Y220C mutation, further study of this alteration in PDAC is warranted. Research Sponsor: None.

Age, number of genomic alterations (GA) per tumor, and alteration frequency of select commonly altered genes based on *TP53* Y220C status.

	TP53 Y220C+ (n = 488)	*TP53* Y220C- (n = 26,889)	P Value
Median Age, years (range)	66 (36-89+)	66 (21-89+)	NS
GA/tumor	5.2	5.0	NS
KRAS	96.7%	92.7%	0.007
CDKN2A	60.1%	56.6%	NS
CDKN2B	28.6%	29.5%	NS
MTAP	24.4%	23.8%	NS
ARID1A	7.8%	8.7%	NS
ATM	1.4%	4.0%	0.0008
BRCA2	2.3%	3.1%	NS

NS = non-significant.

A real-world prospective analysis of the relationship between nanoliposomal irinotecan and fluorouracil with folinic acid and neutropenia in patients with pancreatic cancer (NAPOLEON-2 study, NN-2401).

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Background: Nanoliposomal irinotecan and fluorouracil with folinic acid (NFF) is a standard regimen after gemcitabine-based chemotherapy (CTx) for unresectable or recurrent pancreatic cancer (urPC) patients. Neutropenia caused by systemic CTx has been reported to be a prognostic marker in several cancers, however, the relationship between neutropenia and efficacy of NFF in urPC patients is unknown. Therefore, we initiated a prospective observational study to investigate whether neutropenia can be a prognostic marker in the real-world (NN-2401), followed by a retrospective one (NN-2301). **Methods:** We collected the data of urPC patients treated with NFF who had received at least one previous CTx in 17 hospitals in Japan from June 2021 to October 2023. The primary endpoint was overall survival (OS). Secondary endpoints included the progression-free survival (PFS) and overall response rate (ORR). To analyze them, we divided urPC patients into three groups by the grade of the most severe neutropenia during NFF (Cutoff A, grade 0 vs 1-4; Cutoff B, grade 0-1 vs 2-4; Cutoff C, grade 0-2 vs 3-4). Besides that, we compared OS among the following three groups receiving NFF ≥4 cycles in Cutoff C; one showed neutropenia with NFF reduction after the 4th cycle (Cutoff C High-Late), another showed neutropenia with NFF reduction in the 2nd or 3rd cycle (Cutoff C High-Early), and the other did grade 0-2 neutropenia (Cutoff C Low). **Results:** Among the 150 patients, the number of patients with the most severe neutropenia of grade 0, 1, 2, 3, or 4 was 79, 7, 23, 34, and 7, respectively. The median OS of severe neutropenia group in Cutoff A, B, or C was 9.8, 9.8, and 10.6 months, respectively. Both OS and PFS were significantly better in all severe neutropenia groups; hazard ratio [HR] of OS for Cutoff A, B, or C was 0.58, 0.54, and 0.60, respectively; HR of PFS was 0.61, 0.56, and 0.56, respectively. Only Cutoff C showed a significant difference with an odds ratio of 3.06 (95% confidence interval [CI], 1.06-8.80; p=0.04) in ORR. Multivariate Cox regression analysis showed significant difference of OS in all Cutoffs; adjusted HR was 0.58 for Cutoff A, 0.53 for Cutoff B, and 0.52 for Cutoff C. Logistic regression analysis showed a significant difference in ORR of Cutoff C with an odds ratio of 4.02 (95%CI, 1.29-12.6; p=0.02). In addition, the numbers of patients were 73, 14, and 19 in Cutoff C Low, High-Late, and High-Early, respectively. The median OS of Cutoff C High-Late was significantly better than that of Cutoff C High-Early or Cutoff C Low (15.0 vs 10.4 or 9.1 months, $p<0.05$). **Conclusions:** In NFF treatment, neutropenia showed the usefulness as a prognostic marker for both OS and PFS, and also a predictive one for tumor response in urPC patients. The results are similar to our previous data, so we conclude that neutropenia can be a useful biomarker of NFF. Clinical trial information: UMIN000043939. Research Sponsor: None.

Relationship of circulating tumor DNA (ctDNA) tumor fraction with prognosis in patients with metastatic pancreatobiliary cancer.

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Background: Carcinomas of the pancreas and biliary tract are among the most lethal cancers. Currently, it is very difficult to predict individual patient outcomes with only general prognostic information available. Emerging evidence shows that baseline ctDNA tumor fraction (TF) is a prognostic factor for many tumor types, including metastatic pancreatic ductal adenocarcinoma (mPDAC). With an additional year of follow-up, we refresh our prior study of the prognostic value of ctDNA TF in mPDAC and extend the analysis to biliary tract cancers (BTC). **Methods:** A genomic dataset comprising patients who underwent ctDNA testing using FoundationOne Liquid or FoundationOne Liquid CDx as part of routine care was used to determine the distribution of ctDNA TF across pancreatobiliary cancer subtypes. A subset of patients was also included in the nationwide (~280 US cancer clinics, ~800 sites of care) de-identified Flatiron Health-Foundation Medicine clinico-genomic database, which was used separately for all other analyses. Clinical data were derived from the electronic health record. Real-world overall survival (rwOS) was evaluated by ctDNA TF while controlling for relevant covariates. In parallel, ctDNA TF cutoffs of ≥1% and ≥10% were evaluated. Exploratory analysis of trichotomized ctDNA TF at 1% and 10% was also performed. **Results:** In 8817 patients with pancreatobiliary carcinoma, 46.1% of PDAC and 41.9% of extra-hepatic cholangiocarcinoma patients had detectable levels of ctDNA TF, while patients with gallbladder carcinomas had the highest levels of ctDNA TF (median 0.8%). 470 patients with mPDAC and 89 patients with BTC were included in the outcomes analysis. High ctDNA TF was associated with clinical features linked to poor prognosis in mPDAC: Stage IV at diagnosis, pretherapy opioid prescriptions, high neutrophil-to-lymphocyte ratio and CA19-9 at baseline, and alterations in *KRAS*, *TP53*, *CDKN2A*, or *SMAD4*. In BTC, high ctDNA TF was associated with Stage IV at diagnosis and cholangiocarcinoma histology. Higher ctDNA TF in mPDAC was associated with significantly reduced rwOS for both cutoffs (1%: HR 1.55 [1.25-1.93], $P < 0.001$; 10%: HR 1.85 [1.42-2.41], $P < 0.001$). When ctDNA TF was trichotomized, higher ctDNA TF groups had reduced rwOS stepwise. Similar trends were observed in patients with BTC, but results were not statistically significant. **Conclusions:** ctDNA from pancreatobiliary carcinomas can be frequently detected and characterized in liquid biopsy while also assessing key genomic biomarkers. ctDNA TF was a prognostic biomarker for mPDAC, but not for BTC in this cohort. Detection of ctDNA associated with other clinical factors linked to poor outcomes. Cohorts given uniform treatment could possibly help further evaluate the ability of ctDNA TF to identify patients with more aggressive disease and inform future studies to personalize therapeutic decision-making. Research Sponsor: None.

764 Poster Session

Natural history study of patients with pancreatic acinar cell carcinoma: A 3-year prospective longitudinal analysis.

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Background: Pancreatic acinar cell carcinoma (PACC) is a rare exocrine tumor of the pancreas that comprises 0.2-2% of all pancreatic malignancies. There are no prospective randomized studies in this disease and most data about PACC comes from retrospective database analyses and case reports. **Methods:** The Cancer Moonshot-funded My Pediatric and Adult Rare Tumor (MyPART) network's Rare Solid Tumor Natural History study (NCT03739827) is a prospective, single-institution observational study that enrolls patients of all ages with rare solid tumors (<15 cases per 100,000 people per year), including PACC. Patients can participate remotely (field cohort) or visit the NIH for annual evaluations (in-person cohort). All participants complete medical and family histories, patient reported outcomes (PROs) measures, and provide saliva for DNA analysis. Relevant data such as clinical and family histories, surgical and molecular pathology, and imaging results are extracted from patient medical records and entered in the central study database. Archival tumor samples (when available) are analyzed using a 500+ gene panel (TruSight500, Illumina) and discussed in a molecular tumor board. Patients participating in the in-person cohort undergo additional evaluations including clinical examination, imaging studies, genetic counseling, and blood sample collection for clinical and research purposes (standard clinical labs, germline DNA/RNA, immune phenotypes, cytokines) as indicated. **Results:** From 09/2021 until 09/2024, 17 PACC patients (n=13 for the field cohort, n=4 for the in-person cohort) were accrued. The mean age of diagnosis is 64.7 (48-78) years, and the male to female ratio is 14:3. At the time of diagnosis, 6 patients had Stage I or II, 1 had Stage III, and 10 had Stage IV disease. The median follow-up time since accrual is 7.6 months, and 4 patients have died. No patients have been lost to follow-up. Of the 6 patients diagnosed with early-stage disease, 5 had surgery and all received adjuvant chemotherapy, while 1 patient is receiving neoadjuvant chemotherapy. Among the 9 patients diagnosed with advanced disease, 8 received platinum-based regimens as a first-line therapy. Tumor molecular profiling was available for 16 patients. The most common tumor mutations observed were in CDKN2A (29.4%), BRCA2 (29.4%), and SMAD4 (23.5%). Six patients received targeted therapies matched to their tumor molecular profile. **Conclusions:** The demographic profile of our cohort aligns with previous studies, with a mean age of diagnosis in the early-to-mid 60s and a high male to female ratio. Our cohort highlights a shift in first-line treatment to platinum-based therapies over the past decade. Many PACC tumors have targetable mutations found in molecular profiling, and the efficacy of matched therapy warrants further exploration. Clinical trial information: NCT03739827. Research Sponsor: NCI Center for Cancer Research.

765 Poster Session

Clinical and genomic features of the morphological subtypes in advanced pancreatic cancer.

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Background: In resected pancreatic cancer (PDA), morphology from routine histopathology slides provides a rapid inexpensive biomarker that predicts overall survival and correlates with transcriptomic subtypes. In this study, we evaluated clinical and genomic associations of morphological subtypes in resected and advanced disease and validated the consistency of subtypes in patient-derived organoids (PDO) and mouse xenografts (PDXs) during *in vitro* and *in vivo* modeling. **Methods:** Our cohort included PDA tumor tissues from 152 resectable (stage I/II) and 228 advanced cases (Stage III/IV). Hematoxylin and eosin-stained slides were blindly reviewed by two pathologists and classified into subtypes based on Kalimuthu (1). Morphological subtypes were correlated with clinical and genomic data from whole-genome and transcriptome sequencing. Histological preparations from PDOs and PDXs obtained from pancreatic resections and metastases were reviewed using the same criteria. **Results:** Morphological subtypes were significantly associated with clinical patterns. Locally advanced PDA exhibited the highest proportion of glandular tumors. Metastatic tumors were enriched for non-glandular morphologies. The non-glandular morphologies were significantly associated with lower survival rates in both resected and advanced tumors. Furthermore, morphological subtypes were significantly associated with unique genomic alterations. Compared to glandular tumors, non-glandular tumors were associated with increased *KRAS* copy number, *KRAS* imbalances, and polyploid genomes. In advanced settings, glandular and non-glandular tumors mostly exhibited classical and basal-like transcriptional subtypes, respectively. Squamous tumors had the highest mutation burden and the highest proportion of Basal A signature (2). Using differential gene expression, we identified transcriptional signatures of the morphological subtypes. PDOs and PDXs maintained the morphological subtypes, although non-glandular tumors had a lower success rate for PDO establishment. **Conclusions:** Morphological subtypes have distinct clinical and genomic associations across all stages of pancreatic cancer, which can be consistently modeled both *in vitro* and *in vivo*. These results demonstrate that morphological subtyping offers a rapid and biologically-relevant classification of PDA that could be used for drug development and stratification to predict therapy selection. 1. Gut; 69: 317-328 (2020). 2. Nat Gen; 52:231-240 (2020). Research Sponsor: None.

Correlation of mesothelin (MSLN) expression measured by RNA sequencing (RNASeq) and immunohistochemistry (IHC) in MSLN-expressing tumors.

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Background: BASECAMP-1 (NCT04981119) is a pre-screening study to identify patients with tumor-associated human leukocyte antigen (HLA)-A*02 loss of heterozygosity (LOH) for interventional studies, such as EVEREST-2 (NCT06051695), a phase 1/2 study of logic-gated chimeric antigen receptor T-cell (CAR T) therapy for MSLN-expressing cancers. MSLN is a cell surface protein expressed in several cancer types, including mesothelioma (MESO), colorectal (CRC), non-small cell lung (NSCLC), ovarian (OVCA), and pancreatic (PANC) cancer, which can be associated with poor prognosis (1). Longitudinal clinical, genomic, and biomarker data were collected from patients enrolled in BASECAMP-1, providing a large dataset for translational discovery and propensity scoring. The aim of this analysis was to determine whether MSLN expression measured by RNAseq and IHC are correlated among patients with solid tumors. **Methods:** In BASECAMP-1, tumor tissue from patients with germline heterozygous HLA-A*02 is tested for HLA-A*02 LOH using an investigational next generation sequencing device codeveloped with Tempus Labs (Lozac'hmeur, et al. *NPJ Precis Oncol.* 2024) that detects somatic alterations, including HLA LOH, and generates RNAseq data (eg, MSLN expression). Gene-level transcripts per million read (TPM) values were determined and the log2 TPM +1 was displayed in a scatter plot (the mean and standard deviation [SD] are provided in the Table). Patients with HLA-A*02 LOH also submit tissue for IHC testing for exploratory biomarkers (eg, MSLN expression using anti-MSLN clone 5B2). Statistical significance was determined with ANOVA or t-tests. **Results:** As of June 1, 2024, 314 patients had been screened for BASECAMP-1 and had RNAseq results; 34 patients also had MSLN IHC results. MSLN expression by RNAseq was consistently higher in patients with PANC or OVCA vs CRC or NSCLC ($P<0.001$). MSLN expression by RNAseq and IHC were correlated overall ($P<0.001$), particularly among patients with PANC and OVCA. Among the 8 patients with OVCA or PANC who were IHC+ for MSLN, only 1 had an RNAseq value below 6 log2 TPM+1; this patient had mesonephric-like ovarian adenocarcinoma, which may account for the low value. The one MESO sample with both RNAseq and IHC available was IHC+ and had a high RNAseq value. **Conclusions:** This analysis demonstrated high correlation between RNAseq and IHC MSLN expression in tumor tissue. 1. Faust, et al. *Cancers.* 2022. Clinical trial information: NCT04981119. Research Sponsor: A2 Bio.

MSLN expression by RNASeq by tumor type and IHC status.

			IHC data available (n=34)			
	Overall (N=314)		MSLN Positive		MSLN Negative	
Tumor	**n**	**Mean TPM[a] (SD)**	**n**	**Mean TPM[a] (SD)**	**n**	**Mean TPM[a] (SD)**
All	314	5.1 (2.3)	20	6.0 (2.1)	14	2.3 (1.7)
adNSCLC	33	3.8 (2.4)	2	4.7 (1.7)	7	1.5 (1.3)
sqNSCLC	4	3.1 (1.8)	0		0	
MESO	4	4.5 (4.2)	1	5.7	0	
CRC	148	4.4 (2.0)	8	5.1 (2.1)	7	3.1 (1.6)
PANC	100	6.1 (1.8)	3	7.8 (0.6)	0	
OVCA	25	7.2 (2.1)	5	6.6 (2.8)	0	

[a]MSLN log2 transcripts per million+1.
ad, adenocarcinoma; sq, squamous.

767 Poster Session

Serum mucin 5AC (MUC5AC) as a predictor of outcomes and microscopic distant metastasis in pancreatic ductal adenocarcinoma (PDA).

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Background: Our prior studies demonstrated that tissue extracellular MUC5AC (EM) detection (EM-D) is associated with early recurrence, and serum MUC5AC (sM) levels have prognostic value in resected PDA following neoadjuvant therapy. This study evaluates the predictive value of post-operative sM before adjuvant therapy (AT). **Methods:** Serum samples, collected from January 2010 to June 2021 and sourced from the Ohio State University biorepository, were analyzed using enzyme-linked immunoassays with a human MUC5AC kit (NBP2-76703). Multivariate (MV) Cox regression models quantified progression-free survival (PFS) and overall survival (OS), while Wilcoxon tests compared group differences. **Results:** Nineteen patients had serum samples available post-operative and pre-AT. The cohort's median age was 65 years (range: 48-83), predominantly Caucasian (n=18), and 58% female. Serum was collected a median of 2 weeks (range: 1-16) after surgery, with a mean sM level of 1.1 ng/mL (range: 0.42 to 3.3). Most patients (17/19) received Gemcitabine-based therapy, one received chemoradiation, and another FOLFIRINOX. Among patients, 3 had no recurrence (Nr), 10 experienced distant metastases only (mets), 3 had local recurrence only (Lr-o), and 3 had both Lr and liver recurrence (Liv/Lr). In the MV model, incorporating sM, EM-D, and AT, higher sM levels (HR=2.83, p=0.04) and negative EM-D (HR=6.85, p=0.02) were associated with increased recurrence risk, whereas AT type did not influence recurrence. Regarding OS, sM was significant (HR=2.95, p=0.04) while other factors were not. Elevated sM levels correlated with metastasis risk (mets vs. Liv/Lr vs. Lr-o vs. Nr = 1.54 vs. 0.61 vs. 0.62 vs. 0.92, p=0.01). Patients with locoregional recurrence had significantly lower sM than those with distant metastasis and no recurrence (mets vs. Liv/Lr+Lr-o vs. Nr = 1.54 vs. 0.62 vs. 0.92, p=0.005). In a separate cohort of 10 metastatic patients (mean sM of 1.8 ng/mL, range: 0.43 to 11.2), liver metastases were linked to significantly lower sM than non-liver metastases (0.54 vs. 3, p=0.02), while peritoneal metastases were associated with higher sM levels (3 vs. 0.6, p=0.04). **Conclusions:** This small study suggests that sM can serve as a biomarker for assessing outcomes and recurrence sites post resection before the initial dose of AT. Elevated sM should prompt suspicion of microscopic metastases, particularly non-liver metastases. Research Sponsor: IRP, Pelotonia.

Predictive methylation signature for gemcitabine sensitivity in pancreatic ductal adenocarcinoma: A pathway to personalized treatment.

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Background: Gemcitabine (Gem) is a key drug used in chemotherapy regimens for treating pancreatic ductal adenocarcinoma (PDA). Identifying a biomarker to predict patient response to Gem could be crucial in personalizing treatment selection between Gemcitabine and fluorouracil (5-FU)-based therapies, thereby improving patient outcomes. We hypothesize that DNA methylation changes in genes linked to Gem resistance may serve as surrogates for the expression of corresponding proteins, thereby predicting treatment outcomes in patients with PDA receiving this therapy. **Methods:** We curated a 93-gene panel through a literature review spanning from January 1970 to April 2024. The proteins encoded by these genes have demonstrated either preclinical evidence (from cell line or mouse models) or clinical evidence (from tissue-based studies) of affecting the response to Gem in PDA. We evaluated the prognostic significance of methylation at specific CpG (cytosine-phosphate-guanine) sites within these genes using data from The Cancer Genome Atlas (TCGA) database, focusing on patients who had received Gem (106 /184 patients). We utilized elastic net multivariate analysis to analyze survival and created Kaplan-Meier plots based on our gene panel to estimate overall survival (OS). We compared RNA and gene expression levels for the clinically significant gene signatures between normal (NT) and PDA tissues using TNMplot.com, a web-based tool developed from data in the Gene Expression Omnibus of the National Center for Biotechnology Information (NCBI-GEO), TCGA, Therapeutically Applicable Research to Generate Effective Treatments (TARGET), and The Genotype-Tissue Expression (GTEx) repositories. This tool employs Mann-Whitney or Kruskal-Wallis tests to determine statistical significance. **Results:** We identified an 8-CpG site methylation signature across eight genes, distinguishing a high-risk subgroup within the Gem-treated cohort in the TCGA database. Patients exhibiting methylation at specific CpG sites in this signature experienced significantly worse OS, with median survival times of 20.35 months compared to 49.38 months (p=0.0068). Additionally, the RNA expression (fold change: 2.11, p=2.12e-55) and gene expression (fold change: 1.42, p=3.14e-18) of these genes were markedly higher in PDA tissues compared to normal tissues. Notably, the candidate genes in our signature do not overlap with those used to classify PDA into classical versus basal subtypes. In an ongoing study, we are also investigating methylation changes of these candidate genes in the blood (cell-free DNA) of PDA patients. **Conclusions:** We present a signature that can predict Gem sensitivity or resistance in patients with PDA. This signature serves as a critical first step toward personalizing treatment strategies. Research Sponsor: None.

769 **Poster Session**

More isn't always better: Expanding and refining prognostic methylation signatures for pancreatic ductal adenocarcinoma.

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Background: In our previous study, we developed protein-informed methylation signatures from a chemoresistance gene panel (RGp, n=122) curated from the literature, identifying 28 signatures with strong prognostic value in The Cancer Genome Atlas (TCGA) pancreatic ductal adenocarcinoma (PDA) database. To enhance the clinical utility of this panel, we incorporated methylation changes from genes with known prognostic (PGp) and diagnostic (DGp) significance in PDA to form a composite gene panel (CGp-PDA) and repeated our analysis. We hypothesized that epigenetic changes in these genes reflect protein expression and that deriving signatures from them could help identify patients at risk of treatment failure. **Methods:** We assembled a composite gene panel (CGp-PDA) of 206 genes (122 RGp + 26 PGp + 82 DGp) through a literature review spanning from January 1970 to April 2024. These PGp and DGp genes are unique, as cell-free DNA methylation changes in them have established prognostic and diagnostic value, respectively in PDA. The same cohort of pancreatic ductal adenocarcinoma (PDA) patients (n=184) from our previous study was analyzed. We applied elastic net multivariate analysis to calculate survival outcomes and generated Kaplan-Meier plots using our gene panel, optimizing values for ? and λ over 100 iterations. **Results:** Our analysis identified 20 signatures with cytosines followed by a guanine residue (CpG) site in genes from our panel. The number of CpG sites in these signatures ranged from 3 to 4248. The most effective signature featured 17 CpG sites (22 months [m] vs. 67m, p=0.03), followed by signatures with 3 (16m vs. 49m, p=0.009) and 12 (17m vs. 36m, p=0.02) CpG sites. The 4248 CpG site signature demonstrated the poorest prognostic value among the signatures (17m vs. 21m, p=0.01). Some signatures included multiple CpG sites within a single gene. Notably, KCNH2 reappeared in several signatures, as in the prior study. Expanding our RGp to CGp-PDA did not produce more (20 vs. 28) signatures or improved prognostic value. **Conclusions:** Targeted methylation signatures, supported by robust preclinical or translational evidence of their impact on treatment resistance, can be clinically valuable for informing treatment selection and prognosis. We may need to shift our focus from machine learning-based broad methylation or transcriptomic analyses to more targeted investigations. Research Sponsor: None.

Investigation of the association of *CTRB2* exon 6 deletion with time to progression and overall survival in pancreatic ductal adenocarcinoma.

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Background: Pancreatic ductal adenocarcinoma (PDAC) is associated with poor prognosis and limited treatment options. A 584 bp deletion in *CTRB2*, which impairs chymotrypsin B2 function, has been linked to increased PDAC risk. This study investigates the impact of this deletion on progression and survival outcomes in patients with PDAC. **Methods:** There were 633 patients in whom *CTRB2ex6* deletion was genotyped; 263 patients received chemotherapy and had time to progression (TTP) information. TTP was calculated from diagnosis date until earliest of 1) progression date (event), 2) surgery, death, or next chemotherapy date (competing risk), or 3) date last known to be alive (censor). Overall survival (OS) was calculated for the cohort from diagnosis until date of death (event) or date last known to be alive (censor). TTP was analyzed using cumulative incidence methods accounting for competing risks; OS was analyzed using Kaplan-Meier methods. Cox proportional hazard models adjusted for age and sex were used to test associations and compute hazard ratios (HR) and 95% confidence intervals (CI). **Results:** Median TTP in those with *CTRB2ex6* deletion negative carrier status (n=209) was 1.3 years vs 5.9 years for those with positive carrier status (n=54). Estimated 3-year progression rates were 54.2% for patients with no *CTRB2ex6* deletions, 43.7% for patients with one deletion (n=48), and 33.3% for patients with two *CTRB2ex6* deletions (n=6). TTP was not associated with *CTRB2* deletion carrier status (HR: 0.8, 95% CI: 0.6-1.3). Median OS for *CTRB2ex6* deletion negative carriers (n=514) was 1.0 year vs 0.9 years for positive carriers (n=119); within deletion carriers, heterozygous carriers (n=110) had a median OS of 0.9 years, and homozygous carriers (n=9) had a median OS of 2.6 years. OS was not associated with *CTRB2* deletion carrier status (HR: 1.0, 95% CI: 0.9-1.3). **Conclusions:** Although we did not detect statistically significant differences in TTP and OS by *CTRB2ex6* deletion carrier status, there was an interesting trend toward longer TTP and OS in the small number of patients who carried homozygous *CTRB2ex6* deletions. Larger studies are warranted to validate these findings. Research Sponsor: None.

p53 pathway genes' mutations as prognostic factors in patients (pts) with metastatic pancreatic ductal adenocarcinoma (mPDAC): A single center study.

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Background: Mutations (muts) in oncosuppressor genes have been associated with poor prognosis in different solid tumors. This study aims to evaluate the prognostic impact of p53 pathway muts in mPDAC pts. **Methods:** For each mPDAC pt enrolled in this study a FFPE tumor tissue specimen was collected and Next Generation Sequencing (NGS) was performed by TSO500HT assay (DNA [523 genes], RNA [55 genes]). Primary endpoint was overall survival (OS). The Kaplan–Meier method was used to estimate efficacy outcome; log-rank test, Peto-Peto test and Cox-regression model were used to compare the differences, considering a statistically significant p value < 0.05. **Results:** A total of 149 mPDAC pts were enrolled; TP53 was mutated in 115/149 (77.2%) pts, CDKN2A in 28 (24.3%), CDKN2B in 8 (5.3%), ATM in 6 (4%), TP53BP1 and MDM2 in 2 pts each (1.3%) and ATR in 1 (0.7%). No mutations were found in MDM4, CHEK1 and CHEK2 genes. Overall, 121 (81.2%) pts had at least one mutation in a gene of p53 signaling pathway (p53sigmut) and 28 pts (19.8%) had none (p53sigwt). The most frequent TP53 alterations were single nucleotide variations (SNVs): R175 (10 pts-8.7%), R248 (9 pts-7.8%) and R273 (8 pts-7%); 27 TP53 muts (18.1%) occurred in exon 7, 26 (17.4%) in exon 5, 19 (12.8%) in exon 8 and 13 (8.7%) in exon 4. Median OS (mOS) was 15 months (mos) (CI 95%: 12.8-18.2) in TP53 mutated pts vs 24.3 mos (CI 95%: 17.3-Not reached) in TP53 wt pts, p=0.03. The site of TP53 alteration was correlated with pts' survival: mOS was 34.9 mos (CI 95%: 31-Nr) in exon 4-mutated pts, 16.9 mos (CI 95%: 11.2-21.2) in exon 7-mut, 13.8 mos (CI 95%: 11.2-17.8) in exon 5-mut and 10.7 mos (CI 95%: 8.2-Nr) in exon 8-mut, p= 0.018. CDKN2A mutations were also associated with survival: mOS was 17.9 mos (CI 95%: 15.9-24.1) in wt pts vs 13.4 (CI 95%: 10.7-22.2) in mutated pts, p= 0.04. CDKN2B, ATM, TP53BP1, MDM2 and ATR mutations were not associated with OS. Overall, patients with at least one mutation in one of the genes of p53 pathway had a mOS of 15 mos (CI 95%: 12.8-18.2) vs 19.6 months (mos) (CI 95%: 17.3-Not reached) in p53sigwt patients. In terms of comutations, TP53 wt pts had a higher frequency of KRAS mutations: 94% vs 64.5%, p<0.001. **Conclusions:** Our study suggests an important prognostic role of p53 signaling muts in mPDAC pts; wt pts resulted in a longer OS than mutated pts, although some mutations seem to be associated with a longer survival. Further studies are needed to investigate these findings, including an in-depth analysis of structural proteogenomic. Research Sponsor: None.

Uncovering the role of oncogenic *HER2* variants as drivers of pancreatic cancer.

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Background: The therapeutic resistance of pancreatic ductal adenocarcinoma (PDAC) to conventional cytotoxic therapy underscores the need for advances in targeted molecular therapeutics. Previous studies have reported approximately 2% of PDAC patients with *HER2* gene amplification. While anti-*HER2* cancer therapy has yielded success in other disease sites, it has had minimal reported benefit in PDAC. Multiple *HER2* isoforms exist that are generated either by loss of exon 16 (*d16 HER2*), or through N-terminal truncations (*p95 HER2*). These isoforms are implicated in differential tumor behavior and response to anti-HER2 therapy in the breast but have not been investigated in PDAC. Herein, we hypothesize that these isoforms may underlie the observed resistance to *HER2* therapy in PDAC. **Methods:** To test for the presence of HER2 structural variants within human PDAC, a tissue microarray was constructed of fifty-one primary tumor samples and two metastatic samples. Multiplex immunohistochemistry (mIHC) was performed for staining of the extracellular N-terminal and intracellular C-terminal HER2 domains. We then utilized our previously described *HER2+* cancer rainbow (Crainbow) mouse model in which human wild-type *HER2* (*WT HER2*), *d16 HER2*, and *p95 HER2* were expressed in the same mouse along with fluorescent protein reporters. *HER2* Crainbow mice were crossed with PDX1-Cre recombinase to initiate recombination and expression of each fluorescently labelled *HER2* isoform during pancreas development. PDX1-Cre / *HER2* Crainbow mice were followed for up to one year and sacrificed to lineage trace each isoform and register *HER2* lineages with histopathology. **Results:** In our human data, four of fifty-three patient samples (three primary and one metastatic) had moderate-to-strong *HER2* staining. All samples identified on mIHC as *HER2* positive stained primarily for the *p95* variant of *HER2*, with absence of the N-terminal domain and preservation of the intracellular domain. In our transgenic mice, 65% of the mouse cohort developed pre-malignant low grade pancreatic intraepithelial neoplasia (PanIN) by eight weeks. At one year, 58.3% of mice demonstrated advanced high grade PanIN, and 25% developed invasive disease. Lineage tracing revealed that while early low grade PanIN began as polyclonal lesions, comprised of *WT*-, *d16*-, and *p95 HER2* expressing cells, advanced PanIN and invasive disease were unanimously monoclonal with one dominant isoform arising. *WT HER2* was represented abundantly in low grade disease, but high grade and invasive disease were predominantly driven by *d16*- and *p95-HER2.* **Conclusions:** Our data implicates the oncogenic *d16*- and *p95 HER2* variants, rather than *WT HER2*, as the primary drivers of *HER2+* PDAC tumorigenesis. In breast *p95 HER2* has been described as a known cause of Trastuzumab resistance. Therefore, future efforts will be aimed at characterizing the mechanisms driving therapeutic resistance in *p95-HER2* PDAC tumors. Research Sponsor: None.

Analysis of serum and urine miRNAs as biomarkers of cancer cachexia in pancreatic cancer.

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Background: Cancer cachexia (CC), characterized by anorexia, weight loss and skeletal muscle wasting, is a common complication in patients (pts) with pancreatic cancer (PC). The association of CC with treatment-related toxicity and decreased quality of life has been reported. Although CC is believed to result from systemic inflammation and cancer-induced metabolic changes, the underlying mechanisms remain poorly understood. Early diagnosis of CC is particularly challenging, highlighting the need for reliable biomarkers. We analyzed microRNAs (miRNAs) in serum and urine as potential biomarkers for the early detection of CC in PC. **Methods:** Clinical information, along with serum and urine samples were collected from pts with PC treated at the National Cancer Center Hospital, Tokyo, from March 2022 to March 2024. CC diagnosis according to Fearon et al: (1) ≥5% weight loss in 6 months, (2) ≥2% weight loss in individuals with a BMI of <20 kg/m^2, or (3) ≥2% weight loss in pts with sarcopenia. In this study, pts were considered to have CC if they met the definition of CC at enrollment or within six months. Serum and urine extracellular vesicle (EV) miRNA profiles were sequenced by NGS and compared by means of counts per million (CPM). Differential expression of miRNAs between pts with and without CC was investigated by DEseq2. **Results:** Of the 84 enrolled cases, 73 had sufficient sample volume for NGS assay. After excluding 4 cases with low miRNA read counts and 10 cases with unknown CC status, 59 cases were analyzed. CC was present in 41 pts (69%). Patients' background was as follows (CC/no CC); median age 70 [IQR:62–75]/ 67 [63–69], male 21/9 pts, cStage {I (7/0), II (1/0), III (18/5), IV (15/13)} pts, median BMI (21.7 [IQR:19.6–24.0]/ 21.8 [20.7–22.4]) kg/m2, respectively. There were 141 common miRNA species between serum and urine samples, 81 miRNAs unique to serum and 33 unique to urine. The correlation between the common miRNAs was moderate (R = 0.60). No significant differences in miRNA expression were observed between pts with and without CC in serum samples. In urine samples, miR-141-5p was significantly downregulated (log2FoldChange = -0.667, padj = 0.0463), while miR-146b-5p was significantly upregulated (log2FoldChange = 1.03, padj = 0.0463) in pts with CC compared to those without. **Conclusions:** The results of this study suggest that urine is a suitable biofluid for detecting CC in patients with PC, as differentially expressed miRNAs were not observed in serum. However, the mechanism underlying the superiority of urine over serum in detecting CC remains unclear and should be addressed in future studies. Research Sponsor: None.

Personalized ctDNA monitoring to assess the recurrence of early stage pancreatic cancer after surgery.

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Background: Up to 80% of patients with resected pancreatic cancer recur within 2 years. Here, we assessed the feasibility and accuracy of a personalized, tumor-informed circulating tumor DNA (ctDNA) test for early warning of recurrence risk during long-term surveillance. **Methods:** This study retrospectively analyzed 43 patients who underwent radical surgery for pancreatic cancer between November 2021 and June 2023 at the Comprehensive Cancer Centre of Drum Tower Hospital, Clinical Cancer Institute of Nanjing University. A personalised panel was constructed to detect ctDNA in plasma based on whole-exon mutation information from tumour tissue. A total of 139 ctDNA samples were obtained from periods including post-operative, during adjuvant therapy, and post-treatment. **Results:** Until the last follow-up time (May 13, 2024), a personalised ctDNA monitoring panel was successfully constructed for 35 (81.4%) stage I-III patients with a median follow-up of 11.82 months (range: 2-17.8 months). A total of 16 patients experienced recurrence within 15.7 months (range: 5.4-30 months), with a median lead time from first positive ctDNA to imaging recurrence of 3.6 months (range: 1-9.3 months). ctDNA positivity was detected in 25.8% (8/31) of ctDNAs at the landmark time point (postoperative prior to adjuvant therapy), with a positive predictive value (PPV) of 75% (6/8) and a negative predictive value (NPV) of 69.6% (16/23). Patients with positive ctDNA at the landmark time point had inferior DFS (6.9 months vs undefined. HR 3.955, P=0.007) and OS (15 months vs undefined. HR 5.599, P=0.001) compared to those with negative ctDNA. When integrating longitudinal time points, PPV and NPV were 68.2% (15/22) and 91.7% (11/12), respectively. For all 16 relapsed patients, the sensitivity of positive ctDNA detected longitudinally was 93.75% (15/16). The ctDNA-MRD status at longitudinal time points had a more significant difference with DFS (undefined vs 12.6 months. HR 5.096, P=0.002) and OS (undefined vs 21 months. HR 6.487, P=0.038). **Conclusions:** Personalized ctDNA surveillance is effective in identifying high recurrence risk after pancreatic surgical resection. Long-term longitudinal warning of arbitrary ctDNA positivity suggests a recurrence probability of up to 93.75%; therefore, full ctDNA dynamic monitoring is essential. And ctDNA monitoring will be further used to guide the decision-making in the clinic to prolong patient survival. Research Sponsor: None.

775 Poster Session

Analysis of lipogenic and metabolic subtypes and survival outcome in patients with pancreatic adenocarcinoma.

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Background: Delayed diagnosis and rapid progression are major drivers of poor survival outcomes for Pancreatic Ductal Adenocarcinoma (PDAC). In previous studies, molecular profiling in tumor tissue has identified Cholesterogenic, Classical, or Exocrine-Like pathological subtypes associated with improved prognosis. Herein, in a clinical trial called Project Survival, we molecularly and clinically characterized a prospective PDAC biomarker cohort. This multicenter (n=6) clinical trial (NCT 02781012) of PDAC combined with high-fidelity longitudinal phenotypic characterization and multi-omic profiling was utilized to identify biomarkers with diagnostic and therapeutic utility. The study included assessment of plasma, serum, buffy coat, urine, saliva, and tumor tissue samples and followed 269 PDAC subjects for up to 7 years (Median survival was 700 days, with IQR of 822 days). **Methods:** We assessed plasma samples to identify circulating biomarkers using proteomics, metabolomics, and lipidomics profiling. Multi-omic comparison of patients with Overall Survival (OS) above the 75th percentile (improved survival) against those with OS below the 25th percentile (poor survival) was performed. **Results:** We identified distinct panels of proteins, metabolites, and lipids that were associated with patient survival outcomes. Protein biomarkers that were increased in improved survival outcomes were associated with cholesterol metabolism (7/14, p-value = 0.00085) and glycolysis/gluconeogenesis (5/12, p-value = 0.00089), however, protein markers associated with immune surveillance and inflammation were associated with poorer survival. For metabolite biomarkers, 19 metabolites were increased in poorer survivors of which 13/19 were carnitine metabolites, and 42 metabolites were increased in patients with improved survival. Interestingly, only two lipid molecular species of the diacylglyceride family were increased in patients with poorer survival. **Conclusions:** Project Survival, a prospective biomarker-driven clinical trial identified distinct metabolic and molecular subtypes of patients which characterize survival outcome in PDAC. These subtypes go beyond tumor genomic characterization and align with distinct cholesterogenic and glucogenic phenotypes associated with survival outcomes, which can potentially be leveraged for therapeutic intervention strategies. Research Sponsor: None.

Validation of PDACai v2.0 in predicting relative benefit from frontline FOLFIRINOX (FFX) and gemcitabine/nab-paclitaxel (GA) for patients (pts) with metastatic pancreatic cancer (mPDAC).

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Background: Novel biomarkers beyond genomic alterations in DDR pathway genes (e.g. *BRCA1/2*) have the potential to optimize frontline and subsequent treatment decisions for mPDAC. This real-world evidence (RWE) study validates PDACai v2.0, a machine learning framework that leverages clinical NGS data to predict progression-free survival (PFS) on FFX vs GA in individual pts with mPDAC. **Methods:** We analyzed outcomes from 774 pts with mPDAC who underwent genomic profiling via the Know Your Tumor program or were referred to Perthera by treating oncologists. Independent training and validation cohorts receiving either 1st line FFX or GA were split (60:40). PDACai v2.0 integrates an updated set of clinical (age < 63, sex) and lab-agnostic molecular features derived from genomic testing reports (specific *KRAS* variants, *TP53* GOF vs LOF, DDR pathway redefined). Relative benefit scores predicted by FFX or GA models were evenly binned into three PDACai signature categories representing lower, middle, and upper tertiles for each independent cohort. Statistical differences in median PFS were evaluated using ordinal Cox regression. **Results:** Median PFS followed PDACai predicted trends for each therapy's training and validation cohorts. The predictive utility of PDACai was confirmed in the independent validation cohorts when comparing PFS on FFX (p = 0.01637; HR = 0.73 [95% CI: 0.56-0.94]) and GA (p = 0.0007506; HR = 0.62 [95% CI: 0.47-0.82]) across tertiles. **Conclusions:** Using RWE, the PDACai v2.0 signature successfully predicted PFS differences in pts treated with 1st line FFX or GA in mPDAC. Prospective validation efforts and additional data-driven insights into 2nd line PFS on FOLFOX vs 5FU/nal-irinotecan are underway. Research Sponsor: Pancreatic Cancer Action Network.

Summary of actual PFS in months on 1st line therapies in pts assigned to lower, middle, and upper thirds based on relative PDACai v2.0 scores.

Independent Cohort (# Pts)	Lower TertilemPFS [95% CI]	Middle TertilemPFS [95% CI]	Upper TertilemPFS [95% CI]
1st line FFX Training (243)	6.5 [5.5-8.1]	9.3 [6.5-13.3]	15.8 [14.1-N/R]
1st line FFX Validation (165)	9.4 [7-13.3]	8.6 [5.6-11.4]	13.2 [9.9-39.6]
1st line GA Training (218)	5.6 [4.7-6.3]	6.9 [5.2-8.2]	11.2 [8.8-13.4]
1st line GA Validation (148)	5.6 [4.2-8.4]	7.5 [6.5-13.1]	8.5 [7.7-11.5]

Frequency and incidence of oncogenic *RAS* mutations in patients with metastatic pancreatic ductal adenocarcinoma: Derived from the real-world evidence database Foundation Medicine Insights.

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Background: Approximately 60,000 patients are diagnosed with pancreatic cancer in the United States annually, with half of all cases diagnosed at the metastatic stage.[1] The most common type, pancreatic ductal adenocarcinoma (PDAC), accounts for ~90% of all pancreatic cancers[2] and is characterized by frequent *KRAS* oncogenic mutations. Newer investigational therapeutics target HRAS and NRAS, in addition to KRAS. To comprehensively characterize the incidence of RAS mutations in patients with PDAC, we analyzed mutational frequency data from the Foundation Medicine Insights (FMI) database. **Methods:** NGS-based comprehensive genomic profiling (CGP) data from 27,298 PDAC tissue samples were accessed from the FMI database (May 7, 2024 data cut). Only 'short variants' in RAS genes (*KRAS*, *NRAS*, and *HRAS)* occurring at amino acid positions G12, G13, or Q61 that had a frequency 0.1% or greater were included. To estimate the number of new US cases of *RAS*-mutant PDAC, pancreatic cancer incidence estimates[3] were adjusted by the derived *RAS* mutational frequency values. All incidence estimates were rounded to the nearest 100. **Results:** Oncogenic *RAS* mutations were identified in 92% of PDAC patients, with nearly all being *KRAS* (91.8%) and the remaining *NRAS* (0.2%). No *HRAS* mutations passed our minimum frequency threshold requirement. Individual *RAS* mutations with frequency >1% included *KRAS* G12D (39.9%), *KRAS* G12V (28.6%), *KRAS* G12R (14.6%), *KRAS* Q61H (4.2%), *KRAS* G12C (1.7%), and *KRAS* Q61R (1.4%). Applying these *RAS* mutation rates to reported PDAC incidence, we estimate that ~56,000 new patients with PDAC in USA per year will have tumors harboring an oncogenic *RAS* mutation, including ~51,000 of those with a *RAS* G12 mutation specifically. **Conclusions:** Oncogenic *RAS* mutations are identified from NGS testing in nearly all patients with advanced PDAC. Both the frequency and diversity of RAS mutations in PDAC suggest that PDAC is largely a RAS-driven cancer and that targeting RAS may be useful broadly in this disease. Research Sponsor: None.

Mutational frequency and incidence at G12, G13, and Q61 amino acid positions in *KRAS* and *NRAS* among patients with PDAC.

		RAS G12	*RAS* G13	*RAS* Q61	*RAS* mutant
Mutational Frequency	KRAS	85.7%	0.8%	6.3%	91.8%
	NRAS	0.1%	Below threshold	0.1%	0.2%
Incidence	KRAS	51,300	600	3,700	55,600
	NRAS	100	Below threshold	100	200

The effect of cachexia and anti-cancer therapies on GDF-15 levels in patients with pancreatic adenocarcinoma.

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Background: Growth differentiation factor 15 (GDF15) is a novel biomarker in the setting of pancreatic adenocarcinoma (PDAC). Preliminary studies have found that downstream effects of GDF-15 promote the progression of pancreatic cancer. PDAC has a high prevalence of cachexia which is commonly treated with enzyme supplementation and nutritional support. Platinum-based therapies often used to treat PDAC increase the likelihood of patients experiencing cachexia. The purpose of this study was to observe the GDF-15 levels in respect to weight change of PDAC patients being treated with cachexia and anti-cancer therapies. **Methods:** Serum and Plasma samples were collected from locally advanced and advanced PDAC patients on 5 IRB approved studies from 2014-2024. Samples were analyzed using the R-PLEX Human GDF-15 Antibody Set. The study objective was compared using two-sample paired t-tests and Pearson correlation matrices. Elevated GDF-15 was determined as >1500 pg/mL. **Results:** Samples from 73 patients were utilized for this study (63% female; median age: 69; BMI: 23.31±4.96 kg/m^2). At baseline 71.8% (46/64) had elevated GDF-15, at Cycle 3 76% (25/33), and at Cycle 5 86% (25/29). In the total sample set, there was a slight negative correlation in the change of weight (-1.75 kg, $p = 0.013$) and GDF-15 (577.5 pg/mL, $p = 0.278$) from C1 to C3 ($r = -0.185$, $p = 0.366$). There was a slight negative correlation ($r = -0.44$, $p = 0.076$) in the difference of weight and GDF-15 from C1 to C3 in patients receiving chemotherapy. A strong negative correlation was observed in the change of GDF15 and weight from C1D1 to C1D8 in patients treated with platinum-based therapy ($r = -0.530$, $p = 0.212$) and a weak negative correlation ($r = -0.421$, $p = 0.065$) for patients with non-platinum-based therapy. There was a strong positive association ($r = 0.993$, $p < 0.01$) between the change in weight and GDF-15 from C1 to C5 in patients not receiving chemotherapy. There was no significant decrease in GDF-15 levels across the patients at all time points. **Conclusions:** Despite cachexia and anti-cancer treatment, the GDF-15 levels of PDAC patients continue to remain elevated in consecutive blood draws. Further research in GDF-15 kinetics is warranted. Clinical trial information: NCT02400398, NCT03207724, NCT04098237, NCT05336266, NCT04634539. Research Sponsor: None.

Weight and GDF-15 levels across different chemotherapy regimens.

	Baseline	Cycle 1		Cycle 3		Cycle 5	
	BMI (kg/m2)	Weight (kg)	GDF15 (pg/mL)	Weight (kg)	GDF15 (pg/mL)	Weight (kg)	GDF15 (pg/mL)
Chemotherapy	23.21	63.83	3243.25	63.44	3498.76	60.14	3201.33
No Chemotherapy	23.77	65.85	3113.73	62.32	2949.56	71.05	2944.54
Platinum Based	20.88	59.89	6527.78	51.7	3600.87	58.1	5152.50
Non-platinum-based chemotherapy	23.86	64.89	2442.14	64.67	3487.41	60.39	2957.43

779 Poster Session

Correlation of tumoral collagen width with overall survival and tumoral collagen straightness with distant recurrence-free survival in patients with pancreatic ductal adenocarcinoma.

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Background: Quantitative measurements of fibrillar collagen characteristics has been shown to correlate with survival in several types of malignancy. The aim of this study was to determine whether these characteristics correlate with survival in pancreatic ductal adenocarcinoma. **Methods:** Patients who underwent pancreatic resection for pancreatic ductal adenocarcinoma at an Allegheny Health Network hospital between 1/1/2019 and 12/31/2023, and for whom archived formalin-fixed paraffin-embedded tumor tissue was available, were included in the study. Clinical and pathologic characteristics were collected from the patient's electronic medical record. Survival data included overall survival (OS), local recurrence-free survival (RFS), and distant RFS. One section of tumor per patient was stained with Picrosirius Red and a representative digital image was obtained with a camera attached to a microscope at 100x magnification. Each image was analyzed with CT-FIRE software to determine averages of collagen fiber width, length, straightness, and angle. Descriptive statistics were initially computed. Log-rank tests were also conducted to compare the OS rate and RFS rate between four different groups (width, length, straightness, angle). Each group was divided into high and low groups, using the mean as cutoff. In the end, multivariate Cox regression was conducted by controlling the confounders including sex, perineural invasion, lymphovascular invasion, and presence of lymph node metastasis to examine the hazard ratios of OS, local RFS, and distant RFS for the four collagen fiber metrics. All statistical analyses were performed via SAS 9.4 under alpha level .05. **Results:** One hundred three patients met the inclusion criteria. Based on the results of log-rank test, OS was significantly different between high and low collagen width groups, p = .0251. The lower width group had higher survival rate than the upper width group. Also, distant recurrence-free survival was significantly different between high and low collagen straightness groups, p = .0053. The lower straightness group had higher distant RFS rate than upper group. The results of multivariate Cox regression were consistent with the Kaplan-Meier curves. Compared to the high collagen width group, the low collagen width group had a 57% lower chance of dying, hazard ratio = 0.43, 95% CI [0.23, 0.78], p = .0056. The low collagen straightness group had a 60% lower chance of distant recurrence than the high collagen straightness group, hazard ratio = 0.40, 95% CI [0.19, 0.85], p =.0163. **Conclusions:** Quantitative collagen fiber metrics could help identify patients with resected pancreatic adenocarcinoma who are at risk of earlier death and distant recurrence. Research Sponsor: None.

Mortality and other outcomes of hospitalizations due to pancreatic cancer.

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Background: Pancreatic cancer continues to be a highly fatal malignancy of the digestive system and imposes a significant financial strain on healthcare systems. Although it is often diagnosed incidentally, several risk factors have been identified, including smoking, family history, age, race, gender, and alcohol use. In this study, we aimed to examine inpatient outcomes and the healthcare burden of patients hospitalized with pancreatic cancer. **Methods:** Patients admitted to the hospital in 2019 and 2020 were identified using the Nationwide Inpatient Sample database and categorized into two groups based on the presence of a pancreatic cancer diagnosis (with relevant ICD-10 codes). To account for confounding factors, multivariate regression analysis was applied when calculating mortality. Statistical analyses were conducted to assess mortality rates, length of stay and hospital charges. **Results: :** A total of 74,584 patients met the inclusion criteria, with a mean age of 68.5 years. The majority were white (69%), while black patients made up 13% of the study population, and 49% were female (36,570 patients). Thrombosis (OR: 2.08, $p < 0.001$), age over 60 (OR: 1.22, $p = 0.042$), and being of black race (compared to white race, OR: 1.45, $p = 0.001$) were linked to higher mortality rates. Thrombosis was also associated with longer hospital stays (+1.9 days, $p<0.001$) and increased costs (+$16,003, $p= 0.001$). However, age over 60 (+0.26 days, $p = 0.088$; +$589, $p = 0.833$) and black race (+1.01 days, $p< 0.001$; +$1,493.7, $p = 0.649$) showed trends toward longer stays and higher costs, these were not statistically significant. On the other hand, pancreatic conditions such as chronic pancreatitis, pancreatic cysts, pseudocysts, or exocrine pancreatic insufficiency (OR: 0.589, $p < 0.001$), female gender (OR: 0.78, $p = 0.002$), and care at urban teaching facilities (OR: 0.37, $p < 0.001$) were associated with lower mortality. While female gender (+0.06 days, p=0.603; -$1699, p=0.448) did not significantly impact hospital length of stay or costs, pancreatic disease (+0.33 days, $p = 0.031$; +$10,826, $p < 0.001$) and being treated at urban teaching facilities (+1.8 days, $p < 0.001$; +$57,106, $p < 0.001$) were linked to longer stays and higher costs. **Conclusions:** Although some unmodifiable risk factors significantly increase mortality, thrombosis contributes not only to higher mortality but also to increased healthcare costs and longer hospital stays. However, it may be preventable with anticoagulation therapy. The inverse relationship between pancreatic disease and mortality could potentially be explained by more frequent imaging and earlier detection of malignancies in these patients. Research Sponsor: None.

Analysis of factors influencing in-hospital mortality, length of hospital stay, and total charges in hospitalized patients.

Data	Mortality	Length of hospital stay	Total Charge
Age>60	1.22(p=0.042)	+0.26 days, p=0.088	$589, p=0.833
Race (black compared to white race)	1.45 (p=0.001)	+1.01 days, p< 0.001	$1493.7, p=0.649
Female gender	0.78 (p=0.002)	+0.06 days, p=0.603	-$1699, p=0.448
Treatment at urban teaching facility	0.37(p<0.001)	+1.8 days, p<0.001	$57,106, p<0.001
Thrombosis	2.08(p<0.001)	+1.9 days, p< 0.001	$16,003, p= 0.001
Pancreatic conditions	0.589(p<0.001)	+0.33 days, p= 0.031	$10,826, p<0.001

Comparison of mutational landscape in primary and metastatic pancreatic tumors.

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Background: Genomic profiling has brought to the forefront the clinical relevance of intratumoral heterogeneity. Oncogenic *KRAS* mutation is the most common driver event in pancreatic ductal adenocarcinoma (PDAC), but little is known about how *KRAS* alteration may cooperate with other genetic lesions to support dissemination and growth of metastatic disease. Therefore, an improved understanding of molecular events affecting the clinical course of *KRAS*-altered PDAC is needed to identify biomarkers of early disease and expand the repertoire of targeted therapies. **Methods:** We retrospectively analyzed tissue samples from PDAC patients using the OncoExTra tumor-normal whole exome and whole transcriptome test between April 2018 and May 2024. The prevalence of actionable somatic alterations was determined overall and by specimen location (primary vs. metastatic). Fisher's exact test with FDR-based correction for multiple testing was used to identify co-alterations with *KRAS* and 19 DNA damage response (DDR) gene alterations. **Results:** PDAC samples from 434 patients (51.2% female; mean age 65.4±10.9 years) were analyzed. *KRAS* alterations were detected in 369 (85.0%) samples: 198 (45.6%) primary and 239 (55.1%) metastatic. $KRAS^{G12D}$ (32.7%) and $KRAS^{G12V}$ (29.5%) were the most common *KRAS* mutations. The frequency of *KRAS* alterations was 85.9% and 84.5% in primary and metastatic samples, respectively. *TP53* alterations occurred in 68.2% samples overall and in 64.6% and 71.1% of primary and metastatic samples, respectively. Examination of co-alterations in *KRAS*-altered versus *KRAS*-WT samples uncovered enrichment of *TP53* alterations in *KRAS*-altered (277 samples, 75.1%) compared to *KRAS*-WT samples (19 samples, 29.2%) ($p<0.001$). Also, *BRAF* mutations and *SND1:BRAF* fusions exclusively occurred in metastatic *KRAS*-WT samples, although the number of events was small ($n=4$ and $n=2$, respectively). We also found that *MTAP* was significantly co-altered in metastatic *KRAS*-WT samples (n=6, 16.2%) compared to *KRAS*-altered samples (6 samples, 3.0%) ($p=0.004$). A total of 21.9% of all PDAC samples had an alteration in at least 1 DDR gene. No associations were observed between *KRAS* alteration status and alteration in DDR genes. However, *TP53* alterations were more frequent in non-DDR-altered samples ($p<0.001$), while *CDKN2B* alterations were enriched in DDR-altered samples ($p=0.025$). **Conclusions:** We observed expected rates of alterations in known PDAC driver genes and uncovered higher co-occurrence of *KRAS* and *TP53* alteration in metastatic samples. Additionally, *TP53* alterations were less frequent in DDR-altered samples. Our findings suggest that a number of therapy approaches including BRAF, WEE1, PARP, and KRAS inhibitors as well as epigenetic modulators, alone or in combination, could improve clinical outcomes in PDACs with specific mutational profiles. Research Sponsor: Exact Sciences.

Exploring predictive factors for detecting KRAS mutations in pancreatic cancer using liquid biopsy.

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Background: Liquid biopsy offers an alternative to tissue specimens for comprehensive genetic profiling. However, circulating tumor DNA is sometimes insufficient to detect genomic alterations in pancreatic cancer (PC). The detection rate of KRAS mutations was reportedly increased in PC with liver metastasis. In contrast, the predictive value of CA19-9 levels, the timing of liquid biopsy, or the type of liquid biopsy for detecting KRAS mutations remains unclear. We investigated whether these factors could be predictive of detecting KRAS mutations in PC. **Methods:** From September 2021 to August 2024, we retrospectively reviewed patients with PC who underwent liquid biopsy. **Results:** A total of 106 patients were enrolled (median age: 64 years, male: 66). Patient characteristics were as follows: tumor status: locally advanced: 10, metastatic: 90, recurrence: 6; metastatic site: liver: 51, lung: 26, lymph nodes: 20, peritoneum: 38, others: 7; number of metastatic sites: 1: 63, 2: 25, ≥3: 8; treatment line at liquid biopsy: 1: 21, 2: 77, 3:8; median CA19-9 level: 1053.4 U/mL; type of liquid biopsy: FoundationOne Liquid CDx (FL): 75 or Guardant360 (G): 31. The overall rate of KRAS mutations was 47%, with a median variant allele frequency of 0.83 [range:0.1-71.0]. The patterns of KRAS mutations were as follows: G12D: 23, G12V: 18, G12R: 4, and others: 14. The detection rate of KRAS mutations varied by the follows factors: metastatic site: liver: 65%, other organs: 44%; number of metastatic sites: ≥2: 47%, <2: 48%; CA19-9 level: ≥2,000 U/mL: 68%, <2,000: 38%; timing of liquid biopsy: 1st line: 42%, ≥2nd line: 67%; types of liquid biopsy: FL: 49%, G: 42%. In multivariate analysis, liver metastasis and CA19-9 level ≥2,000 U/mL were identified as independent factors associated with improved detection of KRAS mutations. In PC with these two factors, the detection rate of KRAS mutation increased to 76%. **Conclusions:** Our findings indicate that liver metastasis and CA19-9 levels above 2,000 U/mL are predictive factors for detecting KRAS mutations in PC through liquid biopsy. Research Sponsor: None.

Prognostic implication of body composition changes in patients with borderline resectable/locally advanced pancreatic adenocarcinoma (PDAC) treated with mFOLFIRINOX.

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Background: Although cachexia is a known prognostic factor in patients with advanced cancer, individual body composition changes during chemotherapy and their association with survival in PDAC warrant further investigation. We aimed to analyze the prognostic implication of comprehensive body composition, including areas and attenuation of muscle mass, body fat area and its distribution, and body mass index (BMI) during mFOLFIRINOX in patients with PDAC. **Methods:** Patients with borderline resectable or locally advanced PDAC, who were not able to undergo curative surgery and received first-line mFOLFIRINOX between January 2017 and December 2020 from Asan Medical Center, Seoul, Korea, were included in this retrospective study if they had paired CT scans of the abdomen at baseline and after 12 weeks of mFOLFIRINOX. Body composition was measured using artificial intelligence software (AID-UTM, iAID Inc.) from the CT images, and their association with overall survival (OS) were analyzed. **Results:** A total of 377 patients were included. Median age was 63 (range, 56–69) and 214 patients (56.8%) were males and 163 patients (43.2%) were females. During the first 12 weeks, significant changes in the body composition occurred as follows (median changes and interquartile range): skeletal muscle area, -5.8% [-11.1%–0.6%]; normal attenuation muscle area/total attenuation muscle area, -2.8% [-8.7%–3.2%]; visceral fat area (VFA), -6.1% [-23.7%–18.6%]; skeletal fat area (SFA), -12.2% [-27.8%–77.1%]; and BMI, -2.2% [-6.8%–1.5%]. Furthermore, changes in the VFA and BMI was associated with tumor response (Kruskal-Wallis test, $p = 0.012$ for VFA; $p = 0.054$ for BMI). There was no significant relationship between body composition abnormalities including sarcopenia, myosteatosis, or obesity at baseline and OS. However, patients who experienced the greatest decrease in SFA and BMI after 12 weeks of mFOLFIRINOX had poorer OS (tertile 1 [greatest decrease] vs 3 [smallest decrease]: HR 0.67 [95% CI, 0.52–0.87] for SFA, $p = 0.003$; 0.68 [95% CI, 0.49–0.83] for BMI, $p = 0.001$). **Conclusions:** In patients with borderline resectable/locally advanced PDAC, body composition significantly changes during the first 12 weeks. Decrease in the body fat and BMI from baseline were associated with poorer tumor response and OS. Research Sponsor: None.

Clinical outcomes of malignant portal vein thrombosis in patients with pancreatic cancer: A retrospective analysis.

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Background: Portal vein thrombosis (PVT) includes portal vein blockage with a blood clot. Recently, there has been an increase in the incidence of PVT with solid malignancy. Portal vein thrombosis (PVT) is a potentially high-risk complication associated with pancreatic cancer. A hypercoagulable state is frequently observed in pancreatic cancer, with the incidence of thromboembolic events reported to range between 17% and 57%. We wanted to analyze the prevalence of PVT in patients with pancreatic cancer and its impact on hospital outcomes. **Methods:** We performed a retrospective analysis using the National Inpatient Sample Database (NIS) from 2018 to 2020. Patients with the primary diagnosis of pancreatic cancer and age >18 were diagnosed using ICD-10 codes. They were then stratified based on the presence of portal vein thrombosis. The Chi-square test was used to analyze categorical variables, and the T-test was used to analyze continuous variables. The confounders were adjusted using multivariate regression analysis. We observed the outcomes on mortality, length of stay, and total hospital charges. **Results:** A total of 336894 patients were diagnosed with a primary diagnosis of pancreatic cancer. Of them, 0.04% (n=13475 patients) were diagnosed with metastatic portal vein thrombosis. The mean age of the patients was 64 years (p <0.001). Out of them, 47% of patients were female. In comparison, the remaining 53% were male (p-value: 0.5). Almost 71% f the patients were White, 13% were African American and 0.09% were Hispanics (p-value <0.001). The presence of PVT was found to be associated with increased mortality in pancreatic patients (aOR 1.6, p-value <0.001) as compared to those without PVT. It was also associated with an increase in mean length of stay (LOS)(7.1 vs 6 days, p-value 0.009) and total hospital charges ($86,000 vs $73794, p-value <0.001). It was also observed that the presence of PVT was associated with increased odds of secondary outcomes, including venous thromboembolism (p-value <0.001), stroke (p-value 0.5), and acute kidney injury (p-value <0.001). **Conclusions:** The presence of portal vein thrombosis (PVT) in pancreatic cancer patients is associated with significantly worse hospital outcomes. These findings emphasize the importance of early detection and aggressive management of PVT in this patient population. Research Sponsor: None.

Comparison of hospital outcomes between pancreatic cancer patients with and without portal vein thrombosis.

Hospital Outcomes	Pancreatic cancer patients with PVT	Pancreatic cancer patients without PVT	Adjusted Odds Ratio (aOR)	95% CI	P-value
Mortality	9.4% (n=1310)	7.2% (n=23300)	1.3	0.3-1.5	<0.001
LOS (Days)	7.1	6.0	1.6	1.1-2.3	0.009
Venous thromboembolism (VTE)	11% (n=1645)	4.2% (n=13734)	2.9	2.5-3.3	<0.001
Stroke	0.03% (n=5)	0.02 (n=79)	1.8	0.2-14	0.5
Acute Kidney Injury	21% (n=2970)	20% (n=68499)	1.2	1.09-1.3	<0.001

Cohort-level patient-derived organoid assays and prediction of clinical trial outcomes in PDAC.

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Background: Clinical trials are one of the major barriers in contemporary drug development. Improving the number of therapeutic solutions for patients whilst limiting the financial burden of healthcare systems will require dramatically improving clinical trial success rates. Functional assays based on patient-derived organoids (PDO) are a promising new tool for derisking clinical development of new therapies, but their use has been limited due to small cohort sizes and the absence of systematic validation studies. **Methods:** We have assembled a collection of 70 demographically relevant PDAC PDOs. Each PDO was screened with a 22-drug panel including standard of care, chemotherapies and targeted therapies. Using a best-in-class patient response prediction model we generate cohort scale estimations of overall response rates (ORR) and progression-free survival (PFS). The estimations are compared with data from published clinical trials to determine the validity of the approach. **Results:** The PDOs exhibit representative treatment histories in terms of number of lines of treatment (mean = 1.60) and treatment types (prior treatment types: folfirinox = 68.6%, gemcitabine = 50%). All the organoids' mutational profile, past patient treatments and subsequent responses to future lines were analyzed and compared to the distributions observed in the clinic. The cohort-level drug efficacy on our PDO collection matches retrospective clinical trial ORRs reported in the literature. Furthermore, our preclinical assay predicts the relative performance of drugs in different clinical trial arms. **Conclusions:** We report that large-scale PDO-based assays predict clinical ORR and PFS with best-in-class accuracy. This work shows that well-characterized PDO collections can be used to improve the success rates of clinical studies for new potential treatments in PDAC. Research Sponsor: Fondation Jean-Jacques et Felicia Lopez-Loreta pour l'Excellence Académique.

Updated trends in incidence and mortality of pancreatic cancer. an analysis of the Surveillance, Epidemiology, and End Results (SEER) database.

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Background: Pancreatic cancer is the fourth leading cause of cancer deaths in the United States. The prognosis for pancreatic cancer remains poor despite advances in cancer therapy. In this study, we provide updated trends in pancreatic cancer incidence and mortality using the SEER database, with a subset analysis of age and gender. **Methods:** Using the SEER Stat software, we gathered data from the Surveillance, Epidemiology, and End Results (SEER) database (2000–2021). We employed Jointpoint regression to analyze the secular trend of incidence and incidence-based mortality (IBM), which we classified and arranged according to age and sex. **Results:** 235,116 cases met the inclusion criteria for our study. These included diagnosed cases of pancreatic cancer from 2000 to 2001. The total age-adjusted incidence rate was 12.5 per 100,000. 118,791 (50.5%) were males and 116,325 (49.5%) were female. While the annual percentage change (APC) from 2000-2019 was 0.92 ($p < 0.05$), signifying an annual cumulative increase, APC from 2019-2021 was -0.085, signifying an annual cumulative decrease. For males, the APC was 0.94 ($p < 0.05$) from 2000 to 2018 and -0.55 from 2018-2021, and for females, the APC was -0.89 ($p < 0.05$) from 2019-2021. The pancreatic cancer IBM APC from 2000-2022 was 22.60 ($p < 0.05$) and 0.54 ($p < 0.05$) from 2002-2021. For males, IBM APC was 21.10 ($p < 0.05$) from 2000-2002 and 0.58 from 2002-2021. For females, IBM APC was 25.15 from 2000-2002 and 0.42 from 2002-2021. We also studied pancreatic cancer IBM based on age. For 30-49 years, APC from 2000-2002 was 34.65 ($p < 0.05$), and -0.64 ($p < 0.05$) from 2002-2021. For 50-69 years, APC from 2000-2002 was 28.05 ($p < 0.05$) and 0.3 from 2002-2021. For 70+ years, APC from 2000-2002 was 21.28 ($p < 0.05$) and 0.73 from 2002-2021. **Conclusions:** There was a statistically significant rise in incidence and incidence-based mortality across the board, underscoring the need for improved diagnostic and therapeutic modalities of management of pancreatic cancer. Research Sponsor: None.

TPS787 **Trials in Progress Poster Session**

Intra-tumoral CD40 antibody with irreversible electroporation (IRE) in locally advanced pancreas cancer.

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Background: Irreversible electroporation (IRE) is a form of non-thermal tumor ablation that is currently in clinical use for selected patients with locally advanced pancreatic cancer (LAPC). Preclinical and clinical studies have suggested that IRE generates an in situ vaccination effect by releasing tumor neoantigens in the setting of inflammation. Our published data from syngeneic, orthotopic mouse models demonstrate that combination of IRE with local CD40 agonism induces systemic anti-tumor immune effects, including neoantigen-specific T-cell responses and inhibition of liver metastases (1). Mitazalimab, a second-generation CD40 agonistic IgG1 mAb, has demonstrated promising clinical antitumor activity when delivered systemically in combination with modified FOLFIRINOX in metastatic pancreatic cancer (2). **Methods:** This clinical trial (NCT06205849) is a phase I dose-escalation study of an agonistic CD40 antibody (mitazalimab) injected intratumorally at the time of surgical IRE in patients with LAPC who have not demonstrated distant disease progression after a minimum of 4 months of optimal standard of care (SOC) chemotherapy (modified FOLFIRINOX). The doses investigated are 25, 75 (starting dose), and 200 μg/kg, in an interval 3+3 design, with a maximum of 18 subjects. The primary endpoints are the rates of dose limiting toxicities (DLTs) and treatment-emergent adverse events (AEs). The secondary endpoints are progression-free survival (PFS) and overall survival (OS). At a liberal alpha of 25%, appropriate for this proof-of-concept early phase study, we have 80% power to detect an improvement in PFS from 12 months (based on prior studies of IRE alone) to 18 months, with an expected 16 evaluable subjects and an anticipated 11 observed events, using a one-sided log-rank test. Candidate neoantigens will be identified by profiling nucleic acids derived from tumor biopsies obtained intraoperatively prior to IRE using our Identification-Prioritization-Validation (IPV) bioinformatic pipeline (3). Blood samples will be collected pre-operatively and 12-weeks post-operatively to evaluate for evidence of neoantigen-specific T-cell reactivity and other neoantigen-independent analyses of immune response. We have enrolled two patients since the study opened in September 2024. 1. J. Shankara Narayanan et al., JITC 2023. 2. J. Van Laethem et al., Lancet Oncol 2024. 3. A. Miller et al., Sci Transl Med. 2024. Clinical trial information: NCT06205849. Research Sponsor: U.S. National Institutes of Health; R01 CA282439.

TPS788 **Trials in Progress Poster Session**

EVEREST-2: A seamless phase 1/2 study of A2B694, a logic-gated Tmod chimeric antigen receptor T-cell (CAR T) therapy, in patients with pancreatic cancer (PANC) or other mesothelin (MSLN)-expressing solid tumors and human leukocyte antigen (HLA)–A*02 loss of heterozygosity (LOH).

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Background: MSLN is a desirable candidate for targeted therapy given its limited expression in normal tissues and its overexpression in several malignancies, including PANC, mesothelioma, non-small cell lung, and ovarian cancers (TCGA 2022). MSLN-targeted CAR T and T-cell receptor therapies have shown promising clinical activity; however, on-target, off-tumor toxicity, including fatal events, have occurred (1-3). Autologous logic-gated Tmod CAR T therapies address the challenges of on-target, off-tumor toxicity by combining 2 CARs: activating and blocking receptors. The activator recognizes a tumor-associated antigen present on the surface of both tumor and normal cells; the blocker prevents CAR T activity when it binds HLA-A*02, which is present in normal cells and often lost in tumor cells. Thus, in patients with tumor-associated HLA-A*02 LOH, the blocker prevents on-target, off-tumor toxicity on normal cells due to retained HLA-A*02 expression (4). HLA-A*02 LOH is observed in various solid tumors, including ~20% of PANC (5). Two autologous CAR T therapies are currently under investigation: A2B530, targeting carcinoembryonic antigen, in the EVEREST-1 trial, and A2B694, targeting MSLN, in the EVEREST-2 trial, described herein. A2B694 may provide a therapeutic window to treat patients with PANC and other MSLN-expressing solid tumors exhibiting HLA-A*02 LOH due to its unique discriminatory mechanism. **Methods:** EVEREST-2 (NCT06051695) is a first-in-human, phase 1/2, open-label, nonrandomized study to evaluate the safety and efficacy of A2B694 for recurrent unresectable, locally advanced, or metastatic cancers with MSLN expression, including PANC. Eligible patients must have exhausted options of potentially curative therapy and have recurrent disease. Enrollment to EVEREST-2 occurs through the prescreening study BASECAMP-1 (NCT04981119), in which patients with tumor-associated HLA-A*02 LOH are identified via next-generation sequencing (Tempus AI, Inc) and leukapheresis product is collected. A2B694 is manufactured from cryopreserved T cells when clinically appropriate for patients. The primary objective of phase 1 is to evaluate the safety and tolerability of A2B694 and identify the recommended phase 2 dose (RP2D). The dose-expansion phase will confirm RP2D and collect biomarker data to further characterize A2B694. Phase 2 will assess the primary endpoint overall response rate per RECIST v1.1 and secondary endpoints progression-free survival and overall survival. The first patient was dosed on April 24, 2024 and dose escalation is ongoing. 1. Beatty, et al. *Gastroenterology*. 2018. 2. Haas, et al. *Mol Ther*. 2023. 3. Hong, et al. ESMO 2021. Abstr 9590. 4. Hamburger, et al. *Mol Immunol*. 2020. 5. Hecht, et al. *J Clin Oncol*. 2022. Clinical trial information: NCT06051695. Research Sponsor: A2 Bio.

TPS789 **Trials in Progress Poster Session**

Phase II study of olaparib in patients with advanced pancreatic acinar cell carcinoma.

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Background: Pancreatic Acinar Cell Carcinoma (PACC) is a rare and biologically distinct subtype of pancreatic cancer, representing 0.5-1% of all pancreatic malignancies, with a median overall survival of 15-20 months. Significant serum CA 19-9 elevation is uncommon, while elevation of serum lipase is observed in at least 50% of patients. Most patients present at advanced stages and treatment with combination chemotherapies extrapolated from other GI malignancies leads to low response rates. No clinical treatment trials specific for PACC have ever been reported. PACC tumors from patients show evidence of high chromosomal instability, a hallmark of DNA repair deficiency. ID3 is ubiquitously downregulated and was recently shown to play a significant role in homologous recombination repair (HRR). Recent genomic profiling of larger PACC patient cohorts has revealed a high prevalence of mutations in HRR genes, including BRCA1, BRCA2, and PALB2. Olaparib is a Poly-ADP ribose polymerase (PARP)-1 inhibitor that has been FDA approved for treatment of BRCA-mutant HRR deficient cancers. We hypothesize that PACC will be sensitive to PARP inhibition with olaparib. **Methods:** This is an open label, single arm, single center, phase II study of olaparib in patients with advanced previously treated PACC. The primary objective of the trial is to evaluate anti-tumor activity of olaparib (objective response rate). Secondary objectives assess the safety (CTCAE v. 5.0) and alternative measures of efficacy such as clinical outcomes and blood biomarkers (disease control rate, median progression-free survival, median overall survival, best response in serum lipase). Key eligibility criteria include histologic diagnosis of PACC, receipt of at least one line of combination chemotherapy, presence of RECIST measurable disease and adequate organ and bone marrow function. A fixed dose of olaparib at 300 mg will be given orally continuously twice daily during a 28-day cycle, for up to 2 years or until disease progression or intolerable toxicities. In person visits to assess toxicity are required every 4 weeks, and response is determined through serial imaging every 8 weeks (CT or MRI). Exploratory objectives include acquisition of tumor tissue for establishment of new PACC preclinical models, assessment of genetic alterations through genetic mapping technique, assessment of HRR-related gene products as a predictive biomarker, assessment of ID3 protein levels, and assessment of treatment response to tumor profile. Up to 13 evaluable participants will be enrolled. Three participants have been enrolled as of Sep 19, 2024. Clinical trial information: NCT05286827. Research Sponsor: None.

TPS790 **Trials in Progress Poster Session**

A phase 2 study of amplitude-modulated radiofrequency electromagnetic fields (AM RF EMF) in metastatic pancreatic cancer.

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Background: Patients with advanced pancreatic adenocarcinoma (PDC) that progresses after first-line therapy face limited treatment options. Despite advances in systemic therapy, the prognosis for PDC is dismal, with a five-year overall survival (OS) of about 5% . Due to the high morbidity of this disease, there is a critical need for innovative treatments to improve patient outcomes. TheraBionic P1, a hand-held device, delivers tumor specific radio frequencies via a spoon-shaped antenna placed on the patient's tongue. The FDA and EMA have confirmed the safety of the device for amplitude-modulated electromagnetic field (AM-EMF) therapy. Improved efficacy outcomes were reported with AM RF EMF in other solid organ tumors including pancreatic cancer in preliminary studies. Preclinical and clinical evidence suggests that AM-EMF can inhibit tumor growth and was recently approved for use in hepatocellular cancer. The safety and clinical activity observed with AM RF EMF, paired with the unmet medical need for additional effective therapies in pancreatic cancer, provided a strong rationale for further clinical investigation. This protocol evaluates the efficacy and tolerability of AM RF EMF in combination with standard of care frontline therapy, gemcitabine and nab-paclitaxel for metastatic PDC. **Methods:** This is an ongoing, single-center, Phase II, non-randomized study using a Simon two-stage minimax design to assess the efficacy of AM-EMF in combination with gemcitabine/nab-paclitaxel as a first-line treatment in metastatic pancreatic cancer. Gemcitabine and nab-paclitaxel are administered per standard care. AM RF EMF are delivered to patients with a spoon-shaped applicator placed on the tongue three daily 60-minute treatments. Patients will be assessed for response every 8 weeks. Treatment is continued until progression or toxicity. The primary objective of this study is to evaluate the efficacy of the combination of nab-paclitaxel, gemcitabine, and AM RF EMF in improving 6-month overall survival rates in patients with metastatic PDC. Secondary objectives evaluate the safety, tolerability, profession free survival, objective response rate and disease control rate. Exploratory objectives evaluate change in CA19-9 and quality of life using FACT-General (FACT-G) and determine any possible correlation with clinical outcomes. Key eligibility includes treatment-naïve, histologically confirmed metastatic PC, age eighteen years old or greater, ECOG 0-1, and optimal organ function. Clinical trial information: NCT05776524. Research Sponsor: Wake Forest Baptist Health Comprehensive Cancer Center Boot Camp.

TPS791 **Trials in Progress Poster Session**

UNITEPANC: Using organoids to predict efficacy of adjuvant treatment to improve outcome in resectable pancreatic cancer—Prospective, multicenter, proof-of-concept trial of the AIO Pancreatic Cancer Group.

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Background: Pancreatic cancer (PDAC) is a highly heterogeneous disease and there is a lack of predictive biomarker to guide treatment. Patient tumor derived organoids (PDO) have been demonstrated to faithfully recapitulate morphological and genetic features of the parental tumor and also retain patient-specific heterogeneity. Retrospective studies show that PDO can be used to predict treatment sensitivity and can support the selection of an optimal treatment in terms of efficacy and also lower toxicity, e.g. by selecting a double rather than a triple combination. **Methods:** UNITEPANC is a prospective, proof of concept, multicenter IIT of the German AIO PDAC Group and funded by the German Cancer Aid. UNITEPANC examines the feasibility of an organoid-educated adjuvant treatment approach with respect to organoid establishment, expansion and characterization and its potential role for selecting an optimal adjuvant treatment in PDAC in a multicenter setting. Preparatory activities like harmonizing SOPs and round-robin-tests for generation of organoids and organoid-based pharmacotyping in the trial centers and the different Organoid Facilities has been completed and evaluated. UNITEPANC examines selection of adjuvant chemotherapy by pharmacotyping of tumor organoids. Options for adjuvant treatment are gemcitabine, gemcitabine/capecitabine, gemcitabine/nab-paclitaxel or mFOLFIRINOX. Major inclusion criteria are R0 or R1 resected, histologically confirmed PDAC and patients in principle, suitability for mFOLFIRINOX treatment in the adjuvant setting, postoperative Ca19-9<180 U/ml, and a timely PDO-based chemotherapy recommendation by the UNITEPANC Organoid Board within 11 weeks to guarantee a start of the adjuvant treatment within 12 weeks after resection. Design: UNITEPANC is an interventional, prospective, multicenter, single-arm trial. 95 patients shall be allocated to the trial for the generation of organoids, 38 patients shall be enrolled for PDO-based adjuvant treatment and 34 patients need to be analysed as ITT population in in the follow-up period. Efficacy: The efficacy endpoint of the trial is purely descriptive and intended to obtain a signal for further development of the strategy. This requires the proof of an efficient 1) generation and 2) expansion of PDO and 3) a clear sign that an organoid-educated treatment selection is superior to standard treatment with mFOLFIRINOX. For 1) and 2) we have chosen a rate of 75% and 60%, respectively, according to the literature. For 3): The approach would be considered promising, if the true 18-months-DFS rate is 77% (corresponding to a hazard ratio of 0.5 compared to Prodige-24, 80% power, one-sided type I error 0.1. HR-QoL data will be analyzed (QLQ-C30, Pan26). Additionally, an extensive translational program is enclosed. Research Sponsor: German Cancer Aid.

TPS792 **Trials in Progress Poster Session**

Phase I study of autologous CD8+ and CD4+ transgenic T cells expressing high-affinity KRAS G12V mutation-specific T cell receptors (FH-A11KRASG12V-TCR) in patients with metastatic pancreatic, colorectal, and non-small cell lung cancers with KRAS G12V mutations.

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Background: KRAS, the most common driver oncogene in human malignancies, has predictable mutations. 85% of RAS mutations occur at the Glycine (G) 12 position with G12V being amongst the most common. Cancers driven by mutant KRAS include pancreatic (PDAC), colorectal (CRC), and non-small cell lung (NSCLC) malignancies. The prognosis of patients with activating KRAS mutations remains poor. Preliminary anti-tumor activity has been observed with T cell receptor (TCR) T cell therapies targeting KRAS mutations, but durable clinical benefit is lacking. FH-A11KRASG12V-TCR is an autologous CD8+ and CD4+ transgenic T cell product expressing (1) an HLA-A*11-restricted, high affinity KRASG12V mutation-specific T cell receptor (TCR) and (2) the wildtype CD8α/β coreceptor enabling CD4+ T cell recognition of KRAS G12V presented on an MHC class I allele mediating helper T cell activity and cytotoxicity. The TCR sequence is derived from the peripheral blood of a healthy HLA-A*11+ adult based on a strong pre-clinical dataset. **Methods:** This is an ongoing, investigator-initiated, phase 1, first-in-human, single-center, dose escalation study of FH-A11KRASG12V-TCR. The aims are to evaluate the safety, tolerability, maximum tolerated dose (MTD), and preliminary anti-tumor activity of FH-A11KRASG12V-TCR in patients with advanced PDAC, CRC, and NSCLC who express the HLA-A*11-restricted KRAS G12V mutation. Dose modification is guided by the Bayesian Optimal Interval design (BOIN). Patients are initially treated in cohorts of 3 at each dose level (DL). BOIN will be used to identify the recommended dose for subsequent patients. The planned target doses are 5×10^9 transgenic TCR T cells in DL 1 with dose escalation up to 15×10^9 transgenic TCR T cells in DL 3. This study enrolls patients who are 18 years or older, HLA-A*11:01-positive, with KRAS G12V mutated metastatic PDAC, CRC, or NSCLC. Patients must have progressed on or be intolerant of at least one line of prior therapies including any targeted therapies indicated for their current malignancy. Patients undergo leukapheresis prior to treatment and receive lymphodepleting chemotherapy with either cyclophosphamide intravenously (IV) and fludarabine IV on days -6, -5, and -4 or bendamustine IV on days -4 and -3. FHA11KRASG12V-TCR is administered IV on day 0. The dose-limiting toxicity observation period is 28 days. Patients may receive an additional FHA11KRASG12V-TCR IV infusion at the same or lower dose as soon as 28 days or up to 1 year after the first infusion. The study is actively recruiting. Clinical trial information: NCT06043713. Research Sponsor: Affini-T Therapeutics.

TPS793 **Trials in Progress Poster Session**

Randomized phase II trial of olaparib + pembrolizumab vs olaparib alone as maintenance therapy in patients with metastatic pancreatic cancer with germline BRCA1 or BRCA2 mutations: SWOG S2001 NCT04548752.

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Background: Olaparib was approved in 2019 as maintenance therapy for gBRCA1/2+ metastatic pancreatic cancer (mPDA) patients (pts). The POLO trial showed an improvement in median progression-free survival (mPFS) with olaparib compared to placebo (7.4 vs 3.8 months) for platinum sensitive gBRCA1/2+ mPDA pts. Preclinical studies have demonstrated that PARP inhibitors modulate the immune microenvironment by increasing genomic instability, PD-L1 expression and activating the immune inflammatory stimulator of interferon genes (STING) pathway. Results of the phase 2 pembrolizumab and olaparib (POLAR) maintenance trial in patients with germline or somatic BRCA1/2 or PALB2 mutations responding to platinum-based chemotherapy for at least 4 months was presented at ESMO 2024. In 33 patients, a 35% overall response rate with 90% disease control rate at 6 months was observed. S2001 is an important randomized study to better define the impact of the combination. **Methods:** S2001 was developed in collaboration with the Alliance and was activated in SWOG in October 2020. Key eligibility criteria include: mPDA pts with gBRCA1/2 mutations identified with standard of care germline genetic testing and progression-free after receiving at least 4 months of platinum-based chemotherapy (FOLFIRINOX, FOLFOX or gemcitabine/cisplatin +/- nab-paclitaxel). One cycle of gemcitabine and nab-paclitaxel is allowed while waiting for germline testing. Zubrod performance status (PS) 0 or 1 pts are eligible. Pts are stratified according to first line chemotherapy, PS 0 vs 1, and disease status after 1st line treatment. The primary objective of this study is to evaluate the PFS of mPDA pts treated with olaparib + pembrolizumab compared to olaparib alone as maintenance therapy. Based upon the POLO trial, we expect a mPFS of 7 months in the control arm. Targeting a mPFS of 11.7 months in the experimental arm (hazard ratio 0.6) and assuming 15 months follow-up, 80% power and a 1-sided alpha = 0.10, this design requires 78 evaluable pts with a total sample size of 88 pts. Prospective serial blood samples will be collected to bank DNA and RNA for future correlative studies. Support: NIH/NCI grants U10CA180888 and U10CA180819, U10CA180821 and U10CA180868. Clinical trial information: NCT04548752. Research Sponsor: NIH/NCI grants.

TPS794 **Trials in Progress Poster Session**

A phase I, single-center, open label, dose de-escalation and expansion study of Ivosidenib + mFOLFIRINOX in patients with resectable pancreatic adenocarcinoma.

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Background: Pancreatic ductal adenocarcinoma (PDA) carries a dismal prognosis, with a 34% 5-year survival rate for patients with localized disease. Guidelines recommend consideration of neoadjuvant approaches in localized PDA with the goals of controlling microscopic metastatic disease and increasing rates of microscopically margin-negative resections. One of the hallmark characteristics of PDA is a micronutrient poor, highly stromal microenvironment. Surviving in this environment requires enhanced mitochondrial function, which utilizes isocitrate dehydrogenase 1 (IDH1). In pre-clinical studies, ivosidenib, an FDA approved medication for IDH1 mutant AML, induced high levels of reactive oxygen species in PDA cells with wild-type IDH1, and caused tumor regressions and extended survival in implanted KPC mouse models. Here, we investigate the addition of ivosidenib to standard of care neoadjuvant mFOLFIRINOX in patients with resectable PDA to determine safety, pharmacodynamics, and early efficacy signals. **Methods:** This is a phase I, single-center, open label, dose de-escalation and expansion study in patients with resectable PDA. Eligible patients must have histologically confirmed and resectable right-sided PDA based on CT or MRI imaging. Patients receive approximately 10 weeks of neoadjuvant treatment composed of 2 weeks of ivosidenib monotherapy, followed by 6 weeks (3 cycles) of mFOLFIRINOX with ivosidenib, followed by up to 4 weeks of ivosidenib monotherapy until the day of surgery. Ivosidenib is administered once daily at 500mg; it is to be de-escalated to 250mg based on Bayesian Optimal Interval Design with Informative Prior and a target dose limiting toxicity (DLT) rate of 30%. The primary endpoint is to determine the safety and tolerability of ivosidenib in combination with mFOLFIRINOX. Secondary endpoints include RECIST version 1.1 response rates, major pathologic response rates, and biochemical (CA19-9, CEA) response rates. Correlative studies will be performed on surgical samples to evaluate metabolomic profiles. At the time of submission, 10 out of 16 planned patients have been enrolled. There have been no DLT at 500mg of ivosidenib. Clinical trial information: NCT05209074. Research Sponsor: Gateway for Cancer Research Foundation.

TPS795 **Trials in Progress Poster Session**

A phase II study of peri-operative NovoTTF-200T(P) in combination with gemcitabine and nab-paclitaxel for resectable pancreatic adenocarcinoma: Big Ten Cancer Research Consortium (BTCRC-GI21-500).

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Background: The recurrence rates and outcomes in resectable pancreatic ductal adenocarcinoma (R-PDA) is concerningly high. The benefit of neoadjuvant chemotherapy over traditional upfront surgery followed by adjuvant therapy is not clear. Electromagnetic fields generate bi-directional forces on highly polar intracellular components, causing abnormal microtubule polymerization during spindle formation and irregular cleavage furrow formation. We will leverage these anti-proliferative effects in managing R-PDA by tumor treating fields (TTF) with chemotherapy backed by strong in vitro and in vivo evidence. The PANOVA phase II trial (n=40) gave us safety and efficacy data of this combination in advanced PDA. No systemic effects were associated with TTF. The only notable safety issue was a small incidence of grade 3 device-related dermatitis. **Methods:** Our phase II single arm study will investigate peri-operative use of TTF with gemcitabine and nab-paclitaxel (G-NP) combination. Eligible patients R-PDA (visible pancreatic mass, measurable disease, absence of arterial interface, venous interface $\leq 180°$, patent portal splenic confluence, and no metastatic disease, including lymphadenopathy outside the surgical area) will wear TTF and receive three cycles of G-NP before restaging. Patients are expected to wear TTF for $> 80\%$ of the time during this period. If they proceed with resection, additional 3 cycles of Gem-NP with TTF will be given to the patient. This study employs a Bayesian Optimal Phase 2 (BOP2) design with primary endpoints of overall survival (OS) at 2 years and resection rate. The null hypothesis posits a 40% OS rate compared to a target of 60%, and a resection rate null hypothesis of 60% versus a target of 75%. The trial aims to enroll 30 patients, with an interim futility analysis planned after 15 participants. Enrollment will be terminated early if 8 or fewer patients undergo resection in the initial stage, providing 83% power and a type I error rate of 10%. Secondary endpoints include adverse events, overall response rate, TTFields compliance rate, relative dose intensity, and overall survival. The trial started actively recruiting patients in April 2024. The accrual goal is 38 patients. At the time of submission, one patient was enrolled. Clinical trial information: NCT05624918. Research Sponsor: Big Ten Cancer Research Consortium.